

Curve Fit Bench

Least-squares curve fitting in a single HTML file. Drop in x, y and an optional uncertainty, and either let the bench search every model family for the lowest reduced χ^2 , or type your own equations and have those fitted — both print the functional form with parameters, uncertainties and goodness of fit.

442 named physical models, 426 of them carrying physical bounds

20 standard forms · ~ 70 two-component combinations · **30**-term symbolic search

65 functions in the equation language, including the exact BCS gap, true Voigt, Dynes, BTK and Fermi–Dirac integrals

No server, no install, no network. Everything runs in the page.

Curve Fit Bench — Reference Manual, Edition One

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Part I

Getting data in

What the bench is, and what it is not

A least-squares fitter that runs entirely inside one HTML file, with an opinion about which model you should believe.

Curve Fit Bench does one job: you give it measurements, and it gives you a function that describes them, together with an honest statement of how well that function does. It is a single HTML file. Opening it in a browser is the whole installation. There is no server, no account, no upload; the file works with the network cable pulled out, and nothing you load ever leaves the machine it is loaded on.

Two things distinguish it from a plotting package with a fit button bolted on.

The first is the size of the search. In automatic mode the bench does not fit one model, it fits every model it knows: twenty standard forms, roughly seventy sums of two components, a free-form symbolic search over a thirty-term dictionary, a many-band peak fit, and a library of 442 named models from condensed-matter physics, spectroscopy, magnetism, transport, kinetics and optics. It then ranks them all against each other on the same footing.

The second is that it argues with the ranking. The lowest chi-square is very often the wrong answer — a flexible model with many parameters can beat the correct physical law by fitting the noise, and it will tell you nothing. The bench charges every model for the parameters it used *and* for the size of the search it was picked out of, and then awards the top spot to the simplest model that the leader does not significantly beat. Chapter 14 sets out exactly how that decision is made, because you should not trust a ranking whose rule you cannot inspect.

What it will do well

- Fit a named or typed function to x, y data, with or without error bars, and report the parameters with standard errors, χ^2 , reduced χ^2 , R^2 , RMSE and AICc.
- Tell you which of many candidate functional forms your data actually supports.
- Fit a spectrum with a dozen overlapping bands, having found the bands for you.
- Strip a wandering baseline before fitting, and show you the baseline it removed.
- Fit two datasets side by side, from separate columns or from one long-format file.
- Produce a publication-quality figure with every element under your control, saved as PNG or PDF.

What it is not

- **It is not a global optimiser with guarantees.** Non-linear least squares finds a local minimum from a starting point. The bench works hard on the starting points — data-derived guesses, greedy coordinate sweeps, dozens of restarts — but on a badly conditioned model with many

parameters you can still land in the wrong valley. When that matters, give starting values yourself (Chapter 9).

- **It does not fit in two independent variables.** One x , one y .
- **It does not do Bayesian inference, MCMC, or error propagation into derived quantities.** The uncertainties it reports are the classical ones from the covariance matrix, which assume the model is right and the residuals are Gaussian.
- **It does not fit implicit or parametric curves** (circles, ellipses, hysteresis loops that double back in x). Every model is a single-valued $y = f(x)$.
- **It is not a data-cleaning tool.** It will drop unreadable rows and reject outliers if you ask, but it does not reshape, merge or interpolate.

The shape of the window

Everything lives on one screen. Down the left is a rail of panels, in the order you normally need them:

PANEL	WHAT IT IS FOR	CHAPTER
Fitting mode	Search for the best form, or fit equations you type	8
Data	Drop a file, paste points, choose columns, split datasets	2–5
Physics model library	Search 442 named models and insert one with starting values	10
Your equations	The equations you type, in manual mode	9
Model families	Which families the automatic search may use	11–13
Peak finder	Detect every band in a spectrum and fit them together	17
Weighting	How much each point counts; outlier rejection	15
Background	Estimate and subtract a baseline before fitting	16
History	The last twenty fits, reloadable	21
Ranking	The criterion the ranking uses, and the search charge	14

The wide area on the right holds the plot with its own toolbar, the result card for the best fit, an optional second card for a second dataset, an optional card for the best named physical model, and the table of every model tried.

THE ONE RULE WORTH LEARNING FIRST

Nothing is carried over between visits. Closing the page throws away the loaded data and every setting; reopening starts at defaults with the Kepler sample. Only *History* survives, and it lives in this browser only. If a fit matters, export it — do not expect to find it tomorrow.

Getting data in

Four ways in, one parser, and the rules it follows when your file is not as tidy as you thought.

The four routes

Drop a file

Drag a .csv, .txt, .tsv or .dat onto the dashed box at the top of the Data panel. The box highlights when the file is over it.

Browse

Click the same box (or focus it and press `Enter`) to open a file picker.

Paste

Paste straight into the text area underneath. Fitting starts about four tenths of a second after you stop typing, so you can edit numbers in place and watch the fit follow.

Sample sets

Five chips load real or realistic datasets with sensible axis labels and scales. They exist so you can see the program work before trusting it with your own numbers. They are described in Chapter 22.

A dropped or browsed file is also copied into the paste box, up to four hundred thousand characters, so you can see and edit what was read. Beyond that size the box holds a truncated copy but the fit still uses the whole file.

What the parser accepts

One record per line. The delimiter is decided per line, in this order:

1. a tab anywhere in the line → split on tabs;
2. otherwise a semicolon → split on semicolons (European CSV);
3. otherwise a comma → split on commas;
4. otherwise any run of whitespace.

Each cell is then trimmed, and leading and trailing quotes — single or double — are stripped. A single empty cell at the end of a line is discarded, which is what forgives the trailing comma that spreadsheets like to leave behind.

Lines that are skipped

- Blank lines.
- Lines whose first character is # or %, and lines starting //. Use these freely for provenance: instrument, date, slit width, whatever you will want to know in a year.

- Any line with no readable number at all, once a header has already been found.

Values that become "missing"

An empty cell, or the literal text NA, nan or NaN, is read as missing. A leading + is tolerated. Anything else that is not a number is also missing — but if a whole column is text, the parser keeps the text, because that is how it recognises a long-format file (Chapter 5).

How the header is recognised

The first line that contains fewer than two readable numbers is taken as the header, and only the first such line. So `t_s,counts,sigma` is a header; `0.0,1840,42.9` is data. A file with no header is fine — columns are then called *column 1*, *column 2* and so on. Header names are worth having: they become the axis labels, they are how the bench recognises an uncertainty column, and they name your series in the legend.

Ragged rows

Rows may have different lengths. The table width is the longest row; short rows simply have missing values in the columns they lack, and a row missing its *x* or *y* is skipped and counted.

What "skipped" means in the status box

After every load the status box under the column pickers reports the number of usable points, the range in *x* and in *y*, the weighting in force, and — if any rows were unusable — how many were skipped. A row is skipped when its *x* or its *y* is missing, or when a σ column is in use and that row's σ is missing, zero or negative. A zero uncertainty would demand infinite weight, so it cannot be used.

IF THE COUNT IS ZERO

"Need at least three usable rows" almost always means the column assignment is wrong — most often that *x* is pointing at a column of text. Check the three pickers. If your file is in long format with a name column first, the bench now sets this up for you (Chapter 5), but a file that mixes text and numbers in one column will still need manual help.

Worked examples of acceptable input

```
# Absorption spectrum, sample B3, 2026-03-11
# monochromator slit 0.5 nm, 10 scans averaged
wavelength_nm,transmittance,sigma
340.0,0.0118,0.0009
341.0,0.0125,0.0009
342.0,0.0141,0.0010
```

```
0.387    0.241
0.723    0.615
1.000    1.000
1.524    1.881
```

```
"T (K)";"rho (u0hm cm)"
2,0;1,842
```

| 4,0;1,851

The third example is a European CSV. Semicolons separate the fields, but note that the decimal commas inside the numbers are *not* understood — 1,842 reads as missing. Convert decimal commas to points before loading.

Size

There is no hard limit. Fifty thousand points load and fit; the search takes longer and the bench thins the data internally for the exploratory passes, but the reported χ^2 is always computed on every point you supplied. Chapter 24 explains what is done to keep a large file responsive.

Columns

Which column is x , which is y , which is the uncertainty — and why the bench asks rather than guesses.

Under the paste box are four small pickers: x , y , σ_y and y_2 . They list every column in the file by its header name, or by position if there is no header.

The defaults

On load, x becomes the first column that is not a label column and y the second. A label column is one holding text with only a handful of distinct values — a run name, a sample identifier. This is what lets a file like `dataset, x, y` load correctly instead of pointing x at a column of words and dropping every row.

The uncertainty column

A third column is treated as σ only when the bench is sure, because silently weighting by the wrong column is worse than not weighting at all. Two things can make it sure:

A header that names it

Any column from the third onward whose header begins with, or simply contains, one of these is taken as the uncertainty on y :

sigma	sigma	sd	std	stdev	stddev	err	error	errors
unc	uncert	uncertainty	dy	sy	e_y	yerr	y_err	y_error +-

When that happens the bench selects the column and asks you to confirm, with the reason spelled out: "Using "sigma" as the uncertainty σ on y , so χ^2 is weighted and $\chi^2/\nu \approx 1$ means a good fit." Press **correct** or **no — ignore it**.

You say so

If the file has three or more columns but no header that names an uncertainty, the bench asks the plain question instead: "3 columns found. Is "whatever the third column is called" the uncertainty σ on y ?" Answer **yes — it is σ** or **no — just x and y** . Until you answer, the data is fitted unweighted.

You can also set the picker by hand at any time; the question disappears once you do.

WHY THIS MATTERS MORE THAN IT LOOKS

With a σ column, χ^2 is measured in units of your own error bars and $\chi^2/\nu \approx 1$ becomes a real

statement about whether the model fits. Without one, every point gets unit weight, χ^2 is a plain sum of squared residuals in units of y^2 , and its absolute value means nothing at all — only the ranking between models on the same data carries information. Chapter 19 goes through this properly.

The second y column

The fourth picker, y_2 , appears when the file has three or more columns. Choose a column and the bench fits it as an independent second dataset against the same x : its own model search, its own ranking, its own result card, its own points and curve on the plot. Chapter 6 covers the whole two-dataset workflow.

Setting y_2 to the same column as y is refused — it silently resets to *none*.

Changing columns later

Changing any picker refits immediately and unlocks the axis labels, so they follow the new columns. If you had typed your own axis titles in the plot style editor, those are kept — an explicit title always wins over an inferred one.

Choosing the range to fit

Fit the part of the data that the model is supposed to describe, and leave the rest visible.

Almost every real dataset has a region the model does not claim to describe: the detector noise before the edge, the saturated points at the top, the first few seconds while the shutter was still opening. Fitting those does not make the fit more honest, it makes it worse, because least squares will bend the curve to reduce a residual it should never have been shown.

The **fit range** bar sits directly under the plot toolbar, so it is next to the picture you are judging it by. It has four number boxes — x from and to, y from and to — and two buttons.

The three ways to set a range

Drag across the plot

Press and drag horizontally anywhere inside the plot frame. A shaded band follows the pointer; releasing sets the x limits to the band and refits. This is the fastest way and usually the only one you need.

Type the numbers

Fill any of the four boxes. Leaving one empty means "no limit on that side". A value that is not a number is rejected and the box clears.

Clear it

The **all** button empties all four boxes and refits on everything.

Excluded points stay on the plot

Points outside the range are not deleted. They are drawn smaller and faded, so you can always see what you excluded and reconsider. The bar shows a count — "fitting 412 of 900 — 488 excluded" — whenever anything is being left out.

The zoom button

With **zoom** pressed (the default), the axes rescale to the fitted window, so the region you care about fills the frame. Release it and the axes span all the data, with the fitted window highlighted by the faded points outside it. Zoom affects only what you see; the fit is identical either way.

Limits in y

The y limits exclude points by their value rather than their position — useful for cutting a saturated plateau or a handful of dropouts near zero. Two cautions:

- A y limit is applied *after* background subtraction, so it refers to the baseline-corrected values

you see, not the raw ones.

- Cutting in y biases the fit by construction: you are removing points because of where they landed, which is exactly the thing least squares assumes you did not do. Use it to remove instrument artefacts, not to remove points that disagree with you. To handle genuine outliers properly, use robust weighting (Chapter 15) instead, which down-weights them by how far out they are rather than by a threshold you chose.

A GOOD HABIT

Fit the full range first and look at the residual strip. Its shape tells you where the model stops working, and that is where to put your range limit — rather than choosing the limit first and never finding out.

With two datasets

The x window applies to both series. The y limits apply only to the first, because the second series has its own scale and a limit chosen for one would be meaningless for the other. A note under the pickers says so whenever a second dataset is active.

Long-format files: splitting one file into datasets

When your runs are stacked on top of each other with a name column, rather than side by side in separate columns.

There are two ways to put two measurements in one file. **Wide** format gives each run its own column:

```
E_eV,sample_A,sample_B
1.20,0.051,0.038
1.21,0.052,0.041
```

Long format — the shape that pandas, R, Origin and most instrument software produce — stacks the runs and names them in a column:

```
dataset,x,y
Complex_1,0.2043,1.1047
Complex_1,0.3527,1.7060
...
Complex_2,-5.7984,0.2525
Complex_2,-5.7836,0.1084
```

Long format is the one that used to break: *x* defaulted to column one, every value in it was text, and every row was dropped. The bench now recognises the shape and sets itself up.

What it does automatically

1. Columns are examined for the signature of a *label*: text (or whole numbers) taking between two and twenty-four distinct values, with fewer distinct values than there are rows.
2. *x* and *y* are set to the first two columns that are *not* text labels.
3. If a text label column exists, **split into datasets by** is switched on for it, the first distinct value becomes the first series and the second becomes the second.

Drop the file in and both runs are fitted, separately, with two result cards and two curves.

The controls

Split into datasets by

Lists every column that looks like a label, with the number of distinct values it holds, plus *nothing* — *one dataset* to turn splitting off and fit all the rows together.

`first / second`

Which value goes in each series. *second* also offers *none* — *just the first*, which fits one run and ignores the rest — the way to work through a file of six runs one at a time.

Splitting and the y_2 column are alternatives, not partners: choosing a split column hides the y_2 picker, and choosing a y_2 column clears the split.

The two series are genuinely independent

This is not two curves sharing a fit. Each series gets its own model search, its own ranking, its own parameters, its own χ^2 . In long format they need not even share x values — the runs may have been measured at different points, over different ranges, with different numbers of points — and each fitted curve is drawn only across its own data, never extrapolated over the other's range.

Numeric label columns

A column of whole numbers with few distinct values is offered as a split column too, so `run, x, y` with `run = 1, 2, 3` works the same way. A column with fractional values is never offered: those are measurements, not labels. The limit of twenty-four distinct values keeps a column of, say, integer temperatures from being mistaken for data while still allowing a realistic number of runs.

MORE THAN TWO RUNS

The bench fits two at a time. With six runs in a file, step through the pairs with the two pickers — or set *second* to *none* and work through them one at a time, exporting as you go. Every fit you run lands in History, so you can compare them afterwards without reloading anything.

Fitting two datasets at once

Two series, two independent searches, one figure.

A second dataset comes from either a second y column (wide format, Chapter 3) or a second value of a label column (long format, Chapter 5). From that point on the two behave identically.

What you get

- **Two searches.** The whole model search runs twice, independently. The two series may well end up with completely different models — that is the point.
- **Two cards.** The main result card is the first series. A second card below it carries the second, with the same information: form, fitted formula, reduced χ^2 , every parameter with its standard error, and R^2 , RMSE and AICc.
- **Two series on the plot.** The first in blue circles with an orange curve; the second in green squares with a purple curve. The legend names both, using the column header or the label value.
- **Both in the axis label.** In wide format the y axis is titled with both column names, unless you have set a title of your own.

make this y

The second card has one button. It promotes the second series to be the main one and demotes the first. Use it whenever you want the full apparatus — the ranking table, the overlay checkboxes, the peak finder, the background remover, the equation box — pointed at the other series, since all of those act on the main dataset.

What applies to which series

SETTING	FIRST SERIES	SECOND SERIES
Model families, ranking criterion, search charge	yes	yes
Weighting mode	yes	yes
Fit range in x	yes	yes
Fit range in y	yes	no
σ column	yes	no — unit weights unless the mode says otherwise
Background subtraction	yes	no
Outlier rejection	yes	no
Peak finder	yes	no
Residual strip	yes	no
Ranking table, overlays	yes	no

Anything in the "no" column is available for the second series by pressing **make this y** first.

Manual mode with two datasets

Your typed equations are fitted to both series, independently, so you can compare the same physical model across two samples directly — same form, two sets of parameters, two χ^2 . Parse errors are reported once, from the first series, so one bad line does not print two identical complaints.

COMPARING LIKE WITH LIKE

When you want the two series described by the *same* model rather than the best model for each, switch to manual mode and type the model once. Automatic mode is for finding out what each dataset supports; manual mode is for holding the form fixed and comparing the numbers.

The third way in: a loop

There is one more source of a second dataset, and it needs no columns at all. If x doubles back on itself — the field swept up and then down, the voltage ramped both ways — then the file does not describe a function of x . Two values of y belong to every value of x , one reached going up and one coming down.

This matters more than it sounds. A fitter handed both branches at once has no choice: it runs a single curve between them. On a hysteresis loop that curve is the *anhysteretic* curve, which is a perfectly real object and is not what was measured. The fit is not wrong, and the data are not noisy. The question was malformed.

The bench notices. Direction is read from the row order — the file was written in the order the points were taken — over a window rather than point to point, so that jitter in the field reading does not chop the sweep into a thousand branches. When both directions are a substantial part of the measurement, a checkbox appears under the column pickers: **split the sweep into its two branches**. Tick it and the rising branch becomes the first dataset and the falling branch the second, from which point everything in this chapter applies — two searches, two cards, both drawn.

The library carries hysteresis branch models to meet them: *tanh*, arctangent and error-function branches, two-phase and exchange-biased loops, and the anhysteretic Langevin, Brillouin and Jiles–Atherton curves for the backbone. They come in ascending and descending pairs, shifted by $+H_c$ and $-H_c$, so fitting both branches and differencing the two coercivities gives a number rather than a reading off a graph.

On a synthetic *tanh* loop of 400 points, fitted whole, the best model available misses every point by twenty-one times the measured noise. Split, each branch fits at 0.97.

Part II

Fitting

The two fitting modes

Let the bench find the form, or tell it the form and let it find the parameters.

The **Fitting mode** panel at the top of the rail has two buttons, and they change what the rest of the program is for.

Best form — the automatic search

Every enabled family is fitted and the results are ranked. You are asking a question of the shape "what does this data look like?" This is the mode to start in, even when you already know the answer, because the ranking tells you whether the data actually supports the model you had in mind or merely tolerates it.

The **Model families** and **Ranking** panels are visible in this mode; the equation box is hidden.

My equations — manual fitting

You type one or more functions, and each is fitted. You are asking "how well does *this* describe my data, and what are its parameters?" The equation box appears, and the model-family and ranking panels are hidden because they no longer apply.

Your equations are ranked among themselves, and the headline card always shows the best of *yours*, never an automatic model, even when an automatic model fits better. That is deliberate: in manual mode you asked about your model, and the program should answer the question you asked.

The comparison curve

A checkbox under the equation box, **show the best automatic fit too**, is on by default. It runs the automatic search in the background and draws its winner as a dashed olive curve alongside yours, and lists it in the ranking table as *automatic*. It is the honest counterweight to manual mode: if the dashed curve tracks the data visibly better than yours, your model is missing something. The result is cached, so it costs nothing to leave on while you edit an equation.

Moving between them

Clicking a model in the **Physics model library** writes it into the equation box and switches to manual mode. So does pressing **find peaks & fit**. That is usually how you get to manual mode: not by choosing it, but by choosing a model.

EQUATIONS ARE CLEARED WHEN THE DATA CHANGES

Loading a genuinely different dataset empties the equation box, resets the fit range, and turns background subtraction off. An equation written for one dataset refitted silently against different numbers is a trap, and the program declines to set it. Reloading the *same* data — from History, say — keeps everything.

The equation language

What you can type, exactly. A tokenizer and a recursive-descent parser compiling to closures — no eval, nothing executable.

In manual mode each line of the equation box is one model. x is your data. **Every other letter or word becomes a parameter to be fitted.** There is nothing to declare.

```
| A*exp(-x/tau) + c
```

Three parameters: A, tau, c.

Line structure

One equation per line

Each is fitted separately and appears in the ranking as its own model.

starts a comment

A line beginning with # is ignored. Use it to keep a model you are not currently fitting.

; introduces starting values

$A \cdot \exp(-x/\tau) + c$; $A=1000$, $\tau=10$. Separate several with commas. Any parameter you do not mention starts at 1.

! pins a parameter

$A \cdot \exp(-x/\tau) + c$; $!c=0$ holds c at exactly zero. Pinned parameters are not fitted, do not consume a degree of freedom, and are listed under the fitted formula as "c held at 0".

Operators

OPERATOR	MEANING	NOTES
+ - * /	the usual four	left to right
^ or **	power	binds tighter than *; right operand may be signed, as in x^{-2}
(juxtaposition)	implicit multiplication	$2x$ and $A \exp(-x)$ both work
> >= < <= ==	comparison $\rightarrow$ 1 or 0	$>=$ $<=$ $!=$ also accepted; a lone $=$ reads as a test
!=		
* x	multiplication	for pasting from a paper
- ?	minus	likewise
sqrt	sqrt	likewise

Switching a term on and off

A comparison evaluates to 1 or 0, so multiplying by one turns a term on above a threshold exactly the way a paper writes it:

```
| A*(x>=Eg)*sqrt(x-Eg) + B
```

This is safe even though `sqrt` of a negative number is not a number, because multiplication short-circuits: **zero times anything is zero, and the right-hand side is never evaluated.** Below `Eg` the comparison is 0, the square root is never taken, and the model is exactly `B`. The same protection is what makes `(x<Tc)*(1-x/Tc)^beta` work.

DIVISION DOES NOT SHORT-CIRCUIT

`1/(x-Eg)` becomes infinite at `x = Eg` whatever else is in the expression. If your model has a pole inside the data range, either restrict the fit range or reformulate. A guard of the form `(x>Eg)*(1/(x-Eg))` protects you below the pole but not at it.

Three-way choice

`where(condition, a, b)` — also spelled `if(...)` — returns `a` when the condition is true and `b` otherwise. Both branches are ordinary expressions. This is how the split lineshapes are written:

```
| A*exp(-(x-x0)^2/(2*where(x < x0, s1, s2)^2)) + B
```

which is a Gaussian with one width on the left of its centre and another on the right.

Constants

Only two names are reserved: `pi` (also `PI`) and `e`. This is deliberate and narrow — `tau`, `T`, `A`, `E`, `k` are far more often somebody's parameter than a constant, so they are yours. Write Boltzmann's constant explicitly as `8.617333e-5` when you want energies in eV and temperature in kelvin; the library models all do.

Numbers

Ordinary decimals and scientific notation: `1.5`, `2e-13`, `6.62607015E-34`. A leading `+` is allowed.

Parameter names

A letter or underscore, then letters, digits and underscores: `A`, `Eg`, `tau_1`, `K_L`. Case matters, so `a` and `A` are different parameters — useful, and a source of typos.

DO NOT NAME A PARAMETER AFTER A FUNCTION

The parser checks the function table before it decides something is a parameter, so a bare `gamma` or `step` in your expression is read as a function call and complains about a missing bracket. Use `gam`, `G`, `stp`. The full list of reserved function names is in Chapter 9.

Error messages

Errors appear under the equation box with the offending line above them, and they try to be specific:

MESSAGE	USUALLY MEANS
expected ")"	unbalanced brackets
the expression stops early — something is missing at the end	a trailing operator
unexpected "... " after a complete expression	two expressions run together, often a missing operator
I don't understand the character "... "	a stray symbol, often a smart quote or a stray comma
<i>name()</i> takes N arguments	wrong number of arguments to a function
cannot read "... " — write it as <i>name</i> =value	malformed starting value after ;
the expression is not finite over your x range	it evaluates to infinity or not-a-number everywhere
no parameter values make this expression finite over your data	every start diverged; usually a pole or a runaway exponent
the fit did not converge — try giving starting values	exactly that

Nothing you type is executed

Expressions are tokenised and parsed by hand into a tree of closures. There is no `eval` and no `new Function` in the program. An equation cannot reach the page, the network or the filesystem; the worst a bad one can do is fail to fit.

Every function you can use

Forty-eight of them, including the special functions the harder physics lineshapes need.

Every function below may be used in any equation you type, and the library models use them freely. The argument count is fixed and checked: passing the wrong number is an error, not a silent guess.

The last dozen are the reason several of the harder physics lineshapes can be written down at all. A true Voigt profile, a Frota Kondo resonance or a Dynes density of states cannot be assembled from elementary functions; they are implemented directly and are as much a part of the language as `sin`.

NAME	ARGS	WHAT IT IS
<code>sin</code>	1	sine, radians
<code>cos</code>	1	cosine, radians
<code>tan</code>	1	tangent, radians
<code>asin</code>	1	inverse sine
<code>acos</code>	1	inverse cosine
<code>atan</code>	1	inverse tangent
<code>sinh</code>	1	hyperbolic sine
<code>cosh</code>	1	hyperbolic cosine
<code>tanh</code>	1	hyperbolic tangent — the BCS gap, the Langevin limit
<code>exp</code>	1	e to the power
<code>ln</code>	1	natural logarithm
<code>log</code>	1	natural logarithm (synonym for <code>ln</code>)
<code>log10</code>	1	base-10 logarithm — Tafel slopes, decibels
<code>log2</code>	1	base-2 logarithm
<code>sqrt</code>	1	square root
<code>cbrt</code>	1	cube root
<code>abs</code>	1	absolute value
<code>sign</code>	1	−1, 0 or +1
<code>floor</code>	1	round down
<code>ceil</code>	1	round up
<code>round</code>	1	round to nearest
<code>erf</code>	1	error function — a Gaussian-broadened step
<code>erfc</code>	1	1 − erf — a diffusion profile
<code>lgamma</code>	1	log of the gamma function
<code>gamma</code>	1	the gamma function
<code>step</code>	1	0 below zero, 1 at and above
<code>heaviside</code>	1	as step, but $\frac{1}{2}$ exactly at zero
<code>coth</code>	1	hyperbolic cotangent — Langevin, Manooogian–Woolley
<code>sech</code>	1	hyperbolic secant
<code>csch</code>	1	hyperbolic cosecant
<code>sinc</code>	1	$\sin(z)/z$ with the limit at zero handled — diffraction, Josephson
<code>atan2</code>	2	two-argument arctangent, correct in all quadrants
<code>pow</code>	2	power, as a function

NAME	ARGS	WHAT IT IS
frota	2	frota(ϵ , Γ) — the Frota Kondo lineshape, sharper wings than a Lorentzian
fermi	2	fermi(E , kT) — Fermi–Dirac occupation, both in the same units
min	2	smaller of two
max	2	larger of two — useful to guard a square root
mod	2	remainder, with the sign of the divisor
dynes	3	dynes(E , Δ , Γ) — BCS density of states with lifetime broadening
smearstep	3	smearstep(E , E_0 , kT) — a thermally smeared step
digamma	1	$\psi(z)$ — weak localisation, Hikami–Larkin–Nagaoka
voigt	3	voigt(Δx , σ , γ) — a true Voigt profile, unit area, not a pseudo-Voigt
brillouin	2	brillouin(y , J) — magnetisation for total angular momentum J
langevin	1	$\coth(z) - 1/z$ — classical superparamagnetism
softplus	1	$\ln(1 + e^z)$ — joins a linear ramp to zero smoothly; how the Urbach and Tauc regimes are welded
bose	2	bose(E , kT) — Bose–Einstein occupation, phonon populations
lambertw	1	$W(z)$ — the w that solves $w e^w = z$; what makes the diode equation explicit
lambertwe	2	$\text{lambertwe}(a, b) = W(a e^b)$ without ever forming a e^b — the form every closed-form diode solution needs, and the only one that survives a bias of thirty volts

Names you cannot use for parameters

All of the above, plus `where` and `if`, plus the two constants `pi` and `e`. The parser checks the function table before it decides that something is a parameter, so a bare `gamma` is read as a function call.

Guarding a domain

Three idioms cover almost every case where a function would otherwise be handed an argument it cannot take:

A term that switches on

`(x>=Eg)*sqrt(x-Eg)` — multiplication short-circuits, so the root is never taken below the threshold.

A floor under an argument

`sqrt(max(x-Tb, 1e-9))` — how the Berezinskii–Kosterlitz–Thouless model stays finite below its transition.

A smooth join

`softplus((x-Eg)/Eu)` — equals zero well below the edge and $(x - Eg)/Eu$ well above it, with no kink. Used wherever an exponential tail has to meet a square-root edge without a discontinuity, which a hard switch cannot do.

Physical bounds

A linewidth cannot be negative and a mixing fraction cannot be 1.4. The bench now refuses to report that they are.

Least squares has no idea what your parameters mean. It will report a negative Gaussian width, a Fano mixing fraction of 1.4, a diode ideality factor of 0.3 or a Curie constant below zero, quite happily, if the arithmetic is a shade better there. Such an answer is not slightly wrong. It is meaningless, and worse, it usually *looks* fine on the plot, because the model often depends on the parameter only through its square or its absolute value.

Every model in the library now carries the constraints its physics imposes, and you can write your own.

Writing a bound

Bounds go after the semicolon, in the same comma-separated list as starting values, written the way you would write them on paper:

```
| A*gm^2/((x-x0)^2+gm^2) + c ; A=3, x0=5, gm=0.8, c=0.4, gm>0, A>0
```

FORM	MEANING
$A > 0$	strictly positive
$A \geq 0$	non-negative
$n < 5, n \leq 5$	bounded above
$0 < \text{eta} < 1$	both sides at once
$-1 \leq \text{cb} \leq 0$	likewise, inclusive

A parameter may have a lower bound, an upper bound, both, or neither, and bounds mix freely with starting values and with pinning.

How they are enforced

Not as a penalty, and not by clipping the answer afterwards. Outside the box the model returns not-a-number, χ^2 is infinite, and the Levenberg–Marquardt step is rejected. The wall is infinitely steep, so the search simply never leaves the region you declared physical, and the algorithm itself is unchanged.

Starting points are reflected, not clipped

This detail matters more than it sounds. A random restart may draw a linewidth of -0.8 . The obvious response — move it to the wall, $0 + \epsilon$ — is the worst possible one: it hands the fit a Lorentzian a millionth of a point wide, whose derivatives are zero everywhere, and the search

never recovers. Starting points are therefore **reflected** off the wall instead: -0.8 becomes $+0.8$, which is the same curve and the answer the fit wanted. A two-sided box folds the value repeatedly until it lands inside.

The measured effect of getting this right: on a Lorentzian whose starting width was given as negative, clipping produced $\chi^2/\nu = 786$ and a nonsense fit; reflecting produced $\chi^2/\nu = 0.79$, identical to the unconstrained fit but with a positive width.

When a parameter ends up against a wall

The result card says so, in amber: **at a physical limit: $\eta \leq 1$** .

Read that as information, not as a fault. It means the data is pushing the parameter past where physics allows, and there are only three explanations, all worth knowing:

1. **The true value really is at the limit.** A pseudo-Voigt fitted to a pure Lorentzian rails at $\eta = 1$, correctly.
2. **The model is wrong.** The parameter is absorbing a discrepancy that belongs somewhere else.
3. **Something upstream is wrong** — the weighting, the baseline, the range.

A railed parameter has no meaningful error bar; the curvature of χ^2 at a boundary does not describe a confidence interval. Treat it as a limit, and quote it as one.

BOUNDS ARE AN ASSERTION, NOT A WISH

Writing $A < 0$ on data with a positive peak does not make the peak negative. It makes the fit bad, and it should: you asserted something false about the measurement. If a bound makes a fit much worse, do not widen the bound until the number looks nice — ask which of you is wrong.

What the library constrains

426 of the 442 named models carry bounds. The rules applied are the obvious ones, stated once here rather than repeated through the catalogue:

QUANTITY	CONSTRAINT	WHY
Widths, linewidths, σ , Γ , γ	> 0	a width is a distance
Lifetimes, relaxation times τ	> 0	time runs forwards
Temperatures T_c , Θ_D , T_0 , T_E	> 0	absolute scale
Energies E_g , E_a , E_u , E_B , U_0	> 0	barrier and binding energies are positive as written
Amplitudes of an absorption or a population	> 0	an intensity is not negative
Mixing fractions η , w , weights	$0 \dots 1$	they are fractions
Cole–Cole α , Havriliak–Negami α , β	$0 \dots 1$	defined on that interval
Drude–Smith backscattering c	$-1 \dots 0$	from free Drude to full localisation
Brillouin J	$\geq \frac{1}{2}$	angular momentum
Diode ideality n	≥ 1	below 1 is unphysical
Critical exponent β (order parameter)	$0 \dots 1$	the physically realised range
Avrami n , Weibull m , Hill n	> 0	growth and cooperativity exponents
Curie constant C , susceptibility prefactors	> 0	positive by definition
Transfer coefficient α (Butler–Volmer)	$0 \dots 1$	a partition of the overpotential
Refractive-index backgrounds n_∞ , ϵ_∞	> 0	a medium does not have negative background index

A bonus that fell out of it

Checking that every model's own starting guess obeys its own bounds caught four presets whose data-derived starting values could fall outside them — a Curie constant estimated as negative from a linearisation, an activation energy estimated as negative on data that rises with temperature, a relaxor exponent starting exactly on its upper limit. Each was a latent failure that would have shown up as a fit that mysteriously would not converge. They are fixed.

The physics model library

Every named model, each carrying starting values read from your data — with materials science one tick away.

The library panel is a searchable list of every named model the bench knows. Type into the search box — it matches the name, the group, the one-line description *and* the formula, so `urbach`, `exponential`, `tail` and `exp((x-E0)/Eu)` all find the Urbach tail.

Clicking a model

Clicking writes it into the equation box, with a comment line naming it and with starting values computed from whatever data is currently loaded, and switches to manual mode. Then it fits. So the whole interaction is one click, and what lands in the box is editable text you can adjust, pin, or copy elsewhere.

```
# Urbach tail
A*exp((x-E0)/Eu) + B ; A=0.114, E0=2.47, Eu=0.0426, B=0.0032
```

Why the starting values matter so much

A five- or ten-parameter physical model almost never converges from a blind start. Every model in the library carries a guess that reads the loaded data and computes a sensible starting point from it. The quantities available to it are:

QUANTITY	MEANING
<code>xmin xmax R</code>	range in x , and its width
<code>ymin ymax yR ybar yMed</code>	range, mean and median in y
<code>xbar</code>	mean x
<code>xEdge</code>	the steepest point, found by a window-averaged slope rather than a single noisy difference — the band edge, the transition, the titration midpoint
<code>xAtYmax</code>	where y is largest
<code>peaks[]</code>	the tallest well-separated local maxima, in descending height
<code>ols(fx, fy)</code>	a straight-line fit in transformed coordinates

The last is the interesting one. An Arrhenius plot is a straight line of $\ln y$ against $1/T$, so the Arrhenius preset does exactly that first and reads the activation energy off the slope. Mott variable-range hopping linearises against $T^{-1/4}$, Efros–Shklovskii against $T^{-1/2}$, Curie–Weiss as $1/\chi$ against T . In each case the starting point is already close, and the non-linear fit only has to polish it. A preset without this treatment can start a thousand times off and never recover.

The library in the automatic search

With **Physics library** ticked in the model families, the automatic search tries every condensed-matter model and brings the best forward. This is covered in Chapter 13, along with the separate card that always shows the best *named physical* model even when a flexible composite scores better.

Materials science is opt-in

The library holds two bodies of models, and only one of them is searched by default.

Condensed matter

Diffraction, photoemission, dielectrics, phonons, devices, transport, magnetism, superconductivity, optical, tunnelling, line shapes, thermal, growth, kinetics and adsorption. Searched automatically.

Materials science

Plasticity, fatigue, creep, corrosion, polymers and rheology, composites, equations of state, sintering. **Off by default.**

The reason is not performance, it is meaning. These models live on different axes and in different units: a Weibull failure probability takes a stress, a Peukert law takes a current, a creep law takes a strain rate. Turned loose on a Raman spectrum they contribute nothing but a longer list of things that did not win, and occasionally something that did win for the wrong reason. Ranking is a competition, and a competition is only informative if the entrants could plausibly be right.

NOTHING IS HIDDEN.

Every model in every group is in the library panel, searchable and clickable, whether or not its library is in the search. Groups that are not being searched are marked `CLICK TO USE` in the panel. Clicking one writes it into your equations with starting values, exactly as for any other model.

To have them ranked as well, tick **Materials science library** under model families. That is the right thing to do when the data really are a creep curve or a fatigue life — and then the condensed-matter models are the ones sitting in the competition for no reason, which costs you nothing but is worth knowing.

Groups

The models are grouped by field. The complete catalogue — every model, its formula, its parameters and what it means — is Part IV of this manual.

GROUP	MODELS
Composites & effective media	8
Corrosion & oxidation	8
Creep & high-temperature	9
Devices	13
Dielectric & impedance	11
Diffraction & structure	17
Equations of state	5
Fatigue & fracture	8
Growth & kinetics	12
Kinetics & adsorption	7

GROUP	MODELS
Line shapes	34
Magnetic hysteresis	22
Magnetism	28
Mechanical & plasticity	17
Non-linear transport & I-V	36
Optical & semiconductor	40
Phonons & luminescence	11
Photoemission & core levels	6
Polymers & rheology	22
Quantum oscillations	29
Sintering & microstructure	11
Superconductivity	27
Thermal	15
Transport	36
Tunnelling & STM	10
Total	442

The standard families

Polynomials, powers, exponentials, peaks and waves — the twenty forms that describe most data.

Three of the family checkboxes cover the classical shapes. They are cheap, they are fitted directly in y -space, and between them they account for the correct answer more often than anything more elaborate.

MODEL	FAMILY	K	FITTED BY
Linear	standard	2	exact linear solve
Quadratic	standard	3	exact linear solve
Cubic	standard	4	exact linear solve
Quartic	standard	5	exact linear solve
Quintic	standard	6	exact linear solve
Logarithmic	standard	2	exact linear solve
Reciprocal	standard	2	exact linear solve
Square root	standard	2	exact linear solve
Power law	standard	2	Levenberg–Marquardt
Power law + offset	standard	3	Levenberg–Marquardt
Exponential	standard	2	Levenberg–Marquardt
Exponential + offset	standard	3	Levenberg–Marquardt
Double exponential	peaks	4	Levenberg–Marquardt
Saturating $ax/(b+x)$	standard	2	Levenberg–Marquardt
Rational (1,1)	standard	3	Levenberg–Marquardt
Logistic	standard	4	Levenberg–Marquardt
Gaussian	peaks	4	Levenberg–Marquardt
Lorentzian	peaks	4	Levenberg–Marquardt
Sinusoid	periodic	4	Levenberg–Marquardt
Damped sinusoid	periodic	5	Levenberg–Marquardt

Standard

Polynomials up to fifth order (quartic and quintic only when there are enough points), logarithmic, reciprocal, square root, power law with and without an offset, exponential with and without an offset, a saturating hyperbola, a first-order rational function and a logistic step.

Several of these are **linear in their parameters** — the polynomials, the logarithm, the reciprocal, the square root. Those are solved exactly, in one step, by weighted linear least squares. There is no iteration, no starting value, and no possibility of landing in the wrong minimum. When a straight line is the answer, the bench does not search for it, it computes it.

Peaks & decays

Gaussian, Lorentzian and the double exponential. The peak models carry a constant, so they fit a peak on a flat background; for a sloping background use a combination (Chapter 12) or a library model.

Their starting values come from the data: the median separates baseline from signal, the larger of the excursion above or below it decides whether it is a peak or a dip, and the position of the extremum becomes the centre.

Periodic

Sinusoid and damped sinusoid. Frequency is the parameter that ruins naive fits, so the starting frequency is not guessed — it is *scanned*. A geometric grid of trial frequencies is swept, and at each one the amplitude and phase are solved exactly by linear least squares, since a sinusoid is linear in its sine and cosine coefficients once the frequency is fixed. The best frequency on that grid becomes the starting point.

Two details in that scan earn their keep. The grid is linear in *cycles across the range*, not geometric: a ten per cent step sounds fine until the data hold twenty cycles, where ten per cent is two whole cycles of phase drift and the true frequency falls between two grid points that are both meaningless. And the basis carries a quadratic alongside the sine and cosine, so the scan is scored on what remains after a smooth trend is removed. Without that, an oscillation riding on a background loses every time — a magnetoresistance that rises tenfold across the sweep has far more variance in its envelope than in its oscillation, and the scan settles on the lowest frequency it can, which describes the rise and not a single cycle.

AND THE SAME SCAN AGAINST 1/X

A quantum oscillation is periodic in $1/B$, not in B . Against the field it is a chirp: fifteen cycles crowded into the low-field end and two at the top. Every periodogram, every sinusoid guess and every Fourier-based starting value therefore returns a number that describes no part of the data, which is exactly how a trace with fifteen visible oscillations ends up fitted by a smooth polynomial. So the scan is run twice, once against x and once against $1/x$, and the Shubnikov–de Haas, de Haas–van Alphen, magnetophonon and Weiss models start from the second. The number it returns is the Onsager frequency: $\cos(2\pi F/B)$ completes F cycles per unit $1/B$, so if x was in tesla the starting value is in tesla. On a synthetic trace built with $F = 42$ T the scan returns 42.06 T, and the correct model wins at 0.83 times the measured noise in 27 seconds — where before it lost to an interference-fringe model at 6.1 times the noise after 176 seconds.

THE DAMPED SINUSOID HAS A GUARD

A fitted decay time much longer than the data is not a decay: the model has quietly collapsed into a plain sinusoid carrying a spare parameter, which then gets charged against it for nothing. If τ comes out larger than five times the data range, the damped sinusoid is dropped from the results rather than reported as a five-parameter model doing a four-parameter job.

Offsets and why they matter

Both a bare power law and a power law with an offset are offered, and likewise for the exponential. This is not padding. Fitting $y = ax^b$ to data that actually sits on a pedestal forces the exponent to absorb the pedestal, and the exponent is usually the number you wanted. If the two versions

disagree about b , the offset is real.

Combinations

Every sum of two components — peak plus background, two peaks, trend plus oscillation — made affordable by solving the amplitudes exactly.

Real measurements are rarely one thing. A peak sits on a sloping background; a decay has two components; a transport curve is a trend with an oscillation on it. The **Combinations** family fits every sum of two components, each with an optional constant.

Fourteen components are available, and any two of them may be summed:

COMPONENT	AMPLITUDES	SHAPE PARAMETERS	PAIRS WITH ITSELF
constant	c	none	
linear	b	none	
quadratic	b	none	
$\sqrt{x}$	b	none	
log	b	none	
1/x	b	none	
power law	A	b	yes
exponential	A	k	yes
gaussian	A	μ, σ	yes
lorentzian	A	x_0, γ	yes
sinusoid	a, b	ω	
damped sinusoid	a, b	τ, ω	
logistic step	L	k, x_0	
error-function step	A	x_0, w	

Pairs are formed from every combination of the components above, plus each of the four marked *self-pairable* with itself — two Gaussians, two Lorentzians, two exponentials, two power laws. Each pair is tried with and without an added constant. That is on the order of seventy structures.

Why this is affordable: variable projection

Fitting seventy models with up to seven parameters each, by brute force, would be far too slow. The trick is that a sum of components is **linear in its amplitudes** and non-linear only in its shape parameters — positions, widths, rates, frequencies.

$$y = A \cdot \text{shape}_1(x; p_1) + B \cdot \text{shape}_2(x; p_2) + c$$

So the search runs over the shape parameters only, and at every trial point the amplitudes are solved *exactly* by weighted linear least squares. A two-peak-plus-baseline fit becomes a four-dimensional search instead of an eight-dimensional one. This is variable projection, and it is what makes the combination search practical.

The three passes

1. **Triage.** Every pair gets a short Nelder–Mead search — eighty iterations — on a coarse eighty-point subsample, from several starting points drawn from the components' own suggestions. This ranks the pairings; it is not meant to fit them.
2. **Shortlist.** The best fourteen get a proper two-hundred-and-forty-iteration search on a two-hundred-point subsample, then their amplitudes are solved on *all* the data and the χ^2 recorded.
3. **Polish.** The best six get a three-hundred-iteration search on up to eight hundred points. Shape parameters are settled by a few hundred well-spread points; grinding them against every point costs a great deal and changes nothing. Only the scoring ever uses the whole dataset.

The best three survive into the results.

The honesty charges

Two of them, and both matter:

- **Selection cost.** A combination was chosen out of about seventy structures, so it is charged $\log_2(\text{number tried}) \approx 6.1$ *effective* parameters for having been picked. A six-parameter combination is judged as though it had roughly twelve. Without this, a combination would win almost every time.
- **The three-degrees-of-freedom floor.** Any model left with fewer than three effective degrees of freedom is fitted and listed, but flagged *too few points* and never allowed to take the top spot. On a handful of points a five-parameter sum fits anything.

Component boundaries

Each component refuses parameter values that would make it something else: a Gaussian or Lorentzian width outside 10^{-4} to 10 times the data range, a power-law exponent beyond ± 25 , an exponential rate steep enough to overflow, a frequency above the Nyquist limit of your sampling. A rejected trial returns infinity and the search moves on. This is what stops the combination search from "fitting" your data with a component of width 10^{-9} centred on one point.

Seeing the pieces

Press **components** in the plot toolbar and each additive piece is drawn separately, in magenta, under the total. For a peak on a background this immediately shows how much of the peak height is peak and how much is background — which is usually the number you actually wanted.

The free-form, library and multi-peak searches

A symbolic search over a term dictionary, the full physics library, and a fit to every band at once.

Free-form symbolic search

This family does not have a model. It has a dictionary of thirty terms and it looks for the best small sum of them.

On a typical dataset the dictionary holds 30 terms:

x , $x^?$, $x^?$, $1/x$, $1/x^?$, $\sqrt{x}$, $\ln x$, $x \ln x$, $e^{-(x^?/3.72)}$, $e^{(x^?/3.72)}$, $e^{-(x^?/7.43)}$, $e^{(x^?/7.43)}$, $e^{-(x^?/14.9)}$, $e^{(x^?/14.9)}$, $e^{-(x^?/29.7)}$, $e^{(x^?/29.7)}$, $e^{-(x^?/59.4)}$, $e^{(x^?/59.4)}$, $\sin(2^?x/5.94)$, $\cos(2^?x/5.94)$, $\sin(2^?x/7.43)$, $\cos(2^?x/7.43)$, $\sin(2^?x/9.91)$, $\cos(2^?x/9.91)$, $\sin(2^?x/14.9)$, $\cos(2^?x/14.9)$, $\sin(2^?x/29.7)$, $\cos(2^?x/29.7)$, $\sin(2^?x/59.4)$, $\cos(2^?x/59.4)$

Terms that cannot be evaluated over your x — a logarithm with non-positive values, a reciprocal with a zero — are dropped before the search starts. Decay, growth and periodic terms enter at several fixed scales derived from the data range, and the winning ones have their rate or period refined afterwards.

How the search is made tractable

Trying every subset of three from thirty terms, with and without a constant, is about eight thousand fits. The bench first prescreens with **matching pursuit**: each term is scored by how much of the *remaining* residual it explains, the best few are kept, that term's contribution is subtracted, and the process repeats for three rounds keeping eight, then six, then four. Three rounds rather than one matters — it keeps a term that is only useful *alongside* another, which a single-pass ranking would discard. Fourteen terms survive, cutting the exhaustive stage by an order of magnitude for the same answers.

The exhaustive stage then tries every subset of one, two and three of those fourteen, with and without a constant, on at most nine hundred points. Because a sum of fixed terms is linear in its coefficients, each of those is one exact linear solve, not a search. The best six are refined: their rates and periods are tuned by a geometric line search — multiply by 1.25, divide by 1.25, halve the step when neither helps — and rescored on all the data.

Why it is handicapped

The symbolic search is choosing among thousands of candidate formulae, so a form that wins on raw χ^2 alone is very often just noise-fitting. With **charge for the search** on, it pays

$\log_2(\text{dictionary size}) \approx 4.9$ effective parameters *per term*, plus one for each rate or period it tuned. A three-term symbolic form with two tuned rates is judged as though it had about nineteen parameters. It only takes the top spot when it has found real structure the named families miss — which is exactly when you want to hear from it.

The physics library in the search

Every condensed-matter model in the library is tried; the materials-science models join in when that library is switched on, and nine heavy numerical models (marked *click to use*) are left for you to pick by hand. The sweep is in two stages, after a quick triage on 120 points keeps the 64 most promising:

1. **Rank** every model against a thinned copy of the data — three hundred evenly spread points. A model that is going to win does so on three hundred well-chosen points just as clearly as on thirty thousand.
2. **Refit** the four leaders properly on every point.

Ranked by BIC, not by chi-square

Within the library the ranking uses the Bayesian information criterion, which charges $\ln N$ per parameter:

$$\text{BIC} = N \cdot \ln(\chi^2/N) + k \cdot \ln N$$

Reduced χ^2 charges almost nothing for extra parameters, and under it the widest model in the library — three Lorentzians, say, with ten free parameters — quietly wins on any smooth data by fitting the noise, and the "best named physics model" comes back as a Kondo resonance for a straight line. BIC is a selection criterion, which is exactly the job here, and its heavier charge stops that. The same parsimony rule used for the overall ranking is then applied within the library, so a simple named law that the leader does not significantly beat is promoted over it.

Skipping what cannot apply

A four-component lineshape costs about ten times a two-parameter law to fit. Before the sweep the bands in your data are counted once, and the three- and four-component presets are skipped when there are not that many bands. On single-peak data this removes several seconds of pointless work.

The selection charge

A library model was picked out of up to 64 shortlisted candidates, so it pays $\log_2(\text{number scored}) \leq 6$ effective parameters. This is the same accounting applied to the symbolic and combination searches, for the same reason: a model that won a large search has to clear a higher bar than one you specified in advance.

Auto multi-peak

Off by default, because it is the slow one. When on, every band in the data is detected and all of them are fitted together as Lorentzians on a linear background — up to twelve peaks, so up to

thirty-eight parameters in one fit. This is the only route in the automatic search to a many-band spectrum, since the combination search stops at two components.

For anything more demanding — a different lineshape, dips as well as peaks, a quadratic background, more than twelve bands — use the peak finder panel instead (Chapter 17), which gives you the controls and writes an editable equation.

TURNING FAMILIES OFF

Every family has a checkbox and unchecking one removes it from the search immediately. Two good reasons to: speed, and honesty. If you know the data is a spectrum, turning off the polynomials stops a quintic from cluttering the top of the table; if you know it is a decay, turning off the periodic family stops an aliased sinusoid from doing the same.

How the winner is chosen

The most important chapter in this manual. Lowest chi-square is not the rule, and this is the rule that replaces it.

If the bench simply reported the lowest χ^2 , it would be worse than useless: it would confidently hand you a quintic for a straight line and a three-component composite for an exponential, every time. Reduced χ^2 is not enough of a defence, and understanding why is the key to reading the ranking.

Why reduced chi-square is too weak

Dividing by $\nu = N - k$ charges a factor $N/(N-k)$ for extra parameters. On eighty points, going from three parameters to six changes that factor from 1.039 to 1.081 — about four per cent. But adding a meaningless parameter to a fit lowers χ^2 by about one unit of χ^2/ν on average, purely by chance, which on a good fit is also about one part in eighty per parameter. The charge and the free gain are the same size. So on noisy data a quartic routinely beats the true straight line by ten or twenty per cent in reduced χ^2 , having learnt nothing.

The rule the bench actually uses

The honest question is not "which χ^2 is lowest" but "is the improvement bigger than the improvement you would get for free?" That is an F-test in all but name, and it is what the bench does:

THE PARSIMONY RULE

Let *ref* be the model with the lowest χ^2 of all. Compute the free gain per parameter as $\chi^2_{\text{ref}} / (N - k_{\text{ref}})$. Now walk the models from simplest upward. The **first** model simpler than *ref* whose χ^2 exceeds χ^2_{ref} by no more than

$$F \times (k_{\text{ref}} - k_m) \times (\text{free gain per parameter}), \quad F = 2.5$$

takes the top spot. If no simpler model qualifies, *ref* keeps it.

In words: **the simplest model that the leader does not significantly beat wins**. A model must earn its parameters by a factor of 2.5 over what chance would have given it. That threshold was set by testing, not by taste — on sixty controls with known answers it recovers the true model far more reliably than a percentage band, which failed on a third of them.

Effective parameters

The k in that rule is the **effective** parameter count when **charge for the search** is on, which it is by default. A model you specified costs its actual parameters. A model the program found by

searching costs those *plus* the information needed to say which one it picked:

WHERE IT CAME FROM	EFFECTIVE PARAMETERS	TYPICAL SURCHARGE
Standard, peaks, periodic	k	none
Your typed equations	k	none
Combination	$k + \log_2(\text{structures tried})$	≈ 6.1
Physics library	$k + \log_2(\text{models tried})$	≈ 7.5
Symbolic search	$k + m \cdot \log_2(\text{dictionary}) + \text{tuned rates}$	$\approx 4.9 \text{ per term}$
Auto multi-peak	$k + \log_2(\text{peaks found})$	≈ 3.6

This is a minimum-description-length argument. If you had to transmit the model to someone else, you would have to send both the parameters and the identity of the form, and the identity costs bits in proportion to the size of the catalogue it was drawn from. Charging for it puts a specified model and a searched one on the same footing.

Unticking **charge for the search** removes the surcharge and ranks on raw parameter count. It is instructive to toggle it: if a composite only wins with the charge off, it did not really win.

The underdetermined rule

A model left with fewer than three effective degrees of freedom ($N - k_{\text{eff}} < 3$) is fitted, listed and marked *too few points*, but its rank is set to infinity and it can never take the top spot. On eight points, a five-parameter composite carrying a six-parameter selection charge has nothing left to be tested against.

The three criteria

The **Ranking** panel offers three. Changing it re-ranks the existing results instantly; nothing is refitted.

χ^2/ν -- the default

Ranks on effective reduced χ^2 and then applies the parsimony rule above. This is the recommended setting and the only one the parsimony rule runs under.

AICc

$N \cdot \ln(\text{SSR}/N) + 2k + 2k(k+1)/(N-k-1)$. The small-sample-corrected Akaike criterion. Only differences matter: about 2 lower is mildly preferred, about 10 lower is decisive. Negative values are normal and mean nothing on their own.

BIC

$N \cdot \ln(\text{SSR}/N) + k \cdot \ln N$. Charges more per parameter than AICc as soon as $N > 8$, so it prefers simpler models. Use it when you suspect the ranking is still too generous.

Both information criteria use the effective k when the search charge is on, and both are computed from the unweighted sum of squared residuals so that they remain comparable when no σ column is present.

Features narrower than the data

One class of model is disqualified outright, however well it scores. A component whose width collapses below the point spacing spikes through a single sample: between two neighbouring

points the curve shoots to a value nothing measured and comes back, χ^2 falls by exactly the one residual it removed, and the plot grows a hair-thin needle. That is not a peak. It is an artefact of having more freedom than evidence.

Every candidate is therefore walked between its own data points before it is ranked. In each interval the curve is sampled and compared with the straight line joining its own values at the two ends. A smooth model departs from that chord by curvature $\times h^2/8$, which is negligible; a collapsed component departs by its own height. Measured on evenly spaced data, the number is 0.005 for a peak ten spacings wide, 0.054 at one spacing, 0.17 at half a spacing and 0.30 at a fifth — so the threshold sits at 0.12, between "just resolved" and "narrower than the data can see". A sharp step scores 0.027 and a steep exponential 0.013, nowhere near it.

A model that fails is still fitted and still listed, marked `NARROWER THAN THE DATA`, but it can never be the winner. Intervals much wider than the typical spacing are exempt: there the data genuinely say nothing, and that is a property of your sampling rather than a fault of the model.

The same reasoning is applied at the source. When the peak finder or the automatic multi-band pass builds an equation, every width is bounded below at a **full width at half maximum of two point spacings** — one spacing is the Nyquist limit, and a band at exactly that width is still a single sample with a curve drawn through it. The bound is written into the equation you can see, converted for each shape, because a Lorentzian's γ is a half width and a Gaussian's σ is smaller again by $2\sqrt{2 \ln 2}$; a floor applied to the wrong one of those is a floor that does not hold.

The effect, measured on a flat background with one wild sample: unbounded, the fitted width collapses to 0.007 of a point spacing — a needle a hundred and fifty times narrower than the sampling, passing exactly through the one point and driving χ^2 to zero. Bounded, it lands at precisely 2.00 spacings and the needle is gone.

In **manual mode** your own equations are checked too, but not demoted: you asked for that equation and you get it. It is labelled instead, because a hair-thin spike on the plot is otherwise easy to read as real structure rather than as a parameter that ran away.

When nothing fits

Ranking is comparative: it tells you which model is least bad, and says nothing about whether any of them is good. A spectrum with fifteen bands cannot be described by any sum of two components, so the search dutifully hands back a sinusoid tracing the envelope and labels it *best fit*, with the residual strip the only clue that every point is missed by tens of times the noise.

So the bench measures the noise itself. Second differences of a smoothly sampled curve are almost pure noise — the second difference cancels any straight line through three neighbouring points — so the median of $|2y_i - y_{i-1} - y_{i+1}|$, scaled by $1.4826/\sqrt{6}$, is a robust standard deviation that the peaks themselves cannot inflate. Measured against curves with known noise it lands within a few per cent: 13.8 against a true 14 on a fifteen-band spectrum, 0.0196 against 0.02 on a smooth decay.

Every model's **misfit** is then its residual scatter in units of that noise. A correct fit sits near 1. The result card says so plainly above about 3, and the headline fit says it in full: *nothing in the search describes this data*. Where y is positive and runs over decades the comparison is made on fractional residuals instead, because an absolute miss of 0.01 means nothing without knowing whether the value there was 10 or 10^{-9} . Where the data have no scatter at all — a computed curve rather than a measurement — there is nothing to compare against and no misfit is quoted.

THE AUTOMATIC MULTI-PEAK RESCUE

When the best model misses by more than three times the noise *and* the data hold four or more resolved bands, the bench detects every band and fits them together without being asked — even with **Auto multi-peak** switched off. On a fifteen-band spectrum this is the difference between a sinusoid at ten times the noise and a proper fit at 0.94.

If the Lorentzian band set is still outside the noise, the same bands are refitted as Gaussians and as pseudo-Voigts, each seeded from the Lorentzian answer so the search starts most of the way there, and the ranking picks between them. On twelve Gaussian bands that ladder goes $5.1 \rightarrow 3.1 \rightarrow 0.94$. When the first shape already fits, the alternatives are never attempted and cost nothing.

HOW TO READ THE TABLE IN PRACTICE

Look at the top five, not the top one. If they are all the same family with slightly different parameter counts, the family is the answer and the exact member is a detail. If they are five unrelated forms with nearly identical χ^2 , your data does not distinguish them — you need more points, a wider range, or smaller error bars, and no amount of fitting will substitute.

Weighting and outliers

How much each point counts. The single setting most likely to change your answer.

Least squares minimises $\sum w_i(y_i - f(x_i))^2$. The weights decide which points the fit is allowed to miss. Getting them wrong does not produce an error message — it produces a wrong answer that looks fine.

The four modes

`automatic default`

$w = 1/\sigma^2$ when you have supplied and confirmed an uncertainty column. Otherwise unit weights — *unless* y is everywhere positive and spans three decades or more, in which case it switches to relative weighting on its own. Only the σ -column case makes $\chi^2/\nu \approx 1$ meaningful.

`equal weight on every point`

$w = 1$. Every point counts the same in absolute terms, so the largest values dominate. Correct when the scatter is genuinely the same size everywhere — a constant detector noise floor.

`relative --  $1/y^2$`

$w = 1/y^2$, so χ^2 sums squared *fractional* errors. Correct when the scatter is a constant percentage — which is the usual case for anything spanning decades.

`counting statistics --  $1/|y|$`

$w = 1/|y|$, the Poisson assumption for counts, where the variance equals the mean. For raw detector counts without an explicit σ column this is very often right.

Why automatic does not simply use equal weights

Equal weighting is a statement about your data: that the scatter is the same size everywhere. On data running over decades that statement is false, and the consequence is not subtle. Least squares charges nothing for missing a point at 10^{-5} by a factor of ten, because the absolute residual is 10^{-5} , while a one per cent error at $y = 1$ costs a hundred times more. The fit tracks the top decade and walks away from everything below it. On a logarithmic axis this is unmistakable: the curve leaves the data entirely and plunges off the bottom of the plot, while the reported χ^2 looks unremarkable.

So when there is no σ column, every y is positive, and the span is between three and six decades, automatic weighting uses a constant fractional error instead. That is what a log plot already assumes, and it is nearly always what was meant. The status box says so in as many words — *relative $1/y^2$ — chosen automatically, y spans decades* — and setting the menu to anything other than automatic overrides it.

Why relative weighting is floored, and why it stops at six decades

$1/y^2$ has the mirror-image failure of equal weighting, and it is the nastier of the two. The term it minimises is $(1 - f/y)^2$: overshooting a point is punished quadratically, undershooting it saturates at 1. Give the smallest value a weight 10^{40} times the largest and least squares finds the safest possible answer — sit at zero everywhere, never risk an overshoot — and reports it without complaint. On a twenty-decade dataset that is exactly what happens: a flat line along the axis, called *best fit*.

Two things prevent it. Relative weights are floored at 10^{-4} of the largest value, so nothing below a ten-thousandth of full scale can dominate; no measurement is better than some fraction of full scale anyway. And above six decades the mode is not used at all — no per-point weighting in y is meaningful over that range — so the weighting stays equal and the data panel says *y spans too many decades for $1/y^2$ to mean anything*. Fit a restricted range instead, or work in a transformed variable.

The difference is not cosmetic. On a synthetic Arrhenius rise spanning four decades, equal weighting ranks a *quartic polynomial* first and misses the low-temperature end by a factor of twenty-six; automatic weighting ranks *Arrhenius* first and its worst fractional miss anywhere is seven per cent.

THE FAILURE THIS CAUSES, IN FULL

Take a clean exponential decay from 5 down to 0.4 with three per cent *relative* noise, and no σ column. Under equal weighting, the early points carry absolute scatter of about 0.15 and the tail about 0.012. χ^2 is therefore dominated almost entirely by the first few points, and the tail — the part that actually identifies an exponential — contributes almost nothing. The bench will rank a two-step composite above the true exponential, and it will be right to, because under those weights the composite genuinely fits better. Switch to relative weighting and "Exponential + offset" wins outright.

The status box always tells you which mode is in force. When your data spans more than about one decade in y and you have no error bars, read that line before you read the ranking.

Rejecting outliers

The **Reject outliers** checkbox turns on iteratively reweighted least squares with Tukey's biweight. It runs in manual mode, on your typed equations.

1. Fit normally.
2. Compute the residuals and their median absolute deviation, scaled by 1.4826 so it estimates a standard deviation.
3. Give each point the weight $w_0(1 - u^2)^2$ where u is its residual divided by $4.685 \times \text{MAD}$, and **zero** if $|u| \geq 1$.
4. Refit. Repeat once more.

Two passes is enough to shed spikes without quietly discarding real structure. The result card reports how many points were rejected, and — importantly — the χ^2 , R^2 and RMSE are recomputed on the *original* weights excluding only the rejected points. A robust fit that reported its own down-weighted χ^2 would be claiming a quality it never achieved.

ROBUST FITTING IS NOT A LICENCE

If five per cent of your points are rejected, you had five per cent bad points. If forty per cent are, the model is wrong and the biweight is discarding the evidence. Always look at which points went.

Where the uncertainties come from

Parameter standard errors are the square roots of the diagonal of the covariance matrix $(J^T W J)^{-1}$ at the optimum. When you supplied σ , that is used as it stands. When you did not, it is scaled by χ^2/ν — which amounts to inferring the noise level from the scatter about the fit and assuming the model is correct. That assumption is doing real work: if the model is wrong, the errors are optimistic, and no amount of arithmetic will reveal it. The residual strip will.

Background removal

Five ways to estimate a baseline and take it off before fitting — with the baseline always shown.

The **Background** panel estimates a baseline from the data and subtracts it before anything else looks at y . The raw points stay on the plot, faint, and the estimated baseline is drawn as a dashed grey line, so you can always see what was removed and judge whether it was reasonable.

The five modes

`none default`

Fit the data as it is.

`constant offset`

Subtracts the tenth percentile of y . A robust "floor" that ignores the peaks — the mean would be dragged up by them.

`linear baseline`

A straight line fitted with iterative clipping: fit, discard everything above the fit, refit, fourteen times. Peaks fall out of the fit; the trend under them survives.

`polynomial baseline`

The same clipping procedure with an order you set, 2 to 8. The x values are mapped to $-1 \dots 1$ first so that a high order stays numerically stable.

`asymmetric least squares`

The proper tool for a wandering baseline under many peaks. Described below.

Asymmetric least squares

Eilers and Boelens's method finds a smooth curve z minimising

$$\sum w_i (y_i - z_i)^2 + \lambda \sum (\Delta^2 z)^2$$

with weights that are *small* above the current curve and *large* below it, iterated fourteen times. The asymmetry is what makes it ignore peaks: points sticking up are treated as signal and barely pull on the baseline, while points below it pull hard.

Two sliders:

`smoothness --  $\lambda$ , from  $10^2$  to  $10^9$`

How stiff the baseline is. Too small and it climbs into the peaks; too large and it becomes a straight line and misses real curvature. Start at 10^5 and move by whole decades.

`asymmetry --  $p$ , from 0.001 to 0.2`

The weight given to points above the curve. 0.01 is a good default; lower it if the baseline is still

riding up into strong peaks.

The normal equations here are symmetric and pentadiagonal, so the bench solves them with a banded LDL^T factorisation in time proportional to the number of points, not its cube. A ten-thousand-point spectrum is baselined in milliseconds.

What the fit then sees

Everything downstream — the model search, the peak finder, the fit range in y , the residuals — works on the corrected values. The status box notes "bg — als baseline removed" whenever a baseline is in force.

SUBTRACTING A BASELINE IS A MODELLING DECISION

It is not data cleaning; it is asserting that your measurement is signal plus a smooth additive background, and that you know roughly how smooth. If the peak areas are the quantity you care about, they depend directly on that choice. Where you can, fit the background as part of the model instead — a combination (Chapter 12), a library model with a `bg` term, or a peak-finder fit with a linear or quadratic background. Then the background carries error bars like everything else, rather than being fixed by a slider you moved.

Changing the data turns background subtraction off, because a baseline is specific to the measurement it was estimated from.

The peak finder

Detect every band in a spectrum, including shoulders and dips, and fit them all at once.

The automatic combination search stops at two components. A Raman spectrum with a dozen bands needs something else, and this is it: detect the peaks, build a multi-peak equation with starting values from the detection, and fit the whole thing in one go.

How detection works

Not by "is it higher than its neighbours" — on a real spectrum that finds hundreds of noise wiggles. The bench uses **prominence**: for each local maximum, how far must you descend before you can reach a higher peak? That is the only robust measure when the baseline wanders.

Before that, two clean-up steps that affect only the detection and never the fit:

- **Despiking.** Points more than six scaled median absolute deviations from their local median are replaced by that median. Single-sample spikes — cosmic rays, dead pixels — are not bands.
- **Light smoothing.** A short moving average so that single-sample noise cannot be a peak.

A noise level is then estimated from the median absolute successive difference, and the prominence threshold is the larger of your sensitivity setting times the data range and three times that noise level. So the threshold adapts to your data rather than to the units you happened to measure in.

Shoulders

A band that never rises above its neighbour is invisible to a prominence test, but it shows as a sharp minimum in the second derivative. The bench computes a Savitzky–Golay second derivative — a local quadratic fit — and takes minima below six scaled MADs of its own noise as shoulder candidates, provided they sit above the median signal (a shoulder rides on a band; a dip in a hole is not one).

Dips

An absorption line, a transmission dip, a conductance gap. Detection runs the same test on the inverted signal and carries the sign through, so a dip becomes a component with a negative amplitude.

The controls

look for
peaks only, peaks and dips, or dips only.

shape

Lorentzian, Gaussian, or pseudo-Voigt. Lorentzian for lifetime-broadened lines and most Raman work; Gaussian for instrument- or inhomogeneously-broadened lines; pseudo-Voigt when you do not know, since it carries a mixing parameter per peak and can be either.

sensitivity

The prominence threshold as a fraction of the data range, 0.002 to 0.15. Lower finds more. This is the control you will actually use.

min gap

Minimum separation between kept peaks, as a fraction of the number of points. Raise it when one broad band is being reported as three.

max peaks

1 to 40. Detection keeps the most prominent.

background

Constant, linear or quadratic, fitted together with the peaks.

The count next to the panel title updates live — "12 found" — and every detected peak is marked on the plot with a magenta triangle, pointing down for a peak and up for a dip, as you move the sliders. Tune until the triangles sit on the bands you believe in, then fit.

find peaks & fit

Writes a complete equation into the box, with a starting value for every parameter, switches to manual mode and fits. For twelve Lorentzians on a linear background that is thirty-eight parameters:

```
# 12 peaks found automatically
A1*g1^2/((x-x1)^2+g1^2) + A2*g2^2/((x-x2)^2+g2^2) + ...
+ b0 + b1*x ; A1=612, x1=520.3, g1=4.1, A2=298, ...
```

Because it is ordinary text in the equation box, you can then edit it: pin a peak position you know from the literature with `!x3=1332`, delete a component you do not believe, or add one the detector missed. Re-fitting after an edit is immediate.

WHY THIS CONVERGES WHEN A BLIND 38-PARAMETER FIT WOULD NOT

Every starting value comes from the detection — each amplitude from the measured prominence, each centre from the measured position, each width from the measured half-prominence width. When a fit is told where every parameter starts, the bench skips the random-restart machinery entirely and goes straight to polishing. That is the difference between a fit that takes a few seconds and one that never converges.

Realistic expectations

On a twelve-band synthetic spectrum with Poisson noise, the automatic search alone reaches $\chi^2/\nu \approx 1500$ — it simply has no model with enough components. The peak finder reaches ≈ 83 with the same data. Neither is 1, because a real spectrum is never exactly twelve Lorentzians, but the difference between the two numbers is the difference between a description and a shrug.

Reading the result card

Form, fitted formula, every parameter with its error, and six numbers that describe the fit.

The card under the plot is the answer. Top left is what was fitted; top right is how well.

The header

`eyebrow`

"Best fit" when the top-ranked model is showing, "Selected model" when you have clicked another row, "Your equation" in manual mode. It also states the size of the search a model came out of — "chosen from 30 dictionary terms", "chosen from 71 combinations tried" — because that is what its selection charge was based on. A green tag, "also the best named physics model", appears when the winner is itself from the library.

`form`

The functional form with symbols, not numbers. This is what to quote when you say what you fitted.

`fitted`

The same expression with the fitted values substituted in. Copy-and-paste-able into a caption.

`means`

For library models, the one-line physical description — what the model is for and what its parameters mean.

The hero number

Reduced χ^2 , large, with $\nu = N - k$ spelled out underneath so the degrees of freedom are never a mystery.

The parameter grid

Every fitted parameter with its value to six significant figures, its standard error, and the error as a percentage of the value. The percentage is the fastest diagnostic in the program:

RELATIVE ERROR	WHAT IT USUALLY MEANS
under a few per cent	the data determines this parameter
tens of per cent	weakly determined; it is doing little work
comparable to or larger than the value	the data does not determine it at all — consider pinning it or removing it
"no error estimate"	the covariance matrix was singular: two parameters are exactly degenerate, or the fit sits on a boundary

Pinned parameters are listed separately under the fitted formula, as "c held at 0", and are not counted in k .

The statistics bar

$$\chi^2$$

$\sum w(y - f)^2$. With σ , each point's miss is counted in units of its own error bar, so a point one error bar off adds exactly 1.

$$\chi^2/N$$

The same total divided by the raw point count. It does not charge for parameters, so it always improves as you add them. Useful as an average miss per point; useless for choosing between models.

$$R^2$$

The fraction of the spread in y the curve accounts for. It rises with every extra parameter, so it is a description, never a tie-breaker. On a good physical fit it is almost always 0.99-something and tells you nothing.

$$RMSE$$

Root-mean-square residual in your y units — the typical vertical distance from a point to the curve. The one number here that has physical units and an intuitive meaning.

$$AICc$$

Goodness of fit with an explicit per-parameter charge, corrected for small samples. Only differences matter.

$$\text{Quality verdict}$$

With a σ column: green "consistent with the stated errors" for $0.5 < \chi^2/\nu < 1.6$; amber warnings outside. Without one, a reminder of which weighting is in force.

What the verdicts mean

χ^2/ν	READING	WHAT TO DO
≈ 1	The curve misses the points by about as much as the error bars allow.	Nothing. This is the target.
> 1.6	Either the model is wrong, or σ is underestimated.	Look at the residual strip. Structure $\Rightarrow$ wrong model. Featureless scatter $\Rightarrow$ the error bars are too small.
< 0.5	Either σ is overestimated, or you have fitted the noise.	Check the parameter count against the number of points, and check how σ was obtained.

The residual strip

Under the main plot, and the most informative part of the figure. With a σ column it shows the **pull**, $(y - f)/\sigma$, which should scatter randomly in a band of about ± 1 to ± 2 with no visible shape. Without one it shows the plain residual in y units.

Structure in the residuals beats any χ^2 . An arc means the model has the wrong curvature. A sawtooth means a periodic component is missing. A step means something changed part-way through the measurement. A fan means the weighting is wrong. A fit with $\chi^2/\nu = 1.05$ and a

clear arc in its residuals is a worse description than one at 1.4 with none.

The runs test: structure counted, not eyeballed

The paragraph above is the oldest advice in curve fitting, and until now the bench only offered it as advice. It now applies it. χ^2 asks how *far* the curve sits from the points; it never asks whether the misses are *arranged*. Those are different questions and the second one is cheap to answer.

Under a correct model each residual is independently as likely to be positive as negative, so the signs form a random sequence and the number of **runs** — unbroken stretches of one sign — has a known distribution: mean $\mu = 2n_+n_-/n + 1$ and variance $(\mu-1)(\mu-2)/(n-1)$. Structure always *reduces* the count, so only the negative tail of $z = (R - \mu)/\sigma_R$ matters. On 981 points $z = -4$ is a one-in-thirty-thousand accident; the three-arc pattern of a wrong functional form runs to $z = -25$ and beyond.

The card carries the flag **residuals are structured, not random** when two things are true at once: the residual scatter is at least half again the point-to-point noise, so there really is something left over, and $z < -4$, so what is left over is the shape of the model rather than the scatter of the data. Both conditions are needed. A perfect fit to noiseless data can trip a runs test on rounding alone, and a large residual with random signs is honest noise, not a wrong model.

When a polynomial wins. A polynomial has no shape of its own; give it enough terms and it will lie along anything smooth. If one takes the top spot *and* its residuals are structured, the card says so under the heading **interpolation**: no named model matched, and the ranking fell back on the family that can approximate any smooth curve. Its coefficients mean nothing, it must not be extrapolated by a single point, and the shape still visible in the residual strip is the model you actually need. This is the one case where a high R^2 is actively misleading — a quintic at $R^2 = 0.9994$ with a three-lobe residual arc is not a result.

Two buttons

copy result

A plain-text block — model name, source, N , the form, the fitted formula, every parameter with its error, and all six statistics — on the clipboard, ready for a lab notebook.

copy fitted y

CSV of x , y , σ , fit , residual for every point, at ten significant figures. For replotting elsewhere or for further analysis of the residuals.

what do these numbers mean?

At the foot of every card is a collapsible glossary that explains each statistic *with your numbers substituted in* — your N , your k , your ν , your RMSE in your units. It is written for someone who has not met reduced χ^2 before, and it is worth opening once even if you have.

Part III

Numbers, figures, workspace

The statistics, properly

What chi-square is, what it is not, and how to tell a good fit from a lucky one.

This chapter is the theory behind the numbers in Chapter 18. It is worth reading once even if you have met all of it before, because the bench makes specific choices that you should be able to see and disagree with.

Chi-square

$$\chi^2 = \sum_i w_i (y_i - f(x_i))^2$$

With a σ column and the default weighting, $w = 1/\sigma^2$, so every point's miss is counted in units of its own error bar. A point one error bar away from the curve contributes exactly 1. This is the only situation in which the absolute value of χ^2 means anything.

Without a σ column, $w = 1$ and χ^2 is a plain sum of squared residuals with units of y^2 . Change your y from volts to millivolts and it changes by 10^6 . It is then a ranking number and nothing more — comparable between models on the same data, meaningless on its own, and not comparable between datasets.

Degrees of freedom

$$\nu = N - k$$

Your N points minus the k parameters the fit was free to move. The intuition is that each fitted parameter spends one point's worth of information: with two parameters and two points you can fit a line exactly and learn nothing about whether a line was right.

Pinned parameters do not count. Parameters charged for the search — the effective k of Chapter 14 — are used for *ranking*, but the ν reported on the card is the honest $N - k$ with the real parameter count, so the number you quote is the conventional one.

Reduced chi-square

$$\chi^2/\nu$$

With error bars, the expected value for a correct model is 1, with a spread of about $\sqrt{(2/\nu)}$. On thirty degrees of freedom that is ± 0.26 , so anything from about 0.7 to 1.3 is unremarkable. On three hundred it is ± 0.08 and a value of 1.4 is a real problem.

Its ability to choose between models is weak, and the reason is worth stating precisely. Adding one meaningless parameter reduces χ^2 by about 1 on average; dividing by ν instead of $\nu + 1$ inflates χ^2/ν by about a factor $N/(N - k)$. Those two effects are the same size, so reduced χ^2 is

roughly *indifferent* to junk parameters rather than opposed to them. That indifference is why the bench adds the parsimony rule on top.

R-squared

$$R^2 = 1 - \text{SSR} / \text{SST}$$

The fraction of the variance in y the curve accounts for, using unweighted sums. It rises with every extra parameter, it cannot decrease when you add one, and on any decent physical fit it is 0.99-something. Report it if a journal insists; never choose a model with it.

Root-mean-square error

$$\text{RMSE} = \sqrt{(\text{SSR} / N)}$$

The typical vertical distance from a point to the curve, in your y units. The most human number on the card. If your instrument's noise is 0.4 mV and the RMSE is 0.42 mV, the model has captured everything there is to capture, whatever the other numbers say.

Information criteria

$$\text{AICc} = N \cdot \ln(\text{SSR}/N) + 2k + 2k(k+1)/(N - k - 1)$$

$$\text{BIC} = N \cdot \ln(\text{SSR}/N) + k \cdot \ln N$$

Both trade goodness of fit against parameter count; both are meaningful only as differences. The rule of thumb for AICc is that a difference of about 2 is mild evidence, 6 is strong, 10 is decisive. BIC's charge grows with N , so it becomes stricter as you collect more data — and it is consistent, meaning that as N grows it converges on the true model if the true model is in the candidate set. AIC is not consistent but is better at prediction. Neither is a substitute for looking at the residuals. The bench computes both from the *unweighted* SSR so that they remain comparable when there is no σ column, and uses the effective k when the search charge is on.

Parameter uncertainties

At the optimum the curvature of χ^2 in parameter space is the Fisher information, and its inverse is the covariance matrix:

$$C = (J^T W J)^{-1}$$

where J is the Jacobian of the model with respect to the parameters. The standard error on parameter j is $\sqrt{C_{jj}}$. When you supplied σ , that is used directly. When you did not, C is scaled by χ^2/ν , which infers the noise from the scatter about the fit.

Three assumptions are buried in that, and all three can fail:

1. The model is correct. If it is not, the residuals contain systematic structure that gets counted as noise, and the errors come out too small.

2. The errors are Gaussian and independent. Correlated noise — drift, $1/f$, a slowly warming sample — breaks this badly, and always in the optimistic direction.
3. χ^2 is approximately quadratic near the minimum. For strongly non-linear parameters — an exponent, a rate inside an exponential — the true confidence region is banana-shaped and a symmetric $\pm$ understates one side.

A "no error estimate" on the card means the covariance matrix was singular: two parameters are exactly degenerate, or the fit is on a boundary. That is information, not a malfunction.

Correlated parameters

The bench reports standard errors but not the correlation matrix. When two parameters are strongly correlated — the prefactor and the exponent of a stretched exponential, the amplitude and width of a peak on a large background — each may be poorly determined on its own while their combination is known precisely. The symptom is a large percentage error on both, on a fit that visibly matches the data. It is not a failure; it is the data telling you it constrains a product, not the factors.

HOW TO CHECK

Pin one of the pair at a plausible value with ! and refit. If the other's error collapses and χ^2 barely moves, they were trading against each other, and you should quote the combination rather than the parts.

The plot and its toolbar

Axes, overlays, residuals, components, and the second-opinion curve.

The toolbar above the plot carries the controls that change what is drawn without changing what is fitted.

Axis scales

Two pairs of buttons, **x lin/log₁₀** and **y lin/log₁₀**. A log axis is the right choice whenever the data spans decades, and it is how a power law becomes a straight line and an exponential decay becomes one too.

A LOG AXIS CANNOT PLACE A ZERO OR A NEGATIVE NUMBER

Those points are simply not drawn. To make sure you never mistake that for a broken file, the status box counts them: "log — 46 values not shown — zero or negative on a log axis". If an axis has no positive values at all, the plot says so rather than drawing an empty frame.

Axis scales reset to linear when you load a different dataset, for the same reason. Carrying a log axis from one file to the next used to hide half of the next one silently.

The toggles

error bars

Draw $\pm\sigma$ on each point. On by default, and only ever visible when there is a σ column.

residuals

The strip beneath the main panel. On by default. Turning it off gives the main panel its height back for a final figure.

connect

Join the data points with a faint line, in *x* order. Useful for a dense spectrum where individual marks blur; misleading for scattered measurements, because it implies you know what happens between them.

components

Draw each additive term of the current fit separately in magenta. Works for combinations, for library models written as sums, and for your own equations — anything the parser can split at a + or - at the top level. This is how you read off how much of a peak is peak and how much is background.

physics model

Appears when the best *named physical* model is not the overall winner. Pressing it opens the physics card and draws that model as a dashed teal curve. Described below.

edit plot

Opens the style editor, Chapter 21.

The best named physics model

A flexible composite can beat a named physical law and tell you nothing. So the bench always fits the library and always keeps its best model available, whatever wins overall. Three things can happen:

- The library model is the overall winner: the result card is tagged "also the best named physics model" in green and there is nothing more to show.
- It is not the winner: a collapsed bar under the result card names it with its χ^2/ν and its rank, and the **physics model** button appears in the toolbar. Pressing either opens the full card and draws the curve.
- The card has a **use this fit** button that promotes it to the main result, with the overall winner demoted to the dashed comparison curve.

The library's own leader is chosen by BIC and parsimony within the library (Chapter 13), *not* by where library models happen to land in the overall table. Those are different questions and they get different answers.

Overlays

The leftmost column of the ranking table is a checkbox per row. Ticking one holds that model's curve on the plot in its own colour — green, purple, teal, amber — up to four at once; a fifth pushes the oldest off. This is the direct way to answer "do these two models actually differ anywhere I measured?" Very often they do not, which is the real answer to your question.

Clicking anywhere else on a row *selects* that model instead: it becomes the headline, its parameters fill the card, its residuals fill the strip, and the previous winner becomes the dashed comparison curve.

Hovering

Move the pointer near a point and a tooltip gives its x , its y , its σ if there is one, the fitted value there, and the residual or pull. Crosshairs mark it in both panels. This is how you identify the one point that is dragging a fit.

The style editor, export, and history

Twenty-one controls for the figure, three ways to save it, and the last twenty fits.

edit plot

Press it and a five-column panel opens between the toolbar and the plot. Every change redraws immediately.

Labels

plot title

Empty by default. Set it and it is drawn above the frame in the serif face.

x axis, y axis

Default to the column headers. Once you type your own it stays, even when you change columns — an explicit title always wins.

Axes & ticks

tick text

8 to 20 px. Also drives the margins, so a larger size makes room for itself.

tick length

0 to 14 px. Zero gives a clean frame with labels only.

axis title

9 to 24 px.

frame weight

0.5 to 5 px. Major ticks are drawn at 85 per cent of this, so the frame and its ticks stay visually consistent.

tick weight

Normal, bold or heavy for the tick *labels*. Bold is the default because thin numerals disappear when a figure is reduced to a single column.

grid

Major + minor, major only, or none.

Data marks

colour

A colour picker, with a **reset** button to return to the theme colour.

size

1 to 12 px radius. This is a maximum: when points are packed closer together than their own width they are shrunk to the spacing available, because a solid band of markers hides the curve

drawn through it.

shape

Circle, square, diamond, triangle or cross.

error caps

0 to 10 px. Zero gives bare bars.

halo around points

A thin ring in the background colour, so overlapping marks stay countable.

Fit curve

Colour, width 0.5 to 8 px, and solid, dashed or dotted.

Legend & layout

legend

Above the plot, inside any of the four corners, or hidden. Inside, it gets a rounded panel with a border so it stays readable over data.

legend text

8 to 20 px.

plot height

auto (about 0.56 of the width, clamped between 300 and 560 px) or a fixed height up to 760.

residual strip

40 to 220 px.

reset all styling returns everything to defaults without touching the data or the fit.

Drawing order

Worth knowing when you are composing a figure: panels, gridlines, then the data — error bars, the faint raw points and baseline if background subtraction is on, then the marks — then the curves on top: overlays, the physics model, the comparison curve, the components, and finally the selected fit. Curves are drawn over the data deliberately, the way a published figure does it; a curve underneath a few hundred dense markers is invisible.

Each curve is drawn only across its own data. The second series stops where its measurements stop, and no curve is extrapolated into the axis padding — a decaying oscillation drawn beyond its data dives off the frame and reads as a rendering fault.

Saving the figure

PNG

Rendered off-screen at three times resolution. A 900 px-wide plot saves as 2700 px, which is about 230 dpi at column width.

PDF

A single-page PDF assembled by hand, containing the same $3\times$ render as an embedded JPEG at 97 per cent quality, centred on A4 with 36 pt margins, in landscape if the plot is wider than tall. The file is written byte by byte with a correct cross-reference table, so it opens anywhere and needs no network.

copy image

A 2× PNG straight to the clipboard, for pasting into a document or a message.

Filenames are built from the data source and the model: `raman_B3_12-Lorentzian-peaks.png`. Hovering is suppressed during the off-screen render, so no crosshair or tooltip ends up in the saved figure.

THE FIGURE IS VECTOR ON SCREEN AND RASTER IN THE FILE

The plot is drawn on a canvas, so the PNG and the PDF both contain pixels rather than paths. Three times resolution is enough for print at normal figure sizes; if you need true vector output, use **copy fitted y** to get the curve as numbers and redraw it in your own plotting package.

History

Every fit is recorded: the model, its reduced χ^2 , the source name, N , whether there was a σ column, and the time. Twenty entries are kept, newest first, in this browser only.

Clicking an entry restores that fit completely — the data, the column assignments, the fit range, the axis scales, the mode and the equations — and refits. Repeated fits of the same dataset and model fold into one entry rather than logging every keystroke.

Entries store the data itself up to forty thousand characters. Beyond that the settings are kept but the data is not, and the row is dimmed with the note "data was too large to keep — settings only". If storage fills up, the oldest entries lose their stored data first, then the oldest entries go.

clear empties the list. History is the only thing that survives closing the page, and it is deliberately separate from everything else: **reset everything** does not touch it.

reset everything

Top right, and it takes two clicks — the first arms it and the button turns amber and says "click again to clear", the second does it, and it disarms itself after four seconds. It throws away the loaded data and returns every setting to its default. History is kept.

NOTHING PERSISTS ACROSS A RELOAD

Closing and reopening the page starts completely clean, with the Kepler sample loaded. This is deliberate. The program is a bench, not a document: what matters gets exported, and what does not gets thrown away rather than quietly resurfacing next week attached to the wrong dataset.

The sample datasets

Five sets that show the program working before you trust it with your own numbers.

The chips at the bottom of the Data panel load datasets with proper axis labels and sensible axis scales. Each one is chosen to exercise a different part of the program.

Kepler III

The real semi-major axes and orbital periods of the eight planets, in AU and years. Eight points, no error bars, loaded with both axes logarithmic. It is here because it is the cleanest power law in existence and because eight points is few enough to show the underdetermined rule doing its job: several composites fit perfectly and none of them is allowed to win.

Expect $T = a^{1.5}$ to about four significant figures. The exponent is Kepler's third law, and reading it off a fit is a satisfying two seconds.

Radioactive decay

Thirty-three points of counts per minute against elapsed time, generated as $1840 e^{-t/12.4} + 22$ with genuine Poisson scatter, and carrying a $\sigma = \sqrt{N}$ column. This is the set to look at if you want to see what a properly weighted fit looks like: χ^2/ν lands near 1, the verdict turns green, and the residual strip shows pulls scattered in a band of about ± 2 with no shape.

It also demonstrates why the offset matters: fit it with a bare exponential and the decay constant comes out wrong because it has to absorb the background.

Absorption line

Seventy-one points of transmission against detuning, a Lorentzian dip on a flat continuum. A negative-amplitude peak, so it exercises the dip handling in both the automatic search and the peak finder. Try the peak finder with *look for* set to *dips only*.

Torsion pendulum

One hundred and eleven points of angular deflection against time: a damped sinusoid with a small offset. The set to use when you want to see the frequency scan work — a damped sinusoid is the model most likely to be fitted badly by software that guesses its starting frequency instead of scanning for it.

Enzyme kinetics

Seventeen points of rate against substrate concentration, Michaelis–Menten with $V_{\max} = 3.8$ and $K_m = 21$, with a σ column. Non-uniformly spaced in x , spanning more than two decades, so it is a good set for seeing what the weighting mode does to a fit. Compare equal against relative weighting and watch K_m move.

USE THEM TO CHECK A SETTING YOU DO NOT UNDERSTAND

When you are unsure what a control does, load a sample where you know the answer, change the control, and watch the answer change. That is faster and far more convincing than reading about it — including reading this manual.

Worked recipes

Nine situations, and exactly what to do in each.

1. A decay curve with counting statistics

1. Load. Confirm the σ column when asked.
2. Leave the weighting on σ column if present.
3. Read the winner. Expect an exponential with an offset; check that the offset's error is small enough for it to be real.
4. Check χ^2/ν is near 1 and the pulls in the residual strip have no shape.
5. **copy result** into your notes.

2. A spectrum with many overlapping bands

1. Load, and set the fit range to the region you care about by dragging across the plot.
2. If the baseline wanders, set Background to *asymmetric least squares*, $\lambda = 10^5$, and check the dashed line threads under the peaks rather than through them.
3. In the peak finder, set the shape (Lorentzian for lifetime broadening, Gaussian for instrumental) and lower *sensitivity* until the magenta triangles sit on every band you believe in and none you do not.
4. Press **find peaks & fit**.
5. Edit the resulting equation: delete a component you do not accept, or pin a position you know (!x3=1332), and refit.
6. Turn on **components** to see the individual bands.

3. An absorption edge, to extract a band gap

1. Restrict the range to the edge region plus a little either side. Fitting the far continuum drags the fit.
2. Search the library for `tauc`, `urbach` or `elliott` depending on what you believe, and click one to load it with starting values.
3. If the fitted edge is systematically off, look at the residual strip — an arc means the wrong power, a step means a missing tail.
4. Try *Absorption coefficient* (*Urbach* $\rightarrow$ *Tauc*), which joins an exponential tail to a square-root edge continuously. A hard-switched model has a kink your data does not.

4. Data spanning several decades

1. Set both axes to $\log_{10}$ to see the structure.
2. Set the weighting to *relative* — $1/y^2$ unless you have real error bars. This is the single most important step, and skipping it is the most common reason a good model loses.
3. Watch the status box: if it says "log — N values not shown", some of your data is non-positive and is missing from the picture.

5. Comparing two samples

1. Get both into one file, either as two columns or stacked with a label column.
2. Load. If it is long format the split happens automatically; if wide, set y_2 .
3. Run automatic mode first to see what each supports.
4. Then switch to manual mode and type the single model you want both described by, so the comparison is like for like.
5. Read the two cards side by side.

6. You already know the model

1. Manual mode. Type it.
2. Leave **show the best automatic fit too** ticked.
3. If the dashed automatic curve visibly beats yours, your model is missing a term. If it does not, you are done.

7. The fit will not converge

1. Give starting values after a semicolon. Order of magnitude is usually enough.
2. Pin the parameters you actually know with **!**, fit the rest, then unpin one at a time.
3. Reduce the range to a region where the model definitely applies, fit there, then extend.
4. Check for a pole inside your range.
5. Reparameterise: fit $\ln A$ instead of A when A spans decades, or $1/\tau$ instead of τ when τ can approach infinity.

8. A few bad points

1. Hover over them to confirm which they are.
2. Turn on **Reject outliers** in manual mode rather than cutting a range in y .
3. Check the rejected count. A handful is a measurement; a large fraction is a wrong model.

9. Preparing a figure for publication

1. Select the model you want shown by clicking its row.
2. **edit plot**: set the axis titles with units, tick text to 12-13 px, tick weight bold, frame weight about 2, grid to *major only* or *none*.

3. Set the legend inside a corner if the plot has empty space, or above if it does not.
4. Turn off **residuals** if the caption will not discuss them, or leave them on if it will — a figure with a residual panel is a more honest figure.
5. Save as PDF for print, PNG for slides.
6. **copy result** for the caption; **copy fitted y** if a referee asks for the numbers.

Part IV

The complete model catalogue

The complete model catalogue

Every named model in the program: its formula exactly as the fitter reads it, its free parameters, the physical bounds it enforces, and what it is for. 442 models in 25 groups.

Read the formulae as the program reads them: x is your data, every other name is a fitted parameter, $^$ is a power, and a comparison such as $(x \geq E_g)$ evaluates to 1 or 0 so that multiplying by it switches a term on. Constants are written out — 8.617333×10^{-5} is Boltzmann's constant in eV/K, so temperatures are in kelvin and energies in electronvolts wherever it appears.

Every model carries starting values computed from your loaded data; those are not listed here because they depend on the data. *bounds* lists the physical constraints the model enforces (Chapter 10); a model with no entry constrains nothing.

Diffraction & structure

17 MODELS

Peak shapes and line-broadening analysis for powder and thin-film diffraction.

Pearson VII peak

5 parameters

$$A / (1 + ((x-x_0)/w)^{2*(2^{(1/m)}-1)})^m + bg$$

parameters A , x_0 , w , m , bg

XRD peak shape; m interpolates Lorentzian (1) to Gaussian (∞)

True Voigt peak

5 parameters

$$A * \text{voigt}(x-x_0, sg, gm) + bg$$

parameters A , x_0 , sg , gm , bg

exact Voigt (Gaussian $\otimes$ Lorentzian) — instrument σ and sample γ separated

Two Voigt peaks

9 parameters

$$A_1 * \text{voigt}(x-x_1, s_1, g_1) + A_2 * \text{voigt}(x-x_2, s_2, g_2) + bg$$

parameters A_1 , x_1 , s_1 , g_1 , A_2 , x_2 , s_2 , g_2 , bg

a resolved doublet with true Voigt shapes

Williamson–Hall

2 parameters

$$K_L + 4 * \epsilon * x$$

parameters K_L , ϵ

$\beta \cos \theta$ vs $\sin \theta$ — intercept gives size, slope gives strain

Log-normal size distribution

3 parameters

$$A / (x * s * \sqrt{2 * \pi}) * \exp(-(\ln(x) - \mu)^2 / (2 * s^2))$$

parameters A , s , μ

particle or grain size histogram

Modified Williamson–Hall (anisotropic)

3 parameters

$$K_L + 4 * \epsilon * x + q * x^2$$

parameters K_L , ϵ , q

$\beta \cos \theta = K \lambda / D + 4 \epsilon \cdot \sin \theta + q \cdot \sin^2 \theta$ — the quadratic term absorbs dislocation-type anisotropy

Debye–Waller / thermal decay of intensity

3 parameters

$$I_0 * \exp(-2 * B_w * x^2) + c$$

parameters I_0 , B_w , c

$I = I_0 \cdot \exp(-2B \cdot s^2) + c$: the fall-off of Bragg intensity with scattering vector or temperature

Caglioti instrumental resolution

3 parameters

$$\text{sqrt}(\max(U \cdot \tan(x)^2 + V \cdot \tan(x) + W, 0))$$

parameters U, V, W

$\text{FWHM}^2 = U \cdot \tan^2 \theta + V \cdot \tan \theta + W$, with θ in radians — the standard description of a diffractometer resolution function

Scherrer broadening

2 parameters

$$A / \cos(x) + c$$

parameters A, c

$\beta = K\lambda / (D \cdot \cos \theta)$, θ in radians: the whole θ dependence of size broadening in one parameter

 $K\alpha_1/K\alpha_2$ doublet (Pearson VII pair)

6 parameters

$$A * (1 / (1 + ((x - x_0) / w)^{2 * (2^{(1/m)} - 1)})^m + 0.5 / (1 + ((x - x_0 - dk) / w)^{2 * (2^{(1/m)} - 1)})^m) + bg$$

parameters A, x0, w, m, dk, bg

one reflection measured with an unfiltered tube: the $K\alpha_2$ component at fixed 1:2 intensity and a fixed offset, sharing the sample's own width and shape

Three Pearson VII reflections

12 parameters

$$A1 / (1 + ((x - x1) / w1)^{2 * (2^{(1/m)} - 1)})^m + A2 / (1 + ((x - x2) / w2)^{2 * (2^{(1/m)} - 1)})^m + A3 / (1 + ((x - x3) / w3)^{2 * (2^{(1/m)} - 1)})^m + bg + sl * x$$

parameters A1, x1, w1, m, A2, x2, w2, A3, x3, w3, bg, sl

three reflections sharing one shape parameter m, as a Rietveld-style profile does, with independent widths

Size-dependent melting (Gibbs–Thomson)

2 parameters

$$T_{inf} * (1 - \beta / x)$$

parameters Tinf, bet

$T_m(d) = T_{m\infty}(1 - \beta/d)$: surface energy lowers the melting point of a nanoparticle or a confined liquid in proportion to its inverse size

Guinier (small-angle scattering)

3 parameters

$$I_0 * \exp(-x^2 R_g^2 / 3) + bg$$

parameters I0, Rg, bg

$I(q) = I_0 \cdot \exp(-q^2 R_g^2 / 3) + \text{background}$: the radius of gyration from the low-q region, valid while $qR_g \lesssim 1.3$

Debye form factor (Gaussian polymer chain)

3 parameters

```
I0*where(x^2*Rg^2 < 1e-4, 1 - x^2*Rg^2/3, 2*(exp(-x^2*Rg^2) + x^2*Rg^2 - 1)/(x^2*Rg^2)^2) + bg
```

parameters I0, Rg, bg

$P(q) = 2(e^{-u} + u - 1)/u^2$ with $u = q^2 R_g^2$: a flexible coil in a theta solvent, q^{-2} at high q where Guinier has already failed

Beaucage unified fit

5 parameters

```
G*exp(-x^2*Rg^2/3) + Bu*(erf(x*Rg/sqrt(6))^3/x)^P + bg
```

parameters G, Rg, Bu, P, bg

Guinier knee and power-law tail joined without a switch: $G \cdot \exp(-q^2 R_g^2/3) + B \cdot [\text{erf}(q R_g/\sqrt{6})^3/q]^P$.
 $P = 4$ smooth surfaces (Porod), 3–4 rough, 2 Gaussian chains, below 3 mass fractals

Polydisperse spheres (SAXS / SANS)

4 parameters

```
I0*spherepoly(x, R, pd) + bg
```

parameters I0, R, pd, bg

the exact sphere form factor $[3(\sin qR - qR \cos qR)/(qR)^3]^2$ averaged over a Gaussian spread of radii (relative width pd), each weighted by volume². The minima fill in as pd grows

Porod law with background

2 parameters

```
Kp/x^4 + bg
```

parameters Kp, bg

$I(q) = K_p/q^4 + b$ at high q for sharp interfaces; $K_p = 2\pi(\Delta\rho)^2 S/V$ gives the specific surface area

Photoemission & core levels

6 MODELS

Core-level and valence-band lineshapes, including the metallic asymmetry.

Doniach–Šunjić

5 parameters

$$A \cos(\pi \alpha / 2 + (1 - \alpha) \operatorname{atan}((x - x_0) / \gamma)) / ((x - x_0)^2 + \gamma^2)^{((1 - \alpha) / 2)} + b \gamma$$

parameters A , α , x_0 , γ , $b \gamma$

asymmetric core level from metallic screening; α is the asymmetry

Spin–orbit doublet (Voigt pair)

7 parameters

$$A (\operatorname{voigt}(x - x_0, s \gamma, \gamma m) + \text{ratio} \operatorname{voigt}(x - x_0 - d_{\text{so}}, s \gamma, \gamma m)) + b \gamma$$

parameters A , x_0 , $s \gamma$, γm , ratio , d_{so} , $b \gamma$

fixed branching ratio and splitting, as for a 2p or 3d core level

Fermi edge (thermally broadened)

5 parameters

$$A \operatorname{fermi}(x - E_f, kT) + m x + c_0$$

parameters A , E_f , kT , m , c_0

photoemission cut-off; kT in the same units as x

Two Doniach–Šunjić peaks

8 parameters

$$A_1 \cos(\pi a_1 / 2 + (1 - a_1) \operatorname{atan}((x - x_1) / \gamma)) / ((x - x_1)^2 + \gamma^2)^{((1 - a_1) / 2)} + A_2 \cos(\pi a_1 / 2 + (1 - a_1) \operatorname{atan}((x - x_2) / \gamma)) / ((x - x_2)^2 + \gamma^2)^{((1 - a_1) / 2)} + m x + c$$

parameters A_1 , a_1 , x_1 , γ , A_2 , x_2 , m , c

a spin–orbit split core level with the metallic asymmetry on both components

Fermi edge × linear DOS

5 parameters

$$(A + s(x - E_f)) \operatorname{fermi}(x - E_f, kT) + c$$

parameters A , s , E_f , kT , c

the thermally broadened edge multiplying a density of states that is not flat, plus a background

Three Voigt components + Shirley-like background

12 parameters

$$A_1 \operatorname{voigt}(x - x_1, s \gamma, \gamma m) + A_2 \operatorname{voigt}(x - x_2, s \gamma, \gamma m) + A_3 \operatorname{voigt}(x - x_3, s \gamma, \gamma m) + b \gamma + \text{sh} \operatorname{atan}((x - x_s) / w_s)$$

parameters A_1 , x_1 , $s \gamma$, γm , A_2 , x_2 , A_3 , x_3 , $b \gamma$, sh , x_s , w_s

a chemically shifted core level resolved into three states, on the rising inelastic step that a Shirley background models — the arctangent is its usual closed-form stand-in

Dielectric & impedance

11 MODELS

Relaxation and conduction in the frequency domain, from Debye to the fully general Havriliak–Negami form.

Debye relaxation (ϵ'')

2 parameters

$$de \cdot (x \cdot \tau) / (1 + (x \cdot \tau)^2)$$

parameters de, τ

dielectric loss vs angular frequency, single relaxation time

Cole–Cole (ϵ'')

3 parameters

$$de \cdot (x \cdot \tau)^{(1-\alpha)} \cdot \cos(\pi \cdot \alpha / 2) / (1 + 2 \cdot (x \cdot \tau)^{(1-\alpha)} \cdot \sin(\pi \cdot \alpha / 2) + (x \cdot \tau)^{(2-2 \cdot \alpha)})$$

parameters de, τ, α

symmetric broadening of Debye; α is the width parameter

Jonscher universal power law

3 parameters

$$sdc + A \cdot x^n$$

parameters sdc, A, n

$\sigma(\omega) = \sigma_{dc} + A \cdot \omega^n$, the near-universal ac response

Curie–Weiss permittivity

3 parameters

$$C / (x - T_c) + e_0$$

parameters C, T_c, e_0

ferroelectric $\epsilon(T)$ above the transition

Havriliak–Negami (ϵ'')

4 parameters

$$de \cdot (1 + 2 \cdot (x \cdot \tau)^\alpha \cdot \sin(\pi \cdot \alpha / 2) + (x \cdot \tau)^{2 \cdot \alpha})^{-(\beta/2)} \cdot \sin(\beta \cdot \arctan((x \cdot \tau)^\alpha \cdot \cos(\pi \cdot \alpha / 2), 1 + (x \cdot \tau)^\alpha \cdot \sin(\pi \cdot \alpha / 2)))$$

parameters de, τ, α, β

the general relaxation: α broadens symmetrically, β skews it. $\alpha = 1$ gives Cole–Davidson, $\beta = 1$ gives Cole–Cole, both = 1 gives Debye

Cole–Davidson (ϵ'')

3 parameters

$$de \cdot \cos(\arctan(x \cdot \tau))^\beta \cdot \sin(\beta \cdot \arctan(x \cdot \tau))$$

parameters de, τ, β

asymmetric broadening on the high-frequency side only — the glass-former signature

Debye loss + dc conduction

3 parameters

$$\text{de} \cdot (x \cdot \tau) / (1 + (x \cdot \tau)^2) + \text{sdc} / x$$

parameters de, tau, sdc

 $\epsilon'' = \Delta\epsilon \cdot \omega\tau / (1 + \omega^2\tau^2) + \sigma / \epsilon_0\omega$ — the $1/\omega$ tail that buries the relaxation peak at low frequency
Jonscher — two power laws

5 parameters

$$\text{sdc} + A \cdot x^n + B_j \cdot x^m$$

parameters sdc, A, n, B_j, m
 $\sigma_{\text{dc}} + A\omega^n + B\omega^m$: separate low- and high-frequency dispersion regimes
Diffuse ferroelectric (modified Curie–Weiss)

4 parameters

$$1 / (1/\epsilon_m + \text{abs}(x - T_m)^\gamma / C)$$

parameters ϵ_m , T_m , γ , C
 $1/\epsilon - 1/\epsilon_m = (T - T_m)^\gamma / C$. $\gamma = 1$ is a normal ferroelectric, $\gamma \rightarrow 2$ a full relaxor
Vogel–Fulcher relaxor freezing

3 parameters

$$f_0 \cdot \exp(-E_a / (8.617333 \cdot 10^{-5} \cdot (x - T_f)))$$

parameters f_0 , E_a , T_f
 $f = f_0 \cdot \exp[-E_a / k(T_m - T_f)]$: the peak temperature against measuring frequency in a relaxor
Barrett quantum paraelectric

4 parameters

$$C / ((T_1/2) \cdot \coth(T_1/(2 \cdot x)) - T_0) + \epsilon_{\text{inf}}$$

parameters C , T_1 , T_0 , ϵ_{inf}
 $\epsilon(T) = C / [(T_1/2) \coth(T_1/2T) - T_0] + \epsilon_{\infty}$: Curie–Weiss at high T , saturating below T_1 because zero-point motion stops the ferroelectric transition (SrTiO₃, KTaO₃)

Phonons & luminescence

11 MODELS

Phonon populations, anharmonic decay, and the time and temperature dependence of emission.

Exciton-phonon linewidth

4 parameters

$$G0 + gac \cdot x + gL0 \cdot \text{bose}(ELO, 8.617333e-5 \cdot x)$$

parameters $G0$, gac , $gL0$, ELO

$$\Gamma(T) = \Gamma0 + \gamma_{ac} \cdot T + \gamma_{LO}/(\exp(ELO/kT) - 1)$$

Klemens anharmonic phonon shift

3 parameters

$$w0 - C \cdot (1 + 2 \cdot \text{bose}(hw, 8.617333e-5 \cdot x))$$

parameters $w0$, C , hw

Raman peak position vs temperature

Bi-exponential decay

5 parameters

$$A1 \cdot \exp(-x/t1) + A2 \cdot \exp(-x/t2) + c$$

parameters $A1$, $t1$, $A2$, $t2$, c

two-component PL or carrier decay

Tri-exponential decay

7 parameters

$$A1 \cdot \exp(-x/t1) + A2 \cdot \exp(-x/t2) + A3 \cdot \exp(-x/t3) + c$$

parameters $A1$, $t1$, $A2$, $t2$, $A3$, $t3$, c

three decay channels

Exponentially modified Gaussian

5 parameters

$$A \cdot \exp(0.5 \cdot (s/t)^2 - (x-m)/t) \cdot 0.5 \cdot (1 + \text{erf}((x-m-s^2/t)/(s \cdot \sqrt{2}))) + c$$

parameters A , s , t , m , c

an exponential decay convolved with the instrument response

Arrhenius PL quenching

3 parameters

$$I0/(1 + A \cdot \exp(-Ea/(8.617333e-5 \cdot x)))$$

parameters $I0$, A , Ea

integrated intensity vs temperature with one non-radiative channel

Balkanski anharmonic phonon shift

5 parameters

$$\omega_0 - A \cdot (1 + 2 \cdot \text{bose}(h_1, 8.617333e-5 \cdot x)) - B_q \cdot (1 + 3 \cdot \text{bose}(h_2, 8.617333e-5 \cdot x) + 3 \cdot \text{bose}(h_2, 8.617333e-5 \cdot x)^2)$$

parameters ω_0 , A , h_1 , B_q , h_2

$\omega(T)$ with both three- and four-phonon decay channels — the full anharmonic form, of which the Klemens model is the first term

Bimolecular + trap-assisted decay

5 parameters

$$A / (1 + k_2 \cdot x) + B_t \cdot \exp(-x/t_1) + c$$

parameters A , k_2 , B_t , t_1 , c

a $1/(1+kt)$ bimolecular term plus a first-order trapping channel — the usual description of a photoluminescence transient that is not a single exponential

PL intensity against excitation power

2 parameters

$$A \cdot x^{k_p}$$

parameters A , k_p

$I = A \cdot P^k$: $k \approx 1$ excitonic or defect-bound recombination, $k \approx 2$ free-carrier or biexciton, $k < 1$ saturating defect emission

Balkanski anharmonic phonon linewidth

4 parameters

$$G_0 + A \cdot (1 + 2 \cdot \text{bose}(\omega_0/2, 0.695 \cdot x)) + B_q \cdot (1 + 3 \cdot \text{bose}(\omega_0/3, 0.695 \cdot x) + 3 \cdot \text{bose}(\omega_0/3, 0.695 \cdot x)^2)$$

parameters G_0 , A , ω_0 , B_q

$\Gamma(T) = \Gamma_0 + A[1 + 2n(\omega_0/2)] + B[1 + 3n(\omega_0/3) + 3n^2(\omega_0/3)]$, ω_0 in cm^{-1} with $k_B = 0.695 \text{ cm}^{-1}/\text{K}$: cubic and quartic decay. Pin ω_0 to the measured Raman frequency (! $\omega_0 = \dots$)

Soft mode near a structural transition (Cochran)

3 parameters

$$\sqrt{\text{where}(x > T_c, A_q \cdot (x - T_c), 2 \cdot A_q \cdot (T_c - x))} + \omega_{00}$$

parameters T_c , A_q , ω_{00}

$\omega^2 \propto |T - T_c|$, with the Landau factor 2 on the ordered side: the frequency of the mode that condenses at the transition

Devices

13 MODELS

Current–voltage and capacitance–voltage characteristics of diodes, films and barriers.

Shockley diode

4 parameters

$$I0 * (\exp(\min(x / (n * kT), 700)) - 1) + Gsh * x$$

parameters $I0$, n , kT , Gsh

$I(V)$ with ideality factor n ; kT in eV

Mott–Gurney space-charge-limited

2 parameters

$$A * x^2 + B * x$$

parameters A , B

$J \propto V^2$ trap-free SCLC, with an ohmic term

Power-law J–V (trap-limited)

3 parameters

$$A * x^m + B * x$$

parameters A , m , B

$J \propto V^m$ with $m > 2$ signals trap filling

Richardson–Schottky emission

2 parameters

$$A * x^2 * \exp(-\phi / (8.617333e-5 * x))$$

parameters A , ϕ

$\ln(J/T^2)$ vs $1/T$ style barrier extraction

Fowler–Nordheim tunnelling

2 parameters

$$A * x^2 * \exp(-Bf / \text{abs}(x))$$

parameters A , Bf

$J = AV^2 \cdot \exp(-B/V)$ — a straight line on $\ln(J/V^2)$ against $1/V$

Poole–Frenkel emission

2 parameters

$$A * x * \exp(bf * \text{sqrt}(\text{abs}(x)))$$

parameters A , bf

$J = AV \cdot \exp(b\sqrt{V})$: field-assisted release from traps, linear on $\ln(J/V)$ against $\sqrt{V}$

Shockley diode with series resistance

5 parameters

$$I_0 * (\exp((x - R_s * I_0 * (\exp(x / (n_d * kT)) - 1)) / (n_d * kT)) - 1) + G_{sh} * x$$

parameters I_0 , R_s , n_d , kT , G_{sh}

the ideal diode plus an ohmic series drop, solved to first order — what a real I–V actually looks like at high bias

Mott–Schottky (C vs V)

2 parameters

$$C_0 / \sqrt{\text{abs}(1 + x / V_{bi})}$$

parameters C_0 , V_{bi}

$C = C_0 / \sqrt{(1 + V/V_{bi})}$: the depletion capacitance whose $1/C^2$ is linear in bias, giving the doping density and flat-band potential

Carrier freeze-out

2 parameters

$$A * x^{0.75} * \exp(-E_d / (2 * 8.617333e-5 * x))$$

parameters A , E_d

$n(T) \propto T^{3/4} \cdot \exp(-E_d / 2kT)$ in the freeze-out regime of a partially compensated semiconductor

Mobility — phonon and impurity scattering

2 parameters

$$1 / (x^{1.5} / A_i + x^{(-1.5)} / B_i)$$

parameters A_i , B_i

Matthiessen's rule with a $T^{(-3/2)}$ lattice term and a $T^{(3/2)}$ ionised-impurity term: the peak between them locates the crossover

Simmons tunnelling (low bias)

2 parameters

$$a * x + b * x^3$$

parameters a , b

$I = aV + bV^3$ through a thin barrier: the cubic term carries the barrier height and width

Caughey–Thomas mobility against doping

4 parameters

$$\mu_{min} + (\mu_{max} - \mu_{min}) / (1 + (x / N_{ref})^{\alpha})$$

parameters μ_{min} , μ_{max} , N_{ref} , α

$\mu(N) = \mu_{min} + (\mu_{max} - \mu_{min}) / (1 + (N / N_{ref})^{\alpha})$: lattice-limited mobility at low doping falling to the ionised-impurity floor. Silicon electrons: 1414, 68.5 cm²/V·s, 9.2×10¹⁶ cm⁻³, α 0.711

Graphene resistance against gate voltage

6 parameters

$$R_c + Lw / (1.602176634e-19 * \mu * \sqrt{n_0^2 + (C_g * (x - V_D) / 1.602176634e-19)^2})$$

parameters R_c , Lw , μ , n_0 , C_g , V_D

$R = R_c + (L/W) / [e\mu\sqrt{(n_0^2 + (C_g(V - V_D)/e)^2)}]$: contact resistance, mobility (m²/V·s) and the residual puddle density n_0 (m⁻²). C_g is pinned at 1.15×10⁻⁴ F/m² (300 nm SiO₂) and L/W at 1 — set both to your device

Transport

36 MODELS

Resistivity, conductivity and magnetotransport, from band conduction through hopping to quantum oscillations.

Weak localisation (HLN)

3 parameters

$$\alpha * (\text{digamma}(0.5 + B_p / \max(\text{abs}(x), 1e-12)) - \ln(B_p / \max(\text{abs}(x), 1e-12))) + c$$

parameters α , B_p , c

magnetoconductance of a 2D system; $\alpha \approx -0.5$ per coherent channel

Kondo conductance G(T)

4 parameters

$$G_0 / (1 + (2^{(1/s)} - 1) * (x/T_K)^2)^s + G_c$$

parameters G_0 , s , T_K , G_c

empirical Goldhaber-Gordon form; $s = 0.22$ for spin- $\frac{1}{2}$

Two-band conductivity

4 parameters

$$n_1 * m_1 / (1 + (m_1 * x)^2) + n_2 * m_2 / (1 + (m_2 * x)^2)$$

parameters n_1 , m_1 , n_2 , m_2

two carrier types in parallel

Bloch–Grüneisen (n = 5 limit)

3 parameters

$$r_0 + A * x^5 / (1 + (x/T_d)^4)$$

parameters r_0 , A , T_d

phonon-limited resistivity: $\rho_0 + A \cdot T^5$ at low T, linear at high T

Arrhenius

2 parameters

$$A * \exp(-E_a / (8.617333e-5 * x))$$

parameters A , E_a

thermally activated; E_a in eV, x in kelvin

Mott VRH (3D)

2 parameters

$$r_0 * \exp((T_0/x)^{0.25})$$

parameters r_0 , T_0

variable-range hopping, exponent $\frac{1}{4}$

Efros–Shklovskii VRH

2 parameters

$$r0 \cdot \exp((T0/x)^{0.5})$$

parameters $r0, T0$ hopping with a Coulomb gap, exponent $\frac{1}{2}$ **Drude conductivity**

2 parameters

$$s0 / (1 + (x \cdot \tau)^2)$$

parameters $s0, \tau$

$$\sigma(\omega) = \sigma_0 / (1 + \omega^2 \tau^2)$$

Power-law resistivity

3 parameters

$$r0 + A \cdot x^n$$

parameters $r0, A, n$ $\rho = \rho_0 + AT^n$ — $n \approx 2$ Fermi liquid, $n \approx 5$ phonon-limited**Vogel–Fulcher–Tammann**

3 parameters

$$A \cdot \exp(B / (x - T0))$$

parameters $A, B, T0$ glassy / relaxor dynamics diverging at T_0 **Mott VRH — free exponent p**

3 parameters

$$r0 \cdot \exp((T0/x)^p)$$

parameters $r0, T0, p$ $\rho = \rho_0 \cdot \exp[(T_0/T)^p]$. $p = \frac{1}{4}$ is 3D Mott, $\frac{1}{3}$ is 2D, $\frac{1}{2}$ is Efros–Shklovskii — fitting p tells you which regime you are in**Activated + hopping in parallel**

4 parameters

$$A \cdot \exp(-Ea / (8.617333e-5 \cdot x)) + Bh \cdot \exp(-(T0/x)^{0.25})$$

parameters $A, Ea, Bh, T0$ $\sigma = \sigma_{act} \cdot \exp(-Ea/kT) + \sigma_{hop} \cdot \exp[-(T_0/T)^{\frac{1}{4}}]$ — band conduction freezing out into hopping, the usual crossover**Arrhenius — two activation energies**

4 parameters

$$A1 \cdot \exp(-E1 / (8.617333e-5 \cdot x)) + A2 \cdot \exp(-E2 / (8.617333e-5 \cdot x))$$

parameters $A1, E1, A2, E2$

two thermally activated channels in parallel; the kink in an Arrhenius plot

Arrhenius + residual channel

3 parameters

$$A \cdot \exp(-E_a / (8.617333e-5 \cdot x)) + s_0$$

parameters A , E_a , s_0

activated conduction on top of a temperature-independent floor (leakage, shunt, impurity band)

Small-polaron hopping — adiabatic

2 parameters

$$(A/x) \cdot \exp(-E_a / (8.617333e-5 \cdot x))$$

parameters A , E_a

$\sigma = (A/T) \cdot \exp(-E_a/kT)$. The $1/T$ prefactor is exactly what separates polaron hopping from plain Arrhenius — fit both and compare

Small-polaron hopping — non-adiabatic

2 parameters

$$A \cdot x^{-1.5} \cdot \exp(-E_a / (8.617333e-5 \cdot x))$$

parameters A , E_a

$\sigma = A \cdot T^{-3/2} \cdot \exp(-E_a/kT)$; the non-adiabatic limit carries a steeper prefactor

Bloch–Grüneisen — free exponent

4 parameters

$$r_0 + A \cdot x^n / (1 + (x/T_d)^{(n-1)})$$

parameters r_0 , A , n , T_d

$\rho_0 + A \cdot T^n / (1 + (T/\theta)^{(n-1)})$: T^n at low temperature, linear above θ_D . $n = 5$ phonons, 3 s–d scattering, 2 electron–electron

Fermi liquid + phonon resistivity

3 parameters

$$r_0 + A \cdot x^2 + B_p \cdot x^5$$

parameters r_0 , A , B_p

$\rho_0 + AT^2 + BT^5$ — electron–electron and electron–phonon terms separated

Kondo resistivity minimum

3 parameters

$$r_0 + A \cdot x^5 - B_k \cdot \ln(x)$$

parameters r_0 , A , B_k

$\rho_0 + AT^5 - B \cdot \ln T$: the log term rising as T falls is what puts a minimum in $\rho(T)$

Drude–Smith conductivity (Re σ)

3 parameters

$$s_0 \cdot (1 / (1 + (x \cdot \tau)^2) + c_b \cdot (1 - (x \cdot \tau)^2) / (1 + (x \cdot \tau)^2)^2)$$

parameters s_0 , τ , c_b

Drude with one backscattering parameter c : $c = 0$ is free Drude, $c = -1$ is fully localised — the standard THz / nanostructure form

Kohler magnetoresistance

3 parameters

$$r_0 * (1 + A * \text{abs}(x)^m)$$

parameters r_0 , A , m
 $\Delta\rho/\rho_0 = A \cdot B^m$; $m = 2$ is the conventional orbital result, $m < 2$ signals it is not
Two-band magnetoresistance (saturating)

3 parameters

$$r_0 * (1 + A * x^2 / (1 + B_s * x^2))$$

parameters r_0 , A , B_s
 $\rho(B) = \rho_0 (1 + AB^2 / (1 + BB^2))$ — quadratic at low field, saturating at high field
Weak antilocalisation (HLN + B^2 term)

4 parameters

$$\alpha * (\text{digamma}(0.5 + B_p / \max(\text{abs}(x), 1e-12)) - \ln(B_p / \max(\text{abs}(x), 1e-12))) + b_q * x^2 + c_0$$

parameters α , B_p , b_q , c_0
the HLN form with the classical B^2 background that always sits under it in a real magnetoconductance trace
Lifshitz–Kosevich thermal damping

3 parameters

$$A * (a * x) / \sinh(a * x) + c$$

parameters A , a , c
quantum-oscillation amplitude against temperature: $X/\sinh X$ with $X = 2\pi^2 k_B T / \hbar \omega_c$. The fitted a gives the cyclotron effective mass directly
Shubnikov–de Haas oscillation (in $1/B$)

5 parameters

$$A * \sqrt{\text{abs}(x)} * \exp(-b_D * x) * \cos(2 * \pi * (F * x - \text{ph})) + c$$

parameters A , b_D , F , ph , c
plot against $x = 1/B$. F is the oscillation frequency in tesla, so the extremal Fermi-surface area follows from Onsager; the Dingle term gives the scattering time
de Haas–van Alphen / damped oscillation in $1/B$

6 parameters

$$A * \exp(-b_D * x) * \cos(2 * \pi * (F * x - \text{ph})) + m * x + c$$

parameters A , b_D , F , ph , m , c
the same oscillation without the $\sqrt{B}$ prefactor, for magnetisation rather than resistivity
Bloch–Grüneisen (full integral, $n = 5$)

3 parameters

$$r_0 + A * (x/T_d)^5 * \text{bgint}(T_d/x, 5)$$

parameters r_0 , A , T_d
the exact phonon-limited resistivity: $\rho_0 + A(T/\theta)^5 J_5(\theta/T)$, evaluated by quadrature rather than its T^5 and linear limits. θ is the transport Debye temperature

Bloch–Grüneisen (full integral, free n)

4 parameters

$$r_0 + A \cdot (x/T_d)^n \cdot \text{bgint}(T_d/x, n)$$

parameters r_0, A, T_d, n

$n = 5$ electron–phonon, $n = 3$ s–d interband scattering, $n = 2$ electron–electron or magnon. The integral keeps the crossover exact at every temperature

Resistivity saturation (parallel-resistor, Mott–Ioffe–Regel)

4 parameters

$$1/(1/(r_0 + A \cdot x^n) + 1/rsat)$$

parameters $r_0, A, n, rsat$

Wiesmann form $1/\rho = 1/\rho_{ideal} + 1/\rho_{sat}$: resistivity stops rising once the mean free path reaches the lattice spacing, as in A15 compounds and bad metals

Hamann Kondo resistivity (+ phonons)

5 parameters

$$r_0 + rk \cdot 0.5 \cdot (1 - \ln(x/TK) / \sqrt{\ln(x/TK)^2 + \pi^2 S(S+1)}) + A \cdot x^5$$

parameters r_0, rk, TK, S, A

the Hamann expression $\rho_K = \rho_0/2 \cdot [1 - \ln(T/T_K)/\sqrt{(\ln^2(T/T_K) + \pi^2 S(S+1))}]$, which saturates at low T instead of diverging like the bare log. S is pinned at $\frac{1}{2}$; remove the ! to fit it

Weak antilocalisation — two HLN channels

5 parameters

$$a_1 \cdot (\text{digamma}(0.5 + B_1/\max(\text{abs}(x), 1e-12)) - \ln(B_1/\max(\text{abs}(x), 1e-12))) + a_2 \cdot (\text{digamma}(0.5 + B_2/\max(\text{abs}(x), 1e-12)) - \ln(B_2/\max(\text{abs}(x), 1e-12))) + c_0$$

parameters a_1, B_1, a_2, B_2, c_0

top and bottom surfaces (or surface and bulk) each with their own phase-coherence field. α summing to about -1 means two decoupled channels, $-\frac{1}{2}$ means they are coupled

Quadratic-to-linear magnetoresistance crossover

3 parameters

$$r_0 \cdot (1 + A \cdot (\sqrt{1 + (x/B_c)^2} - 1))$$

parameters r_0, A, B_c

$\rho(B) = \rho_0 [1 + A(\sqrt{1 + (B/B^*)^2} - 1)]$: quadratic below B^* , linear above — the non-saturating linear MR of Dirac semimetals and disordered narrow-gap systems

Anomalous Hall effect

3 parameters

$$R_0 \cdot x + R_s \cdot \tanh(x/B_s)$$

parameters R_0, R_s, B_s

$\rho_{xy} = R_0 B + R_s \cdot M(B)$, with the magnetisation as a tanh that saturates at B_s . R_0 gives the carrier density, R_s the anomalous (Berry-curvature or skew-scattering) part

Two-band Hall resistivity (electrons + holes)

4 parameters

$$x * ((sh * mh - se * me) + (sh * me - se * mh) * mh * me * x^2) / ((sh + se)^2 + (sh * me - se * mh)^2 * x^2)$$

parameters sh, mh, se, me

written with each carrier's conductivity $\sigma = ne\mu$ and mobility, which is the same model but far better conditioned than densities of 10^{25} . SI: ρ_{xy} in $\Omega \cdot m$, B in T, σ in S/m, μ in $m^2/V \cdot s$; the densities are $n = \sigma/(e\mu)$. ρ_{xy} alone leaves one combination loose — fit ρ_{xx} too, or pin a mobility

Two-band longitudinal magnetoresistance

4 parameters

$$((sh + se) + (sh * me^2 + se * mh^2) * x^2) / ((sh + se)^2 + (sh * me - se * mh)^2 * x^2)$$

parameters sh, se, me, mh

$\rho_{xx}(B)$ for electrons and holes in the same σ, μ form. A compensated semimetal (equal densities) never saturates — the quadratic rise of WTe_2 and bismuth; any imbalance makes it level off at high field

Chiral-anomaly magnetoconductance (Weyl semimetal)

3 parameters

$$sig0 + Cw * x^2 - Cwal * \sqrt{\text{abs}(x)}$$

parameters sig0, Cw, Cwal

$\sigma(B) = \sigma_0 + C_W \cdot B^2 - C_{WAL} \cdot \sqrt{|B|}$ for B parallel to the current: the positive B^2 term from charge pumping between Weyl nodes, on top of 3D weak antilocalisation

Magnetism

28 MODELS

Magnetisation and susceptibility against field and temperature, plus muon relaxation.

Brillouin magnetisation

3 parameters

$$M_s \cdot \text{brillouin}(a \cdot x, J)$$

parameters M_s, a, J

$M(H)$ for total angular momentum J ; $y = gJ\mu_B H/k_BT$ lumped into a

Modified Curie–Weiss

3 parameters

$$\chi_0 + C/(x - \theta)$$

parameters χ_0, C, θ

with a temperature-independent term χ_0

Critical scaling $M(T)$

3 parameters

$$M_0 \cdot (x < T_c) \cdot (1 - x/T_c)^\beta$$

parameters M_0, T_c, β

order parameter below T_c ; $\beta \approx 0.33\text{--}0.5$

Curie–Weiss

3 parameters

$$C/(x - \theta) + \chi_0$$

parameters C, θ, χ_0

$$\chi(T) = C/(T - \theta) + \chi_0$$

Bloch $T^{3/2}$ magnetisation

2 parameters

$$M_0 \cdot (1 - B \cdot x^{1.5})$$

parameters M_0, B

spin-wave reduction of $M(T)$

Langevin (superparamagnet)

2 parameters

$$M_s \cdot (\coth(ax) - 1/(ax))$$

parameters M_s, a

$$M(H) = M_s [\coth(aH) - 1/(aH)]$$

Curie–Weiss with critical exponent

4 parameters

$$\chi_0 + C/(x - \theta)^g$$

parameters χ_0, C, θ, g

$\chi = \chi_0 + C/(T - \theta)^\gamma$. $\gamma = 1$ is mean field; $\gamma \neq 1$ marks a Griffiths phase or genuine critical behaviour

Brillouin + linear susceptibility

4 parameters

$$M_s \cdot \text{brillouin}(a \cdot x, J) + \chi \cdot x$$

parameters M_s, a, J, χ

a saturating paramagnetic component on top of a linear (para- or diamagnetic) background — the usual $M(H)$ of a real sample

Langevin + linear background

3 parameters

$$M_s \cdot \text{langevin}(a \cdot x) + \chi \cdot x$$

parameters M_s, a, χ

superparamagnetic particles plus a linear matrix contribution

Bloch law with $T^{5/2}$ correction

3 parameters

$$M_0 \cdot (1 - B_b \cdot x^{1.5} - C_b \cdot x^{2.5})$$

parameters M_0, B_b, C_b

$M = M_0(1 - BT^{\{3/2\}} - CT^{\{5/2\}})$; the next spin-wave term, needed once the data reaches a decent fraction of T_c

Bloch law with spin-wave gap

3 parameters

$$M_0 \cdot (1 - B_b \cdot x^{1.5} \cdot \exp(-D/x))$$

parameters M_0, B_b, D

$M = M_0(1 - BT^{\{3/2\}} \cdot e^{\{-\Delta/T\}})$; an anisotropy gap freezes the magnons out and flattens $M(T)$ at low temperature

Law of approach to saturation

4 parameters

$$M_s \cdot (1 - a/x - b/x^2) + \chi \cdot x$$

parameters M_s, a, b, χ

$M = M_s(1 - a/H - b/H^2) + \chi H$ — how $M(H)$ reaches saturation; b carries the anisotropy

Critical scaling — two exponents

4 parameters

$$M_0 \cdot (x < T_c) \cdot (1 - (x/T_c)^{\alpha})^{\beta}$$

parameters M_0, T_c, α, β

$M = M_0[1 - (T/T_c)^{\alpha}]^{\beta}$, the generalised order parameter; α and β separate the shape near T_c from the shape far below it

Kubo–Toyabe (zero-field μ SR)

3 parameters

$$A \cdot (1/3 + (2/3) \cdot (1 - (s \cdot x)^2) \cdot \exp(-(s \cdot x)^2/2)) + b_g$$

parameters A, s, b_g

static Gaussian field distribution: the $1/3$ tail and the dip at $\sqrt{2}/\sigma$ are the fingerprint of randomly oriented nuclear moments

Stretched-exponential muon relaxation

4 parameters

$$A \cdot \exp(-(\text{lam} \cdot x)^{\text{be}}) + \text{bg}$$

parameters A, lam, be, bg

$A \cdot \exp[-(\lambda t)^\beta]$: $\beta = 1$ is dilute moments, $\frac{1}{2}$ is a spin glass, 2 is a dense static distribution

Bleaney–Bowers dimer susceptibility

4 parameters

$$A / (x \cdot (3 + \exp(-2 \cdot J/x))) + \text{Cimp}/x + \text{chi0}$$

parameters A, J, Cimp, chi0

two $S = \frac{1}{2}$ spins coupled by $H = -2J \mathbf{S}_1 \cdot \mathbf{S}_2$, J in kelvin ($J < 0$ antiferromagnetic, singlet ground state), plus a Curie tail from unpaired impurities and χ_0

Bonner–Fisher $S = \frac{1}{2}$ Heisenberg chain

3 parameters

$$A/x \cdot (0.25 + 0.074975 \cdot (J/x) + 0.075235 \cdot (J/x)^2) / (1 + 0.9931 \cdot (J/x) + 0.172135 \cdot (J/x)^2 + 0.757825 \cdot (J/x)^3) + \text{chi0}$$

parameters A, J, chi0

the Estes–Hatfield rational fit to the Bonner–Fisher result for a uniform antiferromagnetic chain ($H = -2J \sum \mathbf{S}_i \cdot \mathbf{S}_{i+1}$, $J = |J|$ in kelvin). The broad maximum sits at $T \approx 1.28 |J|$

Critical isotherm $M \propto H^{(1/\delta)}$

2 parameters

$$D0 \cdot \text{abs}(x)^{(1/dl)} \cdot \text{sign}(x)$$

parameters D0, dl

magnetisation at $T = T_c$. $\delta = 3$ mean field, 4.8 3D Heisenberg, 15 2D Ising — the third exponent that, with β and γ , tests the Widom relation $\delta = 1 + \gamma/\beta$

Heat capacity with ferromagnetic magnons

3 parameters

$$\text{gam} \cdot x + \text{beta} \cdot x^3 + \text{dm} \cdot x^{1.5}$$

parameters gam, beta, dm

$C = \gamma T + \beta T^3 + \delta T^{(3/2)}$: electrons, phonons, and the $T^{(3/2)}$ of quadratic-dispersion spin waves. Plot C/T against T^2 and the magnon term is the curvature at low T

Kittel FMR — field in plane

2 parameters

$$gk \cdot \sqrt{\text{abs}(x) \cdot (\text{abs}(x) + M_{\text{eff}})}$$

parameters gk, Meff

$f = (\gamma/2\pi) \cdot \sqrt{H(H + M_{\text{eff}})}$, H and M_{eff} in tesla, f in GHz, $\gamma/2\pi \approx 28 \text{ GHz/T}$ for $g = 2$. A negative M_{eff} means perpendicular anisotropy outweighs the shape term

Kittel FMR — field out of plane

2 parameters

$$gk \cdot (x - M_{\text{eff}})$$

parameters gk, M_{eff}

$f = (\gamma/2\pi)(H - M_{\text{eff}})$ with the film saturated along its normal: a straight line whose intercept on the field axis is M_{eff}

Gilbert damping from FMR linewidth

3 parameters

$$dH_0 + 2\alpha l \cdot x / gk$$

parameters dH_0, α, gk

$\Delta H_{\text{FWHM}} = \Delta H_0 + 2\alpha f / (\gamma/2\pi)$: the slope of linewidth (T) against frequency (GHz) gives α , the intercept the inhomogeneous broadening. $\gamma/2\pi$ is pinned at 28 GHz/T

Anisotropic magnetoresistance against angle

3 parameters

$$r_p + dr \cdot \cos(\pi \cdot (x - \theta_0) / 180)^2$$

parameters r_p, dr, θ_0

$\rho(\theta) = \rho_{\perp} + (\rho_{\parallel} - \rho_{\perp}) \cos^2(\theta - \theta_0)$, θ in degrees between magnetisation and current. The AMR ratio is $\Delta\rho/\rho_{\perp}$

Metamagnetic / spin-flop jump in M(H)

5 parameters

$$\chi \cdot x + dM / (1 + \exp(-(x - H_c)/w)) + m_0$$

parameters χ, dM, H_c, w, m_0

a linear antiferromagnetic response with a step of height ΔM at the transition field — spin flop, a metamagnetic transition, or a field-induced moment

Mean-field (Weiss) magnetisation M(T)

3 parameters

$$M_0 \cdot \text{mfmag}(x/T_c, J)$$

parameters M_0, T_c, J

the self-consistent Brillouin solution $m = B_J(3J/(J+1) \cdot m \cdot T_c/T)$, solved at every temperature. Mean field gives $\beta = \frac{1}{2}$ near T_c ; a real ferromagnet sits below this curve near T_c and above the spin-wave $T^{3/2}$ far below it

2D Ising spontaneous magnetisation (Onsager)

2 parameters

$$M_0 \cdot \max(1 - \sinh(0.88137359 \cdot T_c/x)^{-4}, 0)^{1/8}$$

parameters M_0, T_c

the exact result $M = M_0 [1 - \sinh^{-4}(2J/k_B T)]^{1/8}$ with $2J/k_B T = 0.8814 \cdot T_c/T$: $\beta = 1/8$ exactly. A layered magnet that follows this is genuinely two-dimensional and Ising-like

NMR spin–lattice relaxation $1/T_1$ (BPP)

4 parameters

$$Cb * (\tau_0 * \exp(E_a/x) / (1 + (w_0 * \tau_0 * \exp(E_a/x))^2) + 4 * \tau_0 * \exp(E_a/x) / (1 + 4 * (w_0 * \tau_0 * \exp(E_a/x))^2))$$
parameters Cb , τ_0 , E_a , w_0

Bloembergen–Purcell–Pound for thermally activated motion, $\tau = \tau_0 \exp(E_a/T)$ with E_a in kelvin. ω_0 (rad/s) is pinned at $2\pi \times 100$ MHz — set it to your spectrometer. The maximum sits where $\omega_0 \tau \approx 0.62$

Korringa + activated relaxation

3 parameters

$$Kr * x + Aa * \exp(-E_a/x)$$
parameters Kr , Aa , E_a

$1/T_1 = K \cdot T + A \cdot \exp(-E_a/T)$: the Korringa law of conduction electrons plus a gapped channel (a spin gap or thermally activated fluctuations), E_a in kelvin

Superconductivity

27 MODELS

Gaps, critical fields, penetration depths, fluctuations and vortex dynamics.

Two-gap α model (superfluid density)

4 parameters

$$w \cdot (1 - (x/T_c)^{p_1}) + (1-w) \cdot (1 - (x/T_c)^{p_2})$$

parameters w, T_c, p_1, p_2

weighted sum of two BCS gaps

Resistive transition (broadened step)

4 parameters

$$R_n / (1 + \exp(-(x - T_c)/dT)) + R_0$$

parameters R_n, T_c, dT, R_0

$R(T)$ through T_c with a finite width

BCS gap $\Delta(T)$

2 parameters

$$D_0 \cdot \tanh(1.74 \cdot \sqrt{\text{where}(x < T_c, T_c/x - 1, 0)})$$

parameters D_0, T_c

interpolation formula; zero above T_c

$H_{c2}(T)$ — Ginzburg–Landau

2 parameters

$$H_{c2} \cdot (1 - (x/T_c)^2)$$

parameters H_{c2}, T_c

$$H_{c2} = H_{c2}(0) \cdot (1 - (T/T_c)^2)$$

Two-fluid penetration depth

2 parameters

$$L_0 / \sqrt{1 - (x/T_c)^4}$$

parameters L_0, T_c

$$\lambda(T) = \lambda_0 / \sqrt{1 - (T/T_c)^4}$$

$H_{c2}(T)$ — free exponent

3 parameters

$$H_{c2} \cdot (1 - (x/T_c)^n)$$

parameters H_{c2}, T_c, n

$H_{c2} = H_{c2}(0)[1 - (T/T_c)^n]$; $n = 2$ is Ginzburg–Landau, other values flag multiband or unconventional pairing

BCS gap — α model (free α)

3 parameters

```
D0*tanh(a1*sqrt(where(x < Tc, Tc/x - 1, 0)))
```

parameters D0, a1, Tc

 $\Delta(T) = \Delta_0 \cdot \tanh[\alpha\sqrt{(T_c/T - 1)}]$; $\alpha = 1.74$ is weak coupling, larger means strong coupling
Penetration depth — power law

4 parameters

```
L0*(1 + a*(x/Tc)^n)
```

parameters L0, a, Tc, n

 $\lambda(T) = \lambda_0[1 + a(T/T_c)^n]$. $n = 2$ is dirty d-wave, $n = 1$ clean d-wave, $n \gtrsim 4$ fully gapped
Superfluid density — linear + quadratic

4 parameters

```
rs0*(1 - a*(x/Tc) - b*(x/Tc)^2)
```

parameters rs0, a, Tc, b

 $\rho_s = 1 - a(T/T_c) - b(T/T_c)^2$; the linear term is the signature of line nodes
Two-step resistive transition

7 parameters

```
Rn*(w/(1 + exp(-(x - Tc1)/d1)) + (1-w)/(1 + exp(-(x - Tc2)/d2))) + R0
```

parameters Rn, w, Tc1, d1, Tc2, d2, R0

grain and inter-grain transitions, or two phases: two broadened steps with independent Tc and width

Aslamazov–Larkin fluctuation conductivity

3 parameters

```
s0 + A*Tc/(x - Tc)
```

parameters s0, A, Tc

excess conductivity above Tc from superconducting fluctuations; diverges as $1/(T - T_c)$ **Berezinskii–Kosterlitz–Thouless resistivity**

3 parameters

```
r0*exp(-bb/sqrt(max(x - Tb, 1e-9)))
```

parameters r0, bb, Tb

 $\rho = \rho_0 \cdot \exp[-b/\sqrt{(T - T_{\text{BKT}})]$ above the vortex-unbinding transition — the signature of a 2D superconductor
Thermally activated flux flow (TAFF)

2 parameters

```
r0*exp(-U0/(8.617333e-5*x))
```

parameters r0, U0

 $\rho = \rho_0 \cdot \exp(-U_0/kT)$: vortex creep over a pinning barrier U_0 , the standard Arrhenius analysis of the resistive tail

Hc2(T) — WHH-like

2 parameters

$$Hc2 * (1 - (x/Tc)^2) / (1 + (x/Tc)^2)$$

parameters Hc2, Tc

$Hc2 \propto (1-t^2)/(1+t^2)$ with $t = T/Tc$: flatter near Tc than Ginzburg–Landau and closer to the Werthamer–Helfand–Hohenberg result

Penetration depth — s-wave, low temperature

2 parameters

$$L0 * (1 + \sqrt{\pi * D0 / (2 * x)}) * \exp(-D0/x)$$

parameters L0, D0

$\Delta\lambda/\lambda_0 = \sqrt{(\pi\Delta_0/2k_{BT})} \cdot \exp(-\Delta_0/k_{BT})$, valid below about $T_c/3$. $D0 = \Delta_0/k_B$ in kelvin; 1.76 T_c is weak-coupling BCS

Penetration depth — dirty d-wave (Hirschfeld–Goldenfeld)

3 parameters

$$L0 + a * x^2 / (x + Ts)$$

parameters L0, a, Ts

$\Delta\lambda = aT^2/(T + T^*)$: linear from line nodes above T^* , quadratic below it where impurity scattering fills the nodes

Ambegaokar–Baratoff $I_c(T)$

3 parameters

$$A * (r * 1.764 * 8.617333e-5 * Tc * \tanh(1.74 * \sqrt{\text{where}(x < Tc, Tc/x - 1, 0)})) * \tanh((r * 1.764 * 8.617333e-5 * Tc * \tanh(1.74 * \sqrt{\text{where}(x < Tc, Tc/x - 1, 0)})) / (2 * 8.617333e-5 * x))$$

parameters A, r, Tc

$I_{cR_n} = (\pi\Delta/2e) \cdot \tanh(\Delta/2k_{BT})$ for a tunnel junction, with $\Delta(T)$ from the BCS interpolation. r is Δ_0 in units of the weak-coupling 1.764 k_{BT_c} , pinned at 1

 $H_{c2}(\theta)$ — anisotropic Ginzburg–Landau (3D)

3 parameters

$$1 / \sqrt{(\cos(\pi * (x - th0) / 180) / H_{par})^2 + (\sin(\pi * (x - th0) / 180) / H_{perp})^2}$$

parameters th0, Hpar, Hperp

$(H \cos\theta / H_{\parallel})^2 + (H \sin\theta / H_{\perp})^2 = 1$, θ in degrees from the plane. A smooth bell; a cusp at θ_0 instead means a 2D film — use the Tinkham form

 $H_{c2}(\theta)$ — Tinkham thin film (2D)

3 parameters

$$2 / (\text{abs}(\sin(\pi * (x - th0) / 180)) / H_{perp} + \sqrt{(\sin(\pi * (x - th0) / 180) / H_{perp})^2 + 4 * (\cos(\pi * (x - th0) / 180) / H_{par})^2})$$

parameters th0, Hperp, Hpar

$|H \sin\theta / H_{\perp}| + (H \cos\theta / H_{\parallel})^2 = 1$, solved for H . The cusp at the parallel orientation is the signature of a superconductor thinner than its coherence length

BCS gap $\Delta(T)$ — exact weak-coupling solution

2 parameters

```
D0*bcsgap(x/Tc)
```

parameters D0, Tc

the gap equation solved numerically at every temperature, not the tanh interpolation: $\Delta(T)/\Delta_0$ is 0.957 at $T_c/2$ and 0.526 at $0.9 T_c$, and $1.7367\sqrt{1 - T/T_c}$ close to T_c . D0 in the units of your y axis

Superfluid density — s-wave BCS (α model, full range)

3 parameters

```
rs0*rhos(x/Tc, a)
```

parameters rs0, Tc, a

$\lambda^{-2}(T)/\lambda^{-2}(0)$ for a single isotropic gap $\Delta_0 = a \cdot k_B T_c$, computed from the clean-limit integral at every temperature. $a = 1.764$ is weak-coupling BCS; larger a is strong coupling, smaller a a minority gap

Superfluid density — two s-wave gaps (α model)

5 parameters

```
rs0*(w*rhos(x/Tc, a1) + (1 - w)*rhos(x/Tc, a2))
```

parameters rs0, w, Tc, a1, a2

$\rho_s = w \cdot \rho_s(a_1) + (1-w) \cdot \rho_s(a_2)$, both gaps closing at the same T_c — the standard analysis of MgB_2 and the iron pnictides. The small gap shows as a shoulder at low T

Superfluid density — clean d-wave

3 parameters

```
rs0*rhod(x/Tc, a)
```

parameters rs0, Tc, a

$\Delta(\phi) = a \cdot k_B T_c \cdot \delta(T) \cdot |\cos 2\phi|$ averaged over the Fermi surface: linear in T at low temperature from the line nodes, with slope set by a (2.14 for weak-coupling d-wave)

Penetration depth — s-wave BCS over the full range

3 parameters

```
L0/sqrt(max(rhos(x/Tc, a), 1e-9))
```

parameters L0, Tc, a

$\lambda(T) = \lambda_0/\sqrt{\rho_s(T)}$ from the full BCS integral, valid all the way to T_c rather than only below $T_c/3$

Electronic heat capacity C/T — BCS α model

4 parameters

```
gn*cesbcs(x/Tc, a)/(x/Tc) + beta*x^2
```

parameters gn, Tc, a, beta

$C/T = \gamma_n \cdot C_{es}(T/T_c)/(\gamma_n T) + \beta T^2$, the Padamsee α model: the entropy of Bogoliubov quasiparticles differentiated numerically. $a = 1.764$ gives the BCS jump $\Delta C/\gamma T_c = 1.43$; a larger jump means stronger coupling

Upper critical field — WHH (dirty limit)

2 parameters

$$H_0 * whh(x/T_c)$$

parameters H_0 , T_c

the Werthamer–Helfand–Hohenberg orbital limit, $\ln(1/t) = \psi(\frac{1}{2} + h/2t) - \psi(\frac{1}{2})$, solved at every temperature. $H_0 = 0.693 \cdot T_c \cdot |dH_c2/dT|$ at T_c ; a measured H_c2 above this curve points to multiband or Pauli physics

McMillan T_c against coupling λ

2 parameters

$$TD/1.45 * \exp(-1.04 * (1 + x) / (x - \mu * (1 + 0.62 * x)))$$

parameters TD , μ

$T_c = (\theta_D/1.45) \cdot \exp[-1.04(1+\lambda)/(\lambda - \mu^*(1 + 0.62\lambda))]$ with x the electron–phonon coupling λ . μ^* is pinned at 0.1; remove the ! to fit it

Growth & kinetics

12 MODELS

Transformation, growth and diffusion against time or position.

Avrami (JMAK)

3 parameters

$$X_{\max} * (1 - \exp(-(k * x)^n))$$

parameters $X_{\max}$, k , n

transformed fraction vs time; n encodes the growth dimensionality

Gompertz growth

4 parameters

$$A * \exp(-\exp(-k * (x - x_0))) + c$$

parameters A , k , x_0 , c

asymmetric sigmoid growth curve

Weibull cumulative

3 parameters

$$A * (1 - \exp(-(x/x_0)^m))$$

parameters A , x_0 , m

failure or transformation statistics; m is the Weibull modulus

Freundlich isotherm

2 parameters

$$K_f * x^{(1/n_f)}$$

parameters K_f , n_f

heterogeneous-surface adsorption

BET isotherm

2 parameters

$$q_m * C_b * x / ((1 - x) * (1 + (C_b - 1) * x))$$

parameters q_m , C_b

multilayer adsorption; x is relative pressure P/P_0

Second-order kinetics

3 parameters

$$C_0 / (1 + k * C_0 * x) + c$$

parameters C_0 , k , c

concentration decay $1/(1 + kt)$

Avrami with incubation time

4 parameters

$$X_{\max} * (x > t_0) * (1 - \exp(-(k * (x - t_0))^n))$$

parameters $X_{\max}$, t_0 , k , n

nothing happens until t_0 , then JMAK kinetics start — nucleation almost always has a delay

Two-component exponential rise

5 parameters

$$A*(1 - w*\exp(-x/t1) - (1-w)*\exp(-x/t2)) + c$$

parameters A, w, t1, t2, c

a fast and a slow approach to the same plateau

Stretched-exponential growth

4 parameters

$$A*(1 - \exp(-(x/\tau)^{be})) + c$$

parameters A, tau, be, c

 $A[1 - \exp(-(t/\tau)^\beta)]$ — a distribution of rates rather than a single one**Gaussian diffusion profile**

3 parameters

$$A*\exp(-x^2/(4*Dt)) + c$$

parameters A, Dt, c

a thin-film source diffusing into a semi-infinite solid: $c(x) \propto \exp(-x^2/4Dt)$. The fitted Dt is the diffusion length squared**Complementary-error-function diffusion profile**

3 parameters

$$c0*\operatorname{erfc}(x/(2*\sqrt{Dt})) + c$$

parameters c0, Dt, c

a constant-source diffusion: $c(x) = c_0 \cdot \operatorname{erfc}(x/2\sqrt{Dt})$ — the shape of a doped or implanted depth profile**Weibull distribution (density)**

3 parameters

$$A*(m/x0)*(x/x0)^{(m-1)}*\exp(-(x/x0)^m)$$

parameters A, m, x0

the failure-rate density rather than its cumulative; m is the Weibull modulus

Optical & semiconductor

40 MODELS

Absorption edges, dielectric functions, dispersion and band gaps. The largest group, because this is where the shapes are most varied.

Elliott (exciton + continuum)

5 parameters

$$\left(\frac{A}{x}\right) * \left(4 * \pi * EB^{(3/2)} * \left(\frac{1}{\pi}\right) * \left(\frac{\Gamma}{2}\right) / \left((x - E_g + EB)^2 + \left(\frac{\Gamma}{2}\right)^2\right)\right) + (x \geq E_g) * \left(2 * \pi * \sqrt{EB * (x - E_g)} / \left(1 - \exp(-2 * \pi * \sqrt{EB * (x - E_g)})\right)\right) + B$$

parameters A, EB, Gamma, E_g, B

absorption across an excitonic resonance and the continuum above the gap

Tauc — direct gap

3 parameters

$$A * (x \geq E_g) * \sqrt{x - E_g} / x + B$$

parameters A, E_g, B

$(\alpha h\nu)^2$ linear above the gap; fit α vs photon energy

Tauc — indirect gap

3 parameters

$$A * (x \geq E_g) * (x - E_g)^2 / x + B$$

parameters A, E_g, B

$(\alpha h\nu)^{1/2}$ linear above the gap

Urbach tail

4 parameters

$$A * \exp((x - E_0)/E_u) + B$$

parameters A, E₀, E_u, B

exponential sub-gap absorption; E_u is the Urbach energy

Varshni — gap vs temperature

3 parameters

$$E_{g0} - a * x^2 / (x + b)$$

parameters E_{g0}, a, b

$$E_g(T) = E_{g0} - aT^2 / (T + b)$$

Bose–Einstein gap shrinkage

3 parameters

$$E_{g0} - aB * (1 + 2 / (\exp(T_p/x) - 1))$$

parameters E_{g0}, aB, T_p

phonon-population form of E_g(T)

Sellmeier (2-term)

4 parameters

$$\sqrt{1 + B1 \cdot x^2 / (x^2 - C1) + B2 \cdot x^2 / (x^2 - C2)}$$

parameters B1, C1, B2, C2

refractive index vs wavelength

Cauchy dispersion

3 parameters

$$A + B/x^2 + C/x^4$$

parameters A, B, C

$$n(\lambda) = A + B/\lambda^2 + C/\lambda^4$$

Lorentz oscillator (ε_2)

4 parameters

$$A \cdot G \cdot x / ((x_0^2 - x^2)^2 + (G \cdot x)^2) + B$$

parameters A, G, x0, B

damped-oscillator dielectric loss

Transmission through an absorbing film (Urbach → Tauc)

5 parameters

$$T0 \cdot \exp(-K \cdot \sqrt{\text{Eu} \cdot \text{softplus}((1239.84/x - E_g)/\text{Eu})}) + c$$

parameters T0, K, Eu, Eg, c

x in nm. $T = T0 \cdot \exp(-K \cdot \alpha)$, with α exponential below the gap and $\sqrt{E-E_g}$ above, joined smoothly**Absorption coefficient (Urbach → Tauc)**

4 parameters

$$a0 \cdot \sqrt{\text{Eu} \cdot \text{softplus}((x - E_g)/\text{Eu})} + c$$

parameters a0, Eu, Eg, c

x in eV. $\alpha(E)$, exponential tail joining $\sqrt{E-E_g}$ continuously — fit α directly**Absorption edge — Urbach tail + saturation**

5 parameters

$$A \cdot (1 - \exp(-\exp((x-x_0)/\text{Eu}))) + s \cdot \text{Eu} \cdot \text{softplus}((x-x_0)/\text{Eu}) + c$$

parameters A, x0, Eu, s, c

exponential sub-gap tail rolling smoothly into a saturating edge, plus a linear rise; continuous everywhere

Absorption edge — two-slope (sharper knee)

7 parameters

$$A \cdot (1 - \exp(-\exp((x-x_0)/\text{Eu})))^p + s \cdot w \cdot \text{softplus}((x-x_0)/w) + c$$

parameters A, x0, Eu, p, s, w, c

as above but the tail and the knee get separate widths, for a very abrupt edge

Asymmetric step (generalised logistic)

6 parameters

$$A / (1 + \exp(-(x-x_0)/w))^{\text{nu}} + s \cdot x + c$$

parameters A, x0, w, nu, s, c

a step that turns on faster than it levels off; nu sets the asymmetry

Tauc–Lorentz ϵ_2

4 parameters

$$(x > E_g) * A * E_0 * C * (x - E_g)^2 / (((x^2 - E_0^2)^2 + C^2 * x^2) * x)$$

parameters E_g, A, E_0, C

the standard amorphous-semiconductor dielectric function: a Lorentz oscillator with a Tauc band edge, zero below E_g

Cody–Lorentz ϵ_2

5 parameters

$$(x > E_g) * ((x - E_g)^2 / ((x - E_g)^2 + E_p^2)) * (A * E_0 * C * x / ((x^2 - E_0^2)^2 + C^2 * x^2))$$

parameters E_g, E_p, A, E_0, C

Tauc–Lorentz with the correct $(E - E_g)^2 / [(E - E_g)^2 + E_p^2]$ onset — better behaved right at the edge

Forouhi–Bloomer $k(E)$

4 parameters

$$(x > E_g) * A * (x - E_g)^2 / (x^2 - B_f * x + C_f)$$

parameters E_g, A, B_f, C_f

extinction coefficient of an amorphous film; A, B, C are the standard FB parameters

Elliott — 1s + 2s excitons + continuum

5 parameters

$$(A/x) * (4 * \pi * E_B^{(3/2)} * (1/\pi) * (\Gamma/2) / ((x - E_g + E_B)^2 + (\Gamma/2)^2) + (4 * \pi * E_B^{(3/2)}/8) * (1/\pi) * (\Gamma/2) / ((x - E_g + E_B/4)^2 + (\Gamma/2)^2) + (x > E_g) * (2 * \pi * \sqrt{E_B * (x - E_g)} / (1 - \exp(-2 * \pi * \sqrt{E_B * (x - E_g)}))) + B$$

parameters A, E_B, Γ, E_g, B

the full Elliott lineshape with the $n = 2$ exciton included (weight 1/8, binding $E_B/4$) — needed whenever the 2s shoulder is resolved

Varshni — Pässler p-form

4 parameters

$$E_g(T) = E_{g0} - (\alpha \Theta / 2) * ((1 + (2 * T / \Theta)^p)^{(1/p)} - 1)$$

parameters $E_{g0}, \alpha, \Theta, p$

$E_g(T) = E_{g0} - (\alpha \Theta / 2) [(1 + (2T/\Theta)^p)^{1/p} - 1]$; fixes Varshni's unphysical low-T behaviour and gives a real Debye temperature

Manoogian–Woolley gap shift

5 parameters

$$E_g(T) = E_{g0} - U * T^s - V * \Theta * (\coth(\Theta / (2 * T)) - 1)$$

parameters E_{g0}, U, s, V, Θ

$E_g(T) = E_{g0} - U \cdot T^s - V \cdot \Theta [\coth(\Theta / (2T)) - 1]$: separates lattice dilation from electron–phonon coupling

Urbach energy vs temperature

2 parameters

$$E_u(T) = E_{u0} + E_p * \text{bose}(E_p, 8.617333e-5 * T)$$

parameters E_{u0}, E_p

$E_u(T) = E_{u0} + E_p / [\exp(E_p / kT) - 1]$: the frozen-in (static disorder) part plus the phonon part

Band-tail emission lineshape

4 parameters

$$A \cdot (x > E_g) \cdot \sqrt{x - E_g} \cdot \exp(-(x - E_g)/kT) + c$$

parameters A, E_g, kT, c

$I(E) \propto \sqrt{(E-E_g)} \cdot \exp[-(E-E_g)/kT]$: free-carrier recombination — the asymmetric PL band with a thermal high-energy tail

PL with LO-phonon replicas

6 parameters

$$A \cdot (\exp(-(x-E_0)^2/(2s^2)) + Sh \cdot \exp(-(x-E_0+hw)^2/(2s^2)) + (Sh^2/2) \cdot \exp(-(x-E_0+2hw)^2/(2s^2))) + c$$

parameters A, E₀, s, Sh, hw, c

zero-phonon line plus two Huang–Rhys replicas at fixed phonon spacing — the ladder seen in polar and defect emission

Two Lorentz oscillators (ϵ_2)

7 parameters

$$A1 \cdot G1 \cdot x / ((x1^2 - x^2)^2 + (G1 \cdot x)^2) + A2 \cdot G2 \cdot x / ((x2^2 - x^2)^2 + (G2 \cdot x)^2) + B$$

parameters A₁, G₁, x₁, A₂, G₂, x₂, B

two damped oscillators — the minimum honest description of most measured dielectric loss

Sellmeier (3-term)

6 parameters

$$\sqrt{1 + B1 \cdot x^2 / (x^2 - C1) + B2 \cdot x^2 / (x^2 - C2) + B3 \cdot x^2 / (x^2 - C3)}$$

parameters B₁, C₁, B₂, C₂, B₃, C₃

refractive index with a third resonance — needed across a wide wavelength range

Thin-film interference fringes

4 parameters

$$A + Bf \cdot \cos(4\pi \cdot nd/x + ph)$$

parameters A, Bf, nd, ph

$T(\lambda) = A + B \cdot \cos(4\pi \cdot nd/\lambda + \phi)$: the fringe spacing gives the optical thickness nd directly

Absorption edge + interference fringes

8 parameters

$$(T0 + Bf \cdot \cos(4\pi \cdot nd/x + ph)) \cdot \exp(-K \cdot \sqrt{Eu \cdot \text{softplus}((1239.84/x - E_g)/Eu)}) + c$$

parameters T₀, Bf, nd, ph, K, Eu, E_g, c

a transmission spectrum as measured: the Urbach→Tauc edge with the substrate fringes still on it

Gaussian oscillator (ϵ_2)

3 parameters

$$A \cdot (\exp(-((x - E_0)/sg)^2) - \exp(-((x + E_0)/sg)^2))$$

parameters A, E₀, sg

a Kramers–Kronig-consistent Gaussian absorption band, antisymmetrised in energy as the causality condition requires

Drude free-carrier absorption (ε_2)

2 parameters

$$\omega_p^2 G_d / (x(x^2 + G_d^2))$$

parameters ω_p , G_d
 $\varepsilon_2 = \omega_p^2 \Gamma / [\omega(\omega^2 + \Gamma^2)]$ — the free-carrier tail that rises towards low photon energy in a doped film
Lorentz oscillator (ε_1 , real part)

4 parameters

$$e_{\text{Inf}} + A(x_0^2 - x^2) / ((x_0^2 - x^2)^2 + (Gx)^2)$$

parameters e_{Inf} , A , x_0 , G the dispersive partner of the ε_2 oscillator, with the high-frequency background ε_∞ **Havriliak–Negami (ε' , real part)**

5 parameters

$$e_{\text{Inf}} + d e * (1 + 2 * (x * \tau)^{\alpha} * \sin(\pi * \alpha / 2) + (x * \tau)^{(2 * \alpha)} * (-b e / 2) * \cos(b e * \text{atan2}((x * \tau)^{\alpha} * \cos(\pi * \alpha / 2), 1 + (x * \tau)^{\alpha} * \sin(\pi * \alpha / 2))))$$

parameters e_{Inf} , $d e$, τ , α , $b e$

the dispersion step that accompanies the Havriliak–Negami loss peak

Forouhi–Bloomer $n(E)$

5 parameters

$$n_{\text{Inf}} + (B_0 x + C_0) / (x^2 - B_f x + C_f)$$

parameters n_{Inf} , B_0 , C_0 , B_f , C_f

the refractive index that goes with the Forouhi–Bloomer extinction coefficient, from the same three parameters

Beer–Lambert with Rayleigh scattering

4 parameters

$$A \exp(-\alpha l x) + b_R / x^4 + c$$

parameters A , αl , b_R , c an absorbing species on a λ^{-4} scattering background — a turbid solution or a rough film**O'Donnell–Chen band gap vs temperature**

3 parameters

$$E_{g0} - S_0 d * \hbar \omega * (\coth(\hbar \omega / (2 * 8.617333 e - 5 * x)) - 1)$$

parameters E_{g0} , $S_0 d$, $\hbar \omega$
 $E_g(T) = E_g(0) - S \langle \hbar \omega \rangle [\coth(\langle \hbar \omega \rangle / 2 k_B T) - 1]$: a Huang–Rhys-like coupling S to an average phonon $\langle \hbar \omega \rangle$ in eV. Better behaved at low T than Varshni
Intrinsic carrier density

2 parameters

$$A x^{1.5} \exp(-E_g / (2 * 8.617333 e - 5 * x))$$

parameters A , E_g
 $n_i = A \cdot T^{(3/2)} \cdot \exp(-E_g / 2 k_B T)$: the gap from $\ln(n_i / T^{(3/2)})$ against $1/T$, in eV

Burstein–Moss shift

2 parameters

$$E_{g0} + B_{bm} x^{(2/3)}$$

parameters E_{g0} , B_{bm}

$E = E_{g0} + (\hbar^2/2m^*)(3\pi^2n)^{(2/3)}$: the optical gap of a degenerate semiconductor blue-shifts as $n^{(2/3)}$ because the lowest conduction states are already full

Quantum-dot confinement (Brus)

3 parameters

$$E_g + A_b/x^2 - B_b/x$$

parameters E_g , A_b , B_b

$E(R) = E_g + \hbar^2\pi^2/2\mu R^2 - 1.786e^2/4\pi\epsilon R$: kinetic confinement against the electron–hole Coulomb attraction, R the dot radius

Exciton Rydberg series

2 parameters

$$E_g - R_y/x^2$$

parameters E_g , R_y

$E_n = E_g - R^*/n^2$ with x the principal quantum number: the binding energy R^* and the gap from the ladder of absorption lines

Drude reflectance

3 parameters

$$\text{drudeR}(x, \text{einf}, \text{wp}, \text{gd})$$

parameters einf , wp , gd

normal-incidence $R(\omega)$ of a free-carrier metal or doped semiconductor from $\epsilon = \epsilon_\infty - \omega_p^2/(\omega^2 + i\gamma\omega)$. The plasma edge sits at $\omega_p/\sqrt{\epsilon_\infty}$; ω , ω_p and γ share one unit (cm^{-1} or eV)

Drude–Lorentz reflectance (carriers + one phonon)

6 parameters

$$\text{dlR}(x, \text{einf}, \text{wp}, \text{gd}, \text{So}, \text{w0}, \text{g0})$$

parameters einf , wp , gd , So , w0 , g0

free carriers plus one polar phonon or interband oscillator of strength S at ω_0 — the reststrahlen band and the plasma edge in one reflectance spectrum

Tunnelling & STM

10 MODELS

Differential conductance: gaps, Kondo resonances and Fano interference.

Fano lineshape + linear background

6 parameters

$$A * (q + (x - x_0)/G)^2 / (1 + ((x - x_0)/G)^2) + m * x + c$$

parameters A, q, x_0, G, m, c

Kondo / cotunnelling conductance; q is the asymmetry — the CeCoIn5 dI/dV shape

Frota Kondo resonance

5 parameters

$$A * \text{frota}(x - x_0, G) + m * x + c$$

parameters A, x_0, G, m, c

the correct Kondo lineshape (Frota), sharper wings than a Lorentzian

Dynes superconducting DOS

4 parameters

$$A * \text{dynes}(x, D, G) + c$$

parameters A, D, G, c

BCS gap with lifetime broadening; D is the gap, G the broadening

Two Fano resonances + background

10 parameters

$$A1 * (q1 + (x - x1)/G1)^2 / (1 + ((x - x1)/G1)^2) + A2 * (q2 + (x - x2)/G2)^2 / (1 + ((x - x2)/G2)^2) + m * x + c$$

parameters $A1, q1, x1, G1, A2, q2, x2, G2, m, c$

a hybridisation gap with peaks either side, as in heavy-fermion dI/dV

Dynes DOS + linear background

5 parameters

$$A * \text{dynes}(x, D, G) + m * x + c$$

parameters A, D, G, m, c

the BCS gap with lifetime broadening on a sloping background — what a real dI/dV gap looks like

Fano + Lorentzian (Kondo on a resonance)

9 parameters

$$A * (q + (x - x_0)/G)^2 / (1 + ((x - x_0)/G)^2) + A2 * g^2 / ((x - x2)^2 + g^2) + m * x + c$$

parameters $A, q, x_0, G, A2, g^2, x2, m, c$

an asymmetric Kondo feature riding on a broader resonance, as in adatom or heavy-fermion dI/dV

Dynes DOS with thermal smearing

7 parameters

$$A \cdot \text{dynest}(x - x_0, D, G, kT) + m \cdot x + c$$

parameters A, x_0, D, G, kT, m, c

the Dynes gap convolved with the Fermi-function derivative, so the coherence peaks broaden the way a real dI/dV at finite temperature does. kT in the units of bias ($\approx 86 \mu\text{eV}$ per kelvin)

BTK point-contact Andreev reflection

5 parameters

$$A \cdot \text{btkkt}(x - x_0, D, Z, kT)$$

parameters A, x_0, D, Z, kT

Blonder–Tinkham–Klapwijk conductance $G_{\text{NS}}/G_{\text{NN}}$ with barrier strength Z , thermally smeared. $Z \rightarrow 0$ doubles the conductance inside the gap; $Z \gg 1$ recovers the tunnelling DOS

Inelastic tunnelling (IETS) step pair

5 parameters

$$G_0 + h \cdot (\text{smearstep}(x, E_1, w) + \text{smearstep}(-x, E_1, w)) + m \cdot x$$

parameters G_0, h, E_1, w, m

a conductance step at $\pm E_1$ where an excitation (phonon, spin flip, vibration) opens a new channel, thermally broadened by w

Tunnel current against tip–sample distance

2 parameters

$$I_0 \cdot \exp(-1.025 \cdot \sqrt{\phi} \cdot x)$$

parameters I_0, ϕ

$I = I_0 \cdot \exp(-2\kappa z)$ with $2\kappa = 1.025\sqrt{\phi} \text{ \AA}^{-1}$ for ϕ in eV and z in $\text{\AA}$: the apparent barrier height of an STM junction

Line shapes

34 MODELS

General-purpose peak and multi-peak forms, symmetric and asymmetric, plus the resonance derivatives that spectrometers actually record.

Pseudo-Voigt (single peak)

5 parameters

$$A * (\eta * w^2 / ((x - x_0)^2 + w^2) + (1 - \eta) * \exp(-(x - x_0)^2 / (2 * w^2))) + B$$

parameters A , η , w , x_0 , B

η mixes Lorentzian and Gaussian; the workhorse peak shape

Fano resonance

5 parameters

$$A * (q * G / 2 + x - x_0)^2 / ((x - x_0)^2 + (G / 2)^2) + B$$

parameters A , q , G , x_0 , B

asymmetric line; q is the Fano asymmetry parameter

Two Lorentzians + background

7 parameters

$$A_1 * g_1^2 / ((x - x_1)^2 + g_1^2) + A_2 * g_2^2 / ((x - x_2)^2 + g_2^2) + bg$$

parameters A_1 , g_1 , x_1 , A_2 , g_2 , x_2 , bg

a doublet on a flat background

Three Lorentzians + background

10 parameters

$$A_1 * g_1^2 / ((x - x_1)^2 + g_1^2) + A_2 * g_2^2 / ((x - x_2)^2 + g_2^2) + A_3 * g_3^2 / ((x - x_3)^2 + g_3^2) + bg$$

parameters A_1 , g_1 , x_1 , A_2 , g_2 , x_2 , A_3 , g_3 , x_3 , bg

three overlapping peaks, as in ARPES or Raman band analysis

Two Gaussians + background

7 parameters

$$A_1 * \exp(-(x - x_1)^2 / (2 * s_1^2)) + A_2 * \exp(-(x - x_2)^2 / (2 * s_2^2)) + bg$$

parameters A_1 , x_1 , s_1 , A_2 , x_2 , s_2 , bg

two resolved Gaussian peaks

Three Gaussians + background

10 parameters

$$A_1 * \exp(-(x - x_1)^2 / (2 * s_1^2)) + A_2 * \exp(-(x - x_2)^2 / (2 * s_2^2)) + A_3 * \exp(-(x - x_3)^2 / (2 * s_3^2)) + bg$$

parameters A_1 , x_1 , s_1 , A_2 , x_2 , s_2 , A_3 , x_3 , s_3 , bg

three Gaussian components

Breit–Wigner–Fano (Raman)

5 parameters

$$I_0 * (1 + (x - x_0) / (q * G))^2 / (1 + ((x - x_0) / G)^2) + B$$

parameters I_0 , x_0 , q , G , B

asymmetric Raman band coupled to a continuum

Split Gaussian (asymmetric peak)

5 parameters

$$A * \exp(-(x - x_0)^2 / (2 * \text{where}(x < x_0, s_1, s_2)^2)) + B$$

parameters A , x_0 , s_1 , s_2 , B

independent widths either side of the centre — the simplest honest fit to a skewed band

Split Lorentzian (asymmetric peak)

5 parameters

$$A * \text{where}(x < x_0, g_1, g_2)^2 / ((x - x_0)^2 + \text{where}(x < x_0, g_1, g_2)^2) + B$$

parameters A , x_0 , g_1 , g_2 , B

a Lorentzian with different half-widths on the two sides

Split Pearson VII

6 parameters

$$A / (1 + ((x - x_0) / \text{where}(x < x_0, w_1, w_2))^2 * (2^{(1/m)} - 1))^m + bg$$

parameters A , x_0 , w_1 , w_2 , m , bg

the XRD workhorse made asymmetric — separate width and tail shape on each side

Four Lorentzians + background

13 parameters

$$A_1 * g_1^2 / ((x - x_1)^2 + g_1^2) + A_2 * g_2^2 / ((x - x_2)^2 + g_2^2) + A_3 * g_3^2 / ((x - x_3)^2 + g_3^2) + A_4 * g_4^2 / ((x - x_4)^2 + g_4^2) + bg$$

parameters A_1 , g_1 , x_1 , A_2 , g_2 , x_2 , A_3 , g_3 , x_3 , A_4 , g_4 , x_4 , bg

four overlapping bands — one step beyond the three-peak fit, still few enough to stay meaningful

Four Gaussians + background

13 parameters

$$A_1 * \exp(-(x - x_1)^2 / (2 * s_1^2)) + A_2 * \exp(-(x - x_2)^2 / (2 * s_2^2)) + A_3 * \exp(-(x - x_3)^2 / (2 * s_3^2)) + A_4 * \exp(-(x - x_4)^2 / (2 * s_4^2)) + bg$$

parameters A_1 , x_1 , s_1 , A_2 , x_2 , s_2 , A_3 , x_3 , s_3 , A_4 , x_4 , s_4 , bg

four Gaussian components on a flat background

Two pseudo-Voigt peaks + background

9 parameters

$$A_1 * (e_1 * w_1^2 / ((x - x_1)^2 + w_1^2) + (1 - e_1) * \exp(-(x - x_1)^2 / (2 * w_1^2))) + A_2 * (e_2 * w_2^2 / ((x - x_2)^2 + w_2^2) + (1 - e_2) * \exp(-(x - x_2)^2 / (2 * w_2^2))) + B$$

parameters A_1 , e_1 , w_1 , x_1 , A_2 , e_2 , w_2 , x_2 , B

a doublet where each component mixes Lorentzian and Gaussian independently

BWF + Lorentzian (carbon D and G bands)

8 parameters

$$IG*(1 + (x-xG)/(q*GG))^2/(1 + ((x-xG)/GG)^2) + ID*gD^2/((x-xD)^2+gD^2) + B$$

parameters $IG, xG, q, GG, ID, gD, xD, B$

the standard graphitic Raman fit: an asymmetric Breit–Wigner–Fano G band with a Lorentzian D band

Voigt + linear background

6 parameters

$$A*voigt(x - x0, sg, gm) + m*x + c$$

parameters $A, x0, sg, gm, m, c$ a true Voigt profile on a sloping background — instrument σ and lifetime γ kept separate**First-derivative Lorentzian (EPR)**

5 parameters

$$-2*A*((x-x0)/G)/(1 + ((x-x0)/G)^2)^2 + m*x + c$$

parameters $A, x0, G, m, c$

what a field-modulated EPR spectrometer actually records: the derivative of an absorption line

First-derivative Gaussian (EPR)

5 parameters

$$-A*((x-x0)/s^2)*exp(-(x-x0)^2/(2*s^2)) + m*x + c$$

parameters $A, x0, s, m, c$

the inhomogeneously broadened counterpart of the derivative Lorentzian

Dysonian lineshape

6 parameters

$$A*(G + \alpha*(x - x0))/((x - x0)^2 + G^2) + m*x + c$$

parameters $A, G, \alpha, x0, m, c$ EPR or NMR in a conductor: absorption and dispersion admixed by the skin effect; α sets the asymmetry**Mössbauer quadrupole doublet**

5 parameters

$$c - A*(G^2/((x - x0 + d/2)^2 + G^2) + G^2/((x - x0 - d/2)^2 + G^2))$$

parameters $c, A, G, x0, d$ two absorption dips of equal depth and width, separated by the quadrupole splitting d and centred on the isomer shift x_0 **5 Lorentzians + linear background**

17 parameters

$$A1*g1^2/((x-x1)^2+g1^2) + A2*g2^2/((x-x2)^2+g2^2) + A3*g3^2/((x-x3)^2+g3^2) + A4*g4^2/((x-x4)^2+g4^2) + A5*g5^2/((x-x5)^2+g5^2) + bg + sl*x$$

parameters $A1, g1, x1, A2, g2, x2, A3, g3, x3, A4, g4, x4, A5, g5, x5, bg, sl$

a 5-band region on a sloping baseline — the point where a spectrum stops being a doublet and starts being a spectrum

5 Gaussians + linear background

17 parameters

$$A1 \cdot \exp(-(x-x1)^2/(2*s1^2)) + A2 \cdot \exp(-(x-x2)^2/(2*s2^2)) + A3 \cdot \exp(-(x-x3)^2/(2*s3^2)) + A4 \cdot \exp(-(x-x4)^2/(2*s4^2)) + A5 \cdot \exp(-(x-x5)^2/(2*s5^2)) + bg + s1*x$$

parameters A1, x1, s1, A2, x2, s2, A3, x3, s3, A4, x4, s4, A5, x5, s5, bg, s1

the inhomogeneously broadened counterpart, for 5 overlapping bands on a slope

6 Lorentzians + linear background

20 parameters

$$A1*g1^2/((x-x1)^2+g1^2) + A2*g2^2/((x-x2)^2+g2^2) + A3*g3^2/((x-x3)^2+g3^2) + A4*g4^2/((x-x4)^2+g4^2) + A5*g5^2/((x-x5)^2+g5^2) + A6*g6^2/((x-x6)^2+g6^2) + bg + s1*x$$

parameters A1, g1, x1, A2, g2, x2, A3, g3, x3, A4, g4, x4, A5, g5, x5, A6, g6, x6, bg, s1

a 6-band region on a sloping baseline — the point where a spectrum stops being a doublet and starts being a spectrum

6 Gaussians + linear background

20 parameters

$$A1 \cdot \exp(-(x-x1)^2/(2*s1^2)) + A2 \cdot \exp(-(x-x2)^2/(2*s2^2)) + A3 \cdot \exp(-(x-x3)^2/(2*s3^2)) + A4 \cdot \exp(-(x-x4)^2/(2*s4^2)) + A5 \cdot \exp(-(x-x5)^2/(2*s5^2)) + A6 \cdot \exp(-(x-x6)^2/(2*s6^2)) + bg + s1*x$$

parameters A1, x1, s1, A2, x2, s2, A3, x3, s3, A4, x4, s4, A5, x5, s5, A6, x6, s6, bg, s1

the inhomogeneously broadened counterpart, for 6 overlapping bands on a slope

3 pseudo-Voigt peaks + linear background

14 parameters

$$A1*(e1*w1^2/((x-x1)^2+w1^2) + (1-e1)*\exp(-(x-x1)^2/(2*w1^2))) + A2*(e2*w2^2/((x-x2)^2+w2^2) + (1-e2)*\exp(-(x-x2)^2/(2*w2^2))) + A3*(e3*w3^2/((x-x3)^2+w3^2) + (1-e3)*\exp(-(x-x3)^2/(2*w3^2))) + bg + s1*x$$

parameters A1, e1, w1, x1, A2, e2, w2, x2, A3, e3, w3, x3, bg, s1

each band mixes Lorentzian and Gaussian on its own — the honest default when instrument and lifetime broadening are comparable

4 pseudo-Voigt peaks + linear background

18 parameters

$$A1*(e1*w1^2/((x-x1)^2+w1^2) + (1-e1)*\exp(-(x-x1)^2/(2*w1^2))) + A2*(e2*w2^2/((x-x2)^2+w2^2) + (1-e2)*\exp(-(x-x2)^2/(2*w2^2))) + A3*(e3*w3^2/((x-x3)^2+w3^2) + (1-e3)*\exp(-(x-x3)^2/(2*w3^2))) + A4*(e4*w4^2/((x-x4)^2+w4^2) + (1-e4)*\exp(-(x-x4)^2/(2*w4^2))) + bg + s1*x$$

parameters A1, e1, w1, x1, A2, e2, w2, x2, A3, e3, w3, x3, A4, e4, w4, x4, bg, s1

each band mixes Lorentzian and Gaussian on its own — the honest default when instrument and lifetime broadening are comparable

Three Voigt peaks + linear background

12 parameters

$$A1*voigt(x-x1, sg, g1) + A2*voigt(x-x2, sg, g2) + A3*voigt(x-x3, sg, g3) + bg + s1*x$$

parameters A1, x1, sg, g1, A2, x2, g2, A3, x3, g3, bg, s1

three bands sharing one instrumental width sg, each with its own lifetime width — the correct way to separate resolution from physics across a multiplet

Carbon Raman — D, G and D' bands

12 parameters

$$ID * gD^2 / ((x - xD)^2 + gD^2) + IG * (1 + (x - xG) / (q * GG))^2 / (1 + ((x - xG) / GG)^2) + IDp * gDp^2 / ((x - xDp)^2 + gDp^2) + bg + sl * x$$

parameters ID, gD, xD, IG, xG, q, GG, IDp, gDp, xDp, bg, sl

the full first-order carbon fit: a Lorentzian D band, a Breit–Wigner–Fano G band, and the D' shoulder above it. ID/IG gives the defect density

Carbon Raman — 2D band, four components

13 parameters

$$A1 * g1^2 / ((x - x1)^2 + g1^2) + A2 * g2^2 / ((x - x2)^2 + g2^2) + A3 * g3^2 / ((x - x3)^2 + g3^2) + A4 * g4^2 / ((x - x4)^2 + g4^2) + bg$$

parameters A1, g1, x1, A2, g2, x2, A3, g3, x3, A4, g4, x4, bg

the four-Lorentzian decomposition of the 2D band that distinguishes bilayer graphene from monolayer, where a single Lorentzian suffices

Two Breit–Wigner–Fano bands

10 parameters

$$I1 * (1 + (x - x1) / (q1 * G1))^2 / (1 + ((x - x1) / G1)^2) + I2 * (1 + (x - x2) / (q2 * G2))^2 / (1 + ((x - x2) / G2)^2) + bg + sl * x$$

parameters I1, x1, q1, G1, I2, x2, q2, G2, bg, sl

two bands each coupled to an electronic continuum — metallic nanotubes, heavily doped semiconductors, amorphous carbons

Skewed Voigt (asymmetric true Voigt)

7 parameters

$$A * \text{voigt}(x - x0, sg, gm) * (1 + \text{erf}(al * (x - x0) / (sg * \sqrt{2}))) + bg + sl * x$$

parameters A, x0, sg, gm, al, bg, sl

a Voigt profile multiplied by an error-function skew — the standard way to give a true Voigt an asymmetry without abandoning its physics

Log-normal peak

5 parameters

$$A * \exp(-(\ln(\max(x - x0, 1e-12)) - \mu)^2 / (2 * s^2)) / \max(x - x0, 1e-12) + bg$$

parameters A, x0, mu, s, bg

a peak that is Gaussian in log x: naturally asymmetric with a long high-side tail, the usual shape of a chromatographic band or a size distribution

Doniach–Šunjić convolved with a Gaussian (Voigt-asymmetric)

7 parameters

$$A * \cos(\pi * \alpha / 2 + (1 - \alpha) * \text{atan}((x - x0) / \text{gam})) / ((x - x0)^2 + \text{gam}^2)^{((1 - \alpha) / 2)} * \exp(-(x - x0)^2 / (2 * sg^2)) + bg + sl * x$$

parameters A, alpha, x0, gam, sg, bg, sl

the metallic core-level asymmetry damped by the analyser resolution — an approximation to the proper convolution, but the one that actually converges

Peak on an exponential background

6 parameters

$$A \cdot g^2 / ((x - x_1)^2 + g^2) + B \cdot \exp(-x/tb) + c_0$$

parameters A, g, x_1, B, tb, c_0

a band sitting on a decaying continuum — fluorescence under a Raman line, or a peak on a bremsstrahlung tail

Two peaks with a shared width

7 parameters

$$A_1 \cdot g^2 / ((x - x_1)^2 + g^2) + A_2 \cdot g^2 / ((x - x_2)^2 + g^2) + bg + sl \cdot x$$

parameters $A_1, g, x_1, A_2, x_2, bg, sl$

a doublet whose components must broaden together — a spin–orbit or isotope pair, where letting the widths float independently invents a difference the physics forbids

Thermal

15 MODELS

Heat capacity, relaxation and thermometry.

Low-T heat capacity

2 parameters

$$\text{gam} \cdot x + \text{beta} \cdot x^3$$

parameters gam, beta

$$C = \gamma T + \beta T^3 \text{ (electronic + phonon)}$$

Einstein heat capacity

2 parameters

$$A \cdot (T_e/x)^2 \cdot \exp(T_e/x) / (\exp(T_e/x) - 1)^2$$

parameters A, Te

single-mode Einstein term

Kohlrausch (stretched exponential)

4 parameters

$$A \cdot \exp(-(x/\tau)^\beta) + c_0$$

parameters A, tau, beta, c0

relaxation with a distribution of times; $\beta < 1$

Heat capacity $\gamma T + \beta T^3 + \delta T^5$

3 parameters

$$\text{gam} \cdot x + \text{beta} \cdot x^3 + \text{delta} \cdot x^5$$

parameters gam, beta, delta

the low-temperature expansion with the first anharmonic correction — what a C/T vs T^2 plot needs when it curves

Schottky anomaly (two-level)

2 parameters

$$A \cdot (D/x)^2 \cdot \exp(D/x) / (1 + \exp(D/x))^2$$

parameters A, D

$C = A(\Delta/T)^2 e^{\{\Delta/T\}/(1+e^{\{\Delta/T\}})^2}$: the broad hump from a crystal-field or nuclear two-level split

Electronic + phonon + Einstein mode

4 parameters

$$\text{gam} \cdot x + \text{beta} \cdot x^3 + A_e \cdot (T_e/x)^2 \cdot \exp(T_e/x) / (\exp(T_e/x) - 1)^2$$

parameters gam, beta, Ae, Te

$\gamma T + \beta T^3$ plus one localised (rattler) mode — the standard description of a clathrate or skutterudite $C(T)$

Two stretched exponentials

7 parameters

$$A1 \cdot \exp(-(x/t1)^{b1}) + A2 \cdot \exp(-(x/t2)^{b2}) + c0$$

parameters $A1, t1, b1, A2, t2, b2, c0$

two relaxation processes, each with its own distribution of times

Steinhart–Hart thermistor

3 parameters

$$1/(a + b \cdot \ln(x) + cs \cdot \ln(x)^3)$$

parameters a, b, cs $1/T = a + b \cdot \ln R + c \cdot \ln^3 R$, written for T against R — the standard three-parameter thermistor calibration**Debye heat capacity (full integral)**

3 parameters

$$A \cdot \text{debye}cv(x/TD) + gam \cdot x$$

parameters A, TD, gam $C = A \cdot D(T/\theta_D) + \gamma T$, with the Debye function computed exactly: T^3 at low temperature, Dulong–Petit A (3R per mole of atoms) at high**Debye + Einstein heat capacity (full integral)**

5 parameters

$$Ad \cdot \text{debye}cv(x/TD) + Ae \cdot (TE/x)^2 \cdot \exp(TE/x) / (\exp(TE/x) - 1)^2 + gam \cdot x$$

parameters Ad, TD, Ae, TE, gam

acoustic branches as a Debye continuum, a flat optical or rattling mode as an Einstein term — the description of clathrates, skutterudites and most molecular crystals

Thermal expansion (Grüneisen–Debye)

3 parameters

$$A \cdot \text{debye}cv(x/TD) + a0$$

parameters $A, TD, a0$ $\alpha(T) = \gamma_G \cdot C_V(T)/(3B_{TV})$: with a constant Grüneisen parameter the expansion coefficient follows the Debye heat capacity**Critical heat capacity at a continuous transition**

6 parameters

$$Bc + Lc \cdot x + \text{where}(x < Tc, Am, Ap) \cdot \max(\text{abs}(1 - x/Tc), 1e-6)^{-a1}$$

parameters $Bc, Lc, Tc, Am, Ap, a1$ $C = B + LT + A_{\pm} |1 - T/T_c|^{-\alpha}$ with separate amplitudes above and below. $\alpha \approx 0.11$ 3D Ising, -0.01 3D XY (the λ -cusp), -0.13 3D Heisenberg; A_+/A_- is universal**Lattice thermal conductivity — Callaway**

4 parameters

$$C \cdot \text{callaway}(x, TD, ap, bu)$$

parameters C, TD, ap, bu κ_L from the Debye–Callaway integral with boundary, point-defect (ω^4) and Umklapp ($\omega^2 T \cdot e^{-(\theta/3T)}$) scattering in parallel. ap and bu are those two rates relative to boundary scattering at the Debye frequency: the peak position and height pin them

Seebeck vs carrier density — single parabolic band (Pisarenko)

3 parameters

```
spbseeb(x, ms, Tm, lam)
```

parameters `ms, Tm, lam`

x = carrier density in cm^{-3} , $y = |S|$ in $\mu\text{V/K}$. The Fermi level is found from n through the Fermi–Dirac integral, so it is exact from non-degenerate to degenerate. T_m (K) and the scattering exponent λ , with $\tau \propto E^{(\lambda - \frac{1}{2})}$ (0 acoustic phonons, 2 ionised impurities) are pinned; the fit returns the density-of-states mass m^*

Seebeck vs temperature — single parabolic band

3 parameters

```
spbseeb(nc, ms, x, lam)
```

parameters `nc, ms, lam`

$S(T)$ at fixed carrier density (cm^{-3}). S depends on n only through $n/(m^*T)^{3/2}$, so m^* is pinned at 1 and nc comes back as the density for that mass; fix m^* from optics or Hall data and refit

Kinetics & adsorption

7 MODELS

Langmuir isotherm

2 parameters

$$q_m \cdot K \cdot x / (1 + K \cdot x)$$

parameters q_m, K

monolayer coverage

Hill equation

3 parameters

$$V_{\max} \cdot x^n / (K_d^n + x^n)$$

parameters $V_{\max}, n, K_d$ cooperative binding; n is the Hill coefficient

Dual-site Langmuir

4 parameters

$$q_1 \cdot K_1 \cdot x / (1 + K_1 \cdot x) + q_2 \cdot K_2 \cdot x / (1 + K_2 \cdot x)$$

parameters q_1, K_1, q_2, K_2

two binding sites with different affinities — a single Langmuir will not fit a heterogeneous surface

Butler–Volmer

3 parameters

$$j_0 \cdot (\exp(\min(\alpha \cdot x / V_t, 700)) - \exp(\min(-(1-\alpha) \cdot x / V_t, 700)))$$

parameters j_0, α, V_t the electrode kinetics equation: forward and reverse activation branches with transfer coefficient α ; $V_t = RT/F \approx 0.0257$ V at room temperature

Tafel

2 parameters

$$a + b \cdot \log_{10}(\text{abs}(x))$$

parameters a, b $\eta = a + b \cdot \log_{10} j$; the Tafel slope b identifies the rate-determining step

Michaelis–Menten with substrate inhibition

3 parameters

$$V_{\max} \cdot x / (K_m + x + x^2 / K_i)$$

parameters $V_{\max}, K_m, K_i$ $v = V_{\max} \cdot S / (K_m + S + S^2 / K_i)$: the rate turns over at high substrate rather than saturating

Langmuir–Hinshelwood rate

2 parameters

$$k \cdot K \cdot x / (1 + K \cdot x)^2$$

parameters k, K $r = kKc / (1 + Kc)^2$: a surface reaction whose rate falls again once the surface saturates

Mechanical & plasticity

17 MODELS

Hardening laws, strengthening, hardness and brittle failure statistics. Materials-science library — click to use, or switch it on under Model families.

Hollomon hardening

2 parameters

$$K_h * x^{n_h}$$

parameters K_h , n_h

true stress against true plastic strain. n_h is the strain-hardening exponent, and necking begins where strain equals it

Ludwik hardening

3 parameters

$$s_y + K_h * x^{n_h}$$

parameters s_y , K_h , n_h

Hollomon with a yield stress, so the curve does not start at zero the way a real metal does not

Swift hardening

3 parameters

$$K_h * (e_0 + x)^{n_h}$$

parameters K_h , e_0 , n_h

a pre-strain offset e_0 instead of a yield term — the form used for sheet forming, where the initial slope stays finite

Voce saturation hardening

3 parameters

$$s_s - (s_s - s_y) * \exp(-x/e_c)$$

parameters s_s , s_y , e_c

hardening that saturates rather than continuing as a power law; s_s is where dynamic recovery balances storage

Ludwik–Voce combined

5 parameters

$$s_y + K_h * x^{n_h} + (s_s - s_y) * (1 - \exp(-x/e_c))$$

parameters s_y , K_h , n_h , s_s , e_c

a power-law stage running into a saturating one — what most alloys actually do across a wide strain range

Ramberg–Osgood (strain vs stress)

3 parameters

$$x/E_y + 0.002 * (x/s_y)^{n_r}$$

parameters E_y , s_y , n_r

total strain against stress with the 0.2 % offset built in; n_r is large for a sharp yield and small for a rounded one

Johnson–Cook strain hardening

3 parameters

$$A_{jc} + B_{jc} * x^{n_{jc}}$$

parameters A_{jc} , B_{jc} , n_{jc}

the strain term of the Johnson–Cook law, fitted at one strain rate and temperature

Johnson–Cook strain-rate term

3 parameters

$$s_{ref} * (1 + C_{jc} * \ln(\max(x/\dot{\epsilon}_0, 1e-12)))$$

parameters s_{ref} , C_{jc} , $\dot{\epsilon}_0$ the logarithmic rate sensitivity; C_{jc} is small for f.c.c. metals and large for b.c.c. ones**Hall–Petch**

2 parameters

$$s_0 + k_{HP}/\sqrt{x}$$

parameters s_0 , k_{HP}

yield stress against grain size: the classic strengthening from grain boundaries blocking dislocation pile-ups

Inverse Hall–Petch (nanocrystalline)

3 parameters

$$s_0 + k_{HP}/\sqrt{x} - k_b/x$$

parameters s_0 , k_{HP} , k_b

the breakdown below about 20 nm, where grain-boundary sliding takes over and finer grains become weaker

Nix–Gao indentation size effect

2 parameters

$$H_0 * \sqrt{1 + h^*/x}$$

parameters H_0 , h^* hardness rising at shallow depth because geometrically necessary dislocations dominate; H_0 is the bulk hardness**Oliver–Pharr unloading**

3 parameters

$$a_{lp} * (x - h_f)^{m_{op}}$$

parameters a_{lp} , h_f , m_{op}

the unloading branch of a nanoindentation curve; its slope at maximum load gives the contact stiffness

Hertzian elastic contact

2 parameters

$$(4/3) * E_r * \sqrt{R_s} * x^{1.5}$$

parameters E_r , R_s

load against indentation depth for a sphere on a flat before anything yields — the elastic reference every indentation is judged against

Weibull failure probability (2-parameter)

2 parameters

$$1 - \exp(-(x/s_0w)^{mw})$$

parameters s_0w , mw

brittle strength distribution: mw is the Weibull modulus, and a small one means the flaw population is broad

Weibull failure probability (3-parameter)

3 parameters

$$1 - \exp(-(\max(x - s_u, 0)/s_0w)^{mw})$$

parameters s_u , s_0w , mw

the same with a threshold stress below which nothing fails — worth fitting only when the data reach that low

Griffith / fracture toughness

1 parameter

$$K_{Ic}/\sqrt{\pi x}$$

parameters K_{Ic}

failure stress against flaw size — the inverse square root that says doubling a crack costs 30 % of the strength

Gibson–Ashby foam scaling

3 parameters

$$E_s C_g x^{ng}$$

parameters E_s , C_g , ng

modulus or strength of a cellular solid against relative density; ng near 2 is bending-dominated, near 1 stretch-dominated

Fatigue & fracture

8 MODELS

Life curves and crack-growth laws. Materials-science library.

Basquin S–N curve

2 parameters

$$sfp * x^{bB}$$

parameters sfp , bB

stress amplitude against cycles to failure on log axes — a straight line, with bB the fatigue strength exponent

Basquin with endurance limit

3 parameters

$$se + sfp * x^{bB}$$

parameters se , sfp , bB

the same running down onto a horizontal asymptote, which steels have and aluminium alloys do not

Coffin–Manson

2 parameters

$$efp * x^{cc}$$

parameters efp , cc

plastic strain amplitude against reversals to failure — the low-cycle counterpart of Basquin

Strain–life (Basquin + Coffin–Manson)

5 parameters

$$sfp / Ey * x^{bB} + efp * x^{cc}$$

parameters sfp , Ey , bB , efp , cc

total strain amplitude: the elastic term dominates at long life and the plastic term at short, crossing at the transition life

Paris law crack growth

2 parameters

$$Cp * x^{mp}$$

parameters Cp , mp

da/dN against ΔK in the stable regime; mp is between 2 and 4 for most metals

Paris with threshold (Klesnil–Lukáš)

3 parameters

$$Cp * \max(x^{mp} - dKth^{mp}, 0)$$

parameters Cp , mp , $dKth$

growth that stops below a threshold ΔK rather than continuing as a power law all the way down

Forman crack growth

4 parameters

$$C_p * x^{m_p} / ((1 - R_f) * K_c - x)$$

parameters C_p , m_p , R_f , K_c Paris with the acceleration towards final fracture built in — it diverges as ΔK approaches the toughness**Walker mean-stress correction**

4 parameters

$$C_w * (x * (1 - R_f)^{(g_w - 1)})^{m_w 2}$$

parameters C_w , R_f , g_w , $m_w 2$ collapses growth data taken at different R ratios onto one curve; $g_w = 1$ means no mean-stress sensitivity

Creep & high-temperature

9 MODELS

Strain rate against stress, and strain against time, in all three creep stages. Materials-science library.

Norton power-law creep

2 parameters

$$A_{nc} * x^{n_{nc}}$$

parameters A_{nc} , n_{nc}

steady-state strain rate against stress. n_{nc} near 1 is diffusional creep, 3–5 dislocation climb, above 7 a threshold stress is hiding

Norton–Bailey creep strain

2 parameters

$$A_{nb} * x^{m_{nb}}$$

parameters A_{nb} , m_{nb}

creep strain against time in the primary stage; m_{nb} below 1 is the decelerating transient

Garofalo hyperbolic-sine creep

3 parameters

$$A_{gf} * \sinh(\alpha_f * x)^{n_{gf}}$$

parameters A_{gf} , α_f , n_{gf}

one law across the whole stress range: power-law at low stress, exponential at high, without switching models

Andrade creep

3 parameters

$$e_{0c} + bA * x^{(1/3)} + k_{ss} * x$$

parameters e_{0c} , bA , k_{ss}

the classical decomposition: an instantaneous strain, a transient cube-root term, and a steady-state rate

Primary + secondary creep

4 parameters

$$e_{0c} + e_p * (1 - \exp(-x/t_p)) + k_{ss} * x$$

parameters e_{0c} , e_p , t_p , k_{ss}

a saturating primary stage added to a constant secondary rate — the two-stage curve before tertiary creep sets in

Theta projection creep

4 parameters

$$th_1 * (1 - \exp(-th_2 * x)) + th_3 * (\exp(th_4 * x) - 1)$$

parameters th_1 , th_2 , th_3 , th_4

primary and tertiary stages as two exponentials, which is how creep curves are extrapolated to service lives

Monkman–Grant

2 parameters

$$C_{mg}/x^{b_{mg}}$$

parameters C_{mg} , b_{mg}

time to rupture against minimum creep rate — the empirical link that lets a short test predict a long life

Larson–Miller rupture

2 parameters

$$s_{LM} \exp(-x/CLM)$$

parameters s_{LM} , CLM

stress against the Larson–Miller parameter $T(C + \log t)$; a master curve that collapses every temperature onto one line

Stress relaxation (power law)

3 parameters

$$s_{0r} \cdot (1 + x/t_{0r})^{-nsr}$$

parameters s_{0r} , t_{0r} , nsr

stress decaying under fixed strain; a power law, not an exponential, when a spectrum of relaxation times is active

Corrosion & oxidation

8 MODELS

Scale growth: parabolic, linear, logarithmic and the laws in between. Materials-science library.

Parabolic oxidation (Wagner)

2 parameters

$$\sqrt{k_p \cdot x} + w_0$$

parameters k_p , w_0

a protective scale thickening against its own diffusion resistance — the law a good oxide obeys

Linear oxidation

2 parameters

$$k_l \cdot x + w_0$$

parameters k_l , w_0

a scale that cracks or evaporates as fast as it forms, so the interface never becomes protected

Paralinear oxidation

3 parameters

$$\sqrt{k_p \cdot x} - k_l \cdot x + w_0$$

parameters k_p , k_l , w_0

parabolic growth with simultaneous volatilisation — mass gain peaks and then falls, which pure parabolic cannot do

Cubic oxidation

2 parameters

$$(k_c \cdot x)^{1/3} + w_0$$

parameters k_c , w_0

the intermediate law seen where short-circuit diffusion along grain boundaries in the scale controls growth

Logarithmic oxidation

3 parameters

$$k_e \cdot \ln(1 + x/t_0x) + w_0$$

parameters k_e , t_0x , w_0

very thin films at low temperature, where growth almost stops once the field across the oxide collapses

Inverse-logarithmic (Cabrera–Mott)

2 parameters

$$1/\max(a_{inv} - b_{inv} \cdot \ln(\max(x, 1e-12)), 1e-9)$$

parameters a_{inv} , b_{inv}

the Cabrera–Mott law for field-driven ion transport through a very thin oxide at low temperature

Deal-Grove linear-parabolic

3 parameters

$$(Adg/2)*(sqrt(1 + 4*Bdg*(x + taudg)/Adg^2) - 1)$$

parameters `Adg, Bdg, taudg`

thermal oxidation of silicon across the whole range: reaction-limited when thin, diffusion-limited when thick

Pitting / induction then growth

3 parameters

$$wmax*(1 - exp(-kpit*max(x - tind, 0)))$$

parameters `wmax, kpit, tind`

nothing measurable until the passive film breaks down at tind, then attack that saturates as pits stifle

Polymers & rheology

22 MODELS

Viscosity, shear thinning, viscoelasticity, blends and molar-mass relations. Materials-science library.

Mark–Houwink

2 parameters

$$K_{mh} * x^{amh}$$

parameters K_{mh} , amh

intrinsic viscosity against molar mass; amh is 0.5 in a theta solvent, 0.8 in a good one, and 0 for a hard sphere

WLF shift factor

3 parameters

$$-C1w * (x - Trf) / (C2w + x - Trf)$$

parameters $C1w$, Trf , $C2w$

$\log aT$ for time–temperature superposition above T_g . The universal values $C1 \approx 17$, $C2 \approx 52$ K are a starting point, not a law

Arrhenius shift factor

2 parameters

$$Eaw / 0.0083145 * (1/x - 1/Trf)$$

parameters Eaw , Trf

the shift factor below T_g or for a chemically simple relaxation, where WLF does not apply. Eaw in kJ/mol

Cross viscosity model

4 parameters

$$et_{inf} + (et_0 - et_{inf}) / (1 + (kcr * x)^{mcr})$$

parameters et_{inf} , et_0 , kcr , mcr

shear thinning between two Newtonian plateaus; $1/kcr$ is the shear rate where thinning begins

Carreau

4 parameters

$$et_{inf} + (et_0 - et_{inf}) * (1 + (lamc * x)^2)^{((ncr-1)/2)}$$

parameters et_{inf} , et_0 , $lamc$, ncr

the same shape derived from molecular arguments; ncr is the power-law index of the thinning region

Carreau–Yasuda

5 parameters

$$et_{inf} + (et_0 - et_{inf}) * (1 + (lamc * x)^{acy})^{((ncr-1)/acy)}$$

parameters et_{inf} , et_0 , $lamc$, acy , ncr

Carreau with the width of the transition released — the extra parameter is what fits a broad molar-mass distribution

Power law (Ostwald–de Waele)

2 parameters

$$\tau = K \omega^n$$

parameters K , n shear stress against shear rate with no plateaus; $n < 1$ is shear thinning, > 1 shear thickening**Bingham plastic**

2 parameters

$$\tau = \tau_0 + \eta \dot{\gamma}$$

parameters τ_0 , η

a yield stress and then Newtonian flow — toothpaste, drilling mud, fresh concrete

Herschel–Bulkley

3 parameters

$$\tau = \tau_0 + K \dot{\gamma}^n$$

parameters τ_0 , K , n

a yield stress followed by power-law flow; the three-parameter standard for yield-stress fluids

Casson

2 parameters

$$\sqrt{\tau} = \sqrt{\tau_0} + \sqrt{\eta} \sqrt{\dot{\gamma}}$$

parameters τ_0 , η

the yield-stress model used for blood, chocolate and printing inks, where the square-root plot is the linear one

Sisko

3 parameters

$$\tau = \tau_0 + K \dot{\gamma}^{n-1}$$

parameters τ_0 , K , n

high-shear Newtonian behaviour with a power-law tail — the Cross model where the zero-shear plateau is out of range

Prony series relaxation (2 terms)

5 parameters

$$G = G_e + G_1 \exp(-t/\tau_1) + G_2 \exp(-t/\tau_2)$$

parameters G_e , G_1 , τ_1 , G_2 , τ_2 the standard viscoelastic relaxation modulus; G_e is the equilibrium modulus a crosslinked network keeps and a melt does not**Kelvin–Voigt creep compliance**

2 parameters

$$J(t) = J_0 (1 - \exp(-t/\tau))$$

parameters J_0 , τ

delayed elasticity that fully recovers — a spring and dashpot in parallel, with no permanent flow

Burgers creep compliance

4 parameters

$$J1b + J2b*(1 - \exp(-x/\tau_{ab})) + x/\eta_{ab}$$

parameters $J1b$, $J2b$, τ_{ab} , η_{ab}

instantaneous, delayed and viscous strain in one model — the four-element description of a real polymer under load

Doolittle free-volume viscosity

5 parameters

$$A_d \exp(B_d / (f_{v0} + a_{lv}(x - T_{v0})))$$

parameters A_d , B_d , f_{v0} , a_{lv} , T_{v0}

viscosity from the free volume available for a segment to move into; the divergence is where free volume runs out

Fox equation (blend T_g)

2 parameters

$$1/(x/T_{g1} + (1-x)/T_{g2})$$

parameters T_{g1} , T_{g2}

glass transition of a miscible blend against weight fraction, assuming no specific interaction

Gordon–Taylor (blend T_g)

3 parameters

$$(x*T_{g1} + k_{gt}*(1-x)*T_{g2})/(x + k_{gt}*(1-x))$$

parameters T_{g1} , k_{gt} , T_{g2}

the Fox equation with an interaction parameter; k_{gt} away from 1 means the components are not simply additive

Flory–Fox molar-mass T_g

2 parameters

$$T_{ginf} - K_{ff}/x$$

parameters T_{ginf} , K_{ff}

T_g rising with chain length towards its infinite-mass limit, because chain ends carry extra free volume

Schulz–Zimm distribution

3 parameters

$$A_{sz} * x^{z_{sz}} * \exp(-(z_{sz}+1)*x/M_{nz})$$

parameters A_{sz} , z_{sz} , M_{nz}

the molar-mass distribution of a step-growth or living polymerisation; the dispersity is $(z_{sz}+1)/z_{sz}$

Zimm plot (light scattering)

2 parameters

$$1/M_w + 2*A_{2z}*x$$

parameters M_w , A_{2z}

K_c/R against concentration: the intercept is the weight-average molar mass and the slope the second virial coefficient

Rubber elasticity (neo-Hookean)

1 parameter

$$Gr*(x - 1/x^2)$$

parameters Gr

nominal stress against extension ratio for a crosslinked network; the slope gives the crosslink density directly

Mooney–Rivlin

2 parameters

$$2*(C1m + C2m/x)*(x - 1/x^2)$$

parameters $C1m$, $C2m$

the empirical correction to neo-Hookean that a Mooney plot makes linear; $C2m$ collects the entanglement contribution

Composites & effective media

8 MODELS

Effective properties of two-phase materials, from the bounds to percolation. Materials-science library.

Rule of mixtures (Voigt bound)

2 parameters

$$x \cdot E_f + (1-x) \cdot E_m$$

parameters E_f , E_m

the upper bound: iso-strain loading, which aligned continuous fibres approach along their axis

Inverse rule of mixtures (Reuss bound)

2 parameters

$$1 / (x / E_f + (1-x) / E_m)$$

parameters E_f , E_m

the lower bound: iso-stress loading, approached transverse to the fibres

Halpin–Tsai

3 parameters

$$E_m \cdot (1 + \text{zet} \cdot \text{etah} \cdot x) / (1 - \text{etah} \cdot x)$$

parameters E_m , zet , etah

the interpolation between the two bounds that actually fits short-fibre and particulate composites; zet carries the aspect ratio

Maxwell–Garnett effective permittivity

2 parameters

$$\epsilon_m \cdot (1 + 2 \cdot x \cdot \text{bmg}) / (1 - x \cdot \text{bmg})$$

parameters ϵ_m , bmg

dilute spherical inclusions in a host: valid while the inclusions do not see each other, which fails near percolation

Lichtenecker logarithmic mixing

2 parameters

$$\exp(x \cdot \ln(\max(\epsilon_a, 1e-12))) + (1-x) \cdot \ln(\max(\epsilon_b, 1e-12)))$$

parameters ϵ_a , ϵ_b

the geometric-mean mixing rule for a random two-phase medium with no preferred geometry

Percolation power law

3 parameters

$$\text{sig0} \cdot \max(x - \text{pc}, 0)^{\text{tperc}}$$

parameters sig0 , pc , tperc

conductivity switching on above a critical filler fraction; tperc near 2 is the universal three-dimensional value

Nielsen barrier tortuosity

1 parameter

$$(1 - x)/(1 + a \ln b * x)$$

parameters $a \ln b$

permeability of a filled film against platelet loading; $a \ln b$ is half the aspect ratio, so flakes beat spheres

Maxwell–Eucken thermal conductivity

2 parameters

$$k_m * (2 * k_m + k_d - 2 * x * (k_m - k_d)) / (2 * k_m + k_d + x * (k_m - k_d))$$

parameters k_m, k_d

effective conductivity of a dispersion of spheres; the two-phase analogue of Maxwell–Garnett

Equations of state

5 MODELS

Pressure against compression, and thermal expansion. Materials-science library.

Birch–Murnaghan (3rd order)

2 parameters

$$1.5 * K_0 * (x^{(-7/3)} - x^{(-5/3)}) * (1 + 0.75 * (K_0' - 4) * (x^{(-2/3)} - 1))$$

parameters K_0 , K_0'

pressure against V/V_0 from finite Eulerian strain — the standard equation of state in mineral physics

Birch–Murnaghan (2nd order)

1 parameter

$$1.5 * K_0 * (x^{(-7/3)} - x^{(-5/3)})$$

parameters K_0

the same with K_0' fixed at 4 — worth using when the compression range is too small to determine it

Vinet equation of state

2 parameters

$$3 * K_0 * x^{(-2/3)} * (1 - x^{(1/3)}) * \exp(1.5 * (K_0' - 1) * (1 - x^{(1/3)}))$$

parameters K_0 , K_0'

better behaved than Birch–Murnaghan under strong compression, where the polynomial form starts to misbehave

Murnaghan equation of state

2 parameters

$$K_0 / K_0' * (x^{(-K_0')} - 1)$$

parameters K_0 , K_0'

the simplest integrated form, assuming the bulk modulus rises linearly with pressure

Debye–Grüneisen thermal expansion

3 parameters

$$L_0 * (1 + A_{ex} / (\exp(T_E/x) - 1))$$

parameters L_0 , A_{ex} , T_E

thermal expansion that saturates at high temperature and flattens to zero slope at low — the Einstein-mode form

Sintering & microstructure

11 MODELS

Grain growth, coarsening, densification and thermal analysis. Materials-science library.

Grain growth power law

3 parameters

$$(D0g^{ngr} + kgr*x)^{(1/ngr)}$$

parameters `D0g`, `ngr`, `kgr`

normal grain growth: $ngr = 2$ is ideal curvature-driven growth, and larger values mean something is pinning the boundaries

Parabolic grain growth

2 parameters

$$\sqrt{D0g^2 + kgr*x}$$

parameters `D0g`, `kgr`

the ideal case written out, for when you want to test whether the exponent really is 2

Zener-pinned grain growth

3 parameters

$$Dlim - (Dlim - D0g)*\exp(-x/tzn)$$

parameters `Dlim`, `D0g`, `tzn`

growth stalling at a limiting size set by second-phase particles rather than continuing indefinitely

LSW Ostwald ripening

2 parameters

$$(r0^3 + kls*x)^{(1/3)}$$

parameters `r0`, `kl`

coarsening of a precipitate distribution by diffusion: the cube of the mean radius grows linearly with time

Sintering shrinkage (initial stage)

2 parameters

$$Ash*x^{nsh}$$

parameters `Ash`, `nsh`

linear shrinkage in the initial stage; nsh near 1/2 is volume diffusion, 1/3 grain-boundary diffusion

Master sintering curve

4 parameters

$$\rho0 + (\rho0f - \rho0)/(1 + \exp(-(x - xms)/wms))$$

parameters `rho0`, `rho0f`, `xms`, `wms`

relative density against the work-of-sintering parameter; one sigmoid that collapses every heating schedule onto itself

Recrystallisation (JMAK in time)

3 parameters

$$X_f * (1 - \exp(-(k_{jm} * x)^{n_{jm}}))$$

parameters X_f , k_{jm} , n_{jm}

recrystallised fraction against annealing time; n_{jm} carries the nucleation and growth dimensionality together

TGA single-step decomposition

4 parameters

$$w_f + (w_{0t} - w_f) / (1 + \exp((x - T_d) / w_{td}))$$

parameters w_f , w_{0t} , T_d , w_{td}

a mass-loss step against temperature: T_d is the midpoint and w_{td} the width, so the derivative peak follows without differentiating noisy data

TGA two-step decomposition

7 parameters

$$w_f + dw_1 / (1 + \exp((x - T_{d1}) / w_{t1})) + dw_2 / (1 + \exp((x - T_{d2}) / w_{t2}))$$

parameters w_f , dw_1 , T_{d1} , w_{t1} , dw_2 , T_{d2} , w_{t2}

two overlapping mass-loss events resolved without deconvolving the derivative — usually the more stable way to get both. The two steps are interchangeable, so read them by their fitted temperatures rather than by which is labelled 1

Umklapp thermal conductivity

2 parameters

$$A_{uk} / x * \exp(T_{uk} / x)$$

parameters A_{uk} , T_{uk}

the high-temperature fall of lattice conductivity, where phonon–phonon scattering is the only process that matters. T_{uk} is θ_D/b — the two are not separately measurable from $\kappa(T)$ alone, so they are carried as one parameter

Boundary + Umklapp conductivity

3 parameters

$$1 / (1 / (A_{bd} * x^3) + 1 / (A_{uk} / x * \exp(T_{uk} / x)))$$

parameters A_{bd} , A_{uk} , T_{uk}

the full $\kappa(T)$ hump: boundary scattering giving T^3 at low temperature, Umklapp taking over above the maximum. $T_{uk} = \theta_D/b$, since only the ratio appears

Non-linear transport & I–V

36 MODELS

Current against voltage when it is not a straight line: space-charge and trap-limited conduction, barrier emission and tunnelling, diodes solved in closed form with Lambert–W, negative differential resistance, transistor characteristics, Josephson and vortex dynamics, blockade and Luttinger power laws. Ordered from a cubic correction to Ohm’s law upward.

Ohmic with a cubic non-linearity

2 parameters

$$G1*x + G3*x^3$$

parameters $G1$, $G3$

the first correction to Ohm’s law an inversion-symmetric junction can have — even powers are forbidden by symmetry, so the leading term is cubic. $G3/G1$ measures how far from linear the contact is.

Varistor / power-law conduction

2 parameters

$$kv*abs(x)^{al}*sign(x)$$

parameters kv , al

$I \propto V^\alpha$ with α well above 1 — a ZnO varistor, a grain-boundary barrier, a percolating composite. α is the non-linearity coefficient quoted on surge-arrester datasheets.

Space-charge-limited current (Mott–Gurney) with ohmic leakage

3 parameters

$$G1*x + (9/8)*epmu*x^2/L^3$$

parameters $G1$, $epmu$, L

the trap-free square law, $J = (9/8)\epsilon\mu V^2/L^3$, in parallel with the ohmic channel that dominates at low bias. The crossover voltage gives the free-carrier density.

Trap-limited SCLC (Mark–Helfrich)

3 parameters

$$G1*x + kt*x^{(lt+1)}$$

parameters $G1$, kt , lt

an exponential trap distribution turns the square law into $V^{(l+1)}$, where $l = Tt/T$ is the characteristic trap temperature over the measurement temperature. $l = 1$ recovers Mott–Gurney.

SCLC with a field-dependent mobility

2 parameters

$$ks*x^2*\exp(gm*\sqrt{abs(x)})$$

parameters ks , gm

Mott–Gurney with a Poole–Frenkel mobility $\mu(E) = \mu_0 \exp(\gamma\sqrt{E})$ — the standard description of a disordered organic semiconductor diode above the ohmic regime.

Child–Langmuir law (vacuum space charge)

2 parameters

$$k_c \cdot (\max(x - V_0, 0))^{1.5}$$

parameters k_c , V_0

the three-halves power law for emission-limited current in a vacuum diode. Nothing about it is material-specific: it follows from Poisson's equation and energy conservation alone.

Shockley diode with shunt and series resistance (Lambert W)

4 parameters

$$(\text{nd} \cdot 0.02585 / R_s) \cdot \text{lambertw}e(I_0 \cdot R_s / (\text{nd} \cdot 0.02585), (x + I_0 \cdot R_s) / (\text{nd} \cdot 0.02585)) - I_0 + x / R_{sh}$$

parameters nd , R_s , I_0 , R_{sh}

the diode equation solved exactly rather than iterated. $I = (nV_t/R_s)W[(I_0R_s/nV_t)e^{((V+I_0R_s)/nV_t)}] - I_0$ puts the series resistance inside the exponential where it belongs, so R_s and n stop trading against each other.

Solar cell — single diode with R_s and R_{sh} (Lambert W)

5 parameters

$$(R_{sh} \cdot (I_{ph} + I_0) - x) / (R_s + R_{sh}) - (\text{nd} \cdot 0.02585 / R_s) \cdot \text{lambertw}e(R_s \cdot R_{sh} \cdot I_0 / (\text{nd} \cdot 0.02585 \cdot (R_s + R_{sh})), R_{sh} \cdot (R_s \cdot I_{ph} + R_s \cdot I_0 + x) / (\text{nd} \cdot 0.02585 \cdot (R_s + R_{sh})))$$

parameters R_{sh} , I_{ph} , I_0 , R_s , nd

the five-parameter photovoltaic model — photocurrent, saturation current, ideality, series and shunt resistance — in closed form. Fit it to a light I–V and every figure of merit (V_{oc} , I_{sc} , fill factor, series loss) follows from the parameters.

Two-diode solar cell model

4 parameters

$$I_{ph} - I_{01} \cdot (\exp(\min(x / (0.02585), 700)) - 1) - I_{02} \cdot (\exp(\min(x / (2 \cdot 0.02585), 700)) - 1) - x / R_{sh}$$

parameters I_{ph} , I_{01} , I_{02} , R_{sh}

diffusion current ($n = 1$) and depletion-region recombination ($n = 2$) as separate channels, which is what a real cell shows at low illumination. The two saturation currents separate bulk quality from junction quality.

Back-to-back Schottky barriers

3 parameters

$$I_{sb} \cdot \sinh(\max(\min(x / (2 \cdot n_b \cdot 0.02585), 700), -700)) + G_b \cdot x$$

parameters I_{sb} , n_b , G_b

two identical barriers in series, one always reverse biased: the hyperbolic sine is the signature, and it is symmetric in bias where a single diode is not. Common on unannealed metal contacts to a semiconductor.

Simmons tunnelling — intermediate bias

3 parameters

$$a_1 \cdot (\text{phb} - x/2) \cdot \exp(-b_1 \cdot \sqrt{\max(\text{phb} - x/2, 0)}) - a_1 \cdot (\text{phb} + x/2) \cdot \exp(-b_1 \cdot \sqrt{\max(\text{phb} + x/2, 0)})$$

parameters a_1 , phb , b_1

the full Simmons expression for a rectangular barrier, not its low-bias cubic expansion: the barrier height ϕ comes out directly, and the asymmetry between the two exponentials is what bends the characteristic over.

Brinkman–Dynes–Rowell conductance

4 parameters

$$G_0 * (1 - (A_0 * d_{ph} / \phi_b^{1.5}) * x + (9/128) * (A_0^2 / \phi_b) * x^2)$$

parameters G_0 , A_0 , d_{ph} , ϕ_b

the conductance of a tunnel junction with an ASYMMETRIC barrier, expanded to second order in bias. The linear term is proportional to the barrier asymmetry $\Delta\phi$ and vanishes for a symmetric junction — a parabola with its minimum off zero is the standard test that a junction is a real tunnel barrier and not a pinhole.

Fowler–Nordheim plot ($\ln(I/V^2)$ against $1/V$)

2 parameters

$$a_F - b_F * x$$

parameters a_F , b_F

the linearised form. Plot $\ln(I/V^2)$ against $1/V$ and field emission is a straight line whose slope is $-b\phi^{1.5}/\beta$: a fit here gives the field-enhancement factor β straight off, and curvature is the usual evidence that the emission is not Fowler–Nordheim at all.

Schottky (Richardson) emission against bias

2 parameters

$$a_S * \exp(b_S * \sqrt{\text{abs}(x)})$$

parameters a_S , b_S

image-force lowering of an interface barrier: $\ln I$ against $\sqrt{V}$ is a straight line, and its slope gives the optical dielectric constant. The companion to Poole–Frenkel, which has the same $\sqrt{V}$ but an extra factor of V in front and twice the slope.

Thermionic field emission (Padovani–Stratton)

3 parameters

$$I_{tf} * (\exp(\min(x/E_0, 700)) - 1) + G_{tf} * x$$

parameters I_{tf} , E_0 , G_{tf}

tunnelling through the top of a barrier rather than over it. The characteristic energy $E_0 = E_{00} \cdot \coth(E_{00}/kT)$ is temperature-independent in the pure field-emission limit — an apparent ideality factor that does not fall with T is the fingerprint.

Esaki tunnel diode (with negative differential resistance)

5 parameters

$$I_p * (x/V_p) * \exp(\min(1 - x/V_p, 700)) + I_v * \exp(\min(a_v * (x - V_p), 700)) + I_{sd} * (\exp(\min(x/(0.02585), 700)) - 1)$$

parameters I_p , V_p , I_v , a_v , I_{sd}

the three-term standard model: band-to-band tunnelling that peaks and falls, excess current through gap states, and the ordinary thermal diode current that takes over. The peak-to-valley ratio comes straight out of I_p and I_v .

Resonant tunnelling diode

6 parameters

$$Ar * (\arctan((x - x1)/Gr) + \pi/2) - Ar * (\arctan((x - x2)/Gr) + \pi/2) + Ib * (\exp(\min(x/Vb, 700)) - 1)$$

parameters $Ar, x1, Gr, x2, Ib, Vb$

a Lorentzian transmission resonance swept through the emitter Fermi sea: the current rises when the quasi-bound level enters the window and collapses when it leaves, giving the NDR region a double-atan shape rather than a cubic one.

Cubic negative differential resistance (Gunn)

3 parameters

$$an * x - bn * x^2 + cn * x^3$$

parameters an, bn, cn

the minimal polynomial with a peak and a valley — the Kroemer form for a transferred-electron device. Useful when the microscopic mechanism is not in question and only the NDR region needs parameterising for a circuit model.

Avalanche multiplication (Miller)

3 parameters

$$M0 / (1 - (\text{abs}(x)/Vbr)^m)$$

parameters $M0, Vbr, m$

the empirical multiplication factor that diverges at the breakdown voltage. m between 2 and 6 depending on the material and which carrier initiates; fitting it is how Vbr is extracted from data taken well below breakdown.

Zener / reverse breakdown with forward diode

6 parameters

$$Isd * (\exp(\min(x/(nz * 0.02585), 700)) - 1) - Ibr * \exp(-\min((x + Vz)/Vt2, 700)) + Gz * x$$

parameters $Isd, nz, Ibr, Vz, Vt2, Gz$

both polarities in one expression: thermal injection forward, a sharp reverse knee at $-Vz$, and the shunt in parallel. Fit a full sweep rather than two halves and the leakage floor stops being fitted twice.

MOSFET output characteristic (square law with channel-length modulation)

3 parameters

$$\text{where}(x < Vov, kp * (Vov * x - x^2/2), kp * Vov^2/2 * (1 + lam * x))$$

parameters Vov, kp, lam

the textbook long-channel $I_d(V_{ds})$: quadratic in the triode region, flat in saturation with a slope λ from channel-length modulation. The kink is at $V_{ds} = V_{gs} - V_{th}$, so the fit returns the overdrive without needing to know the threshold separately.

MOSFET transfer characteristic — α -power law

3 parameters

$$kp * (\max(x - Vth, 0))^{\alpha}$$

parameters kp, Vth, α

Sakurai–Newton: velocity saturation drives the exponent down from 2 towards 1 as the channel shortens, so α is a direct measure of how short-channel the device is.

Subthreshold conduction

3 parameters

$$I_{\text{sub}} \cdot \exp(\min(x/(n_s \cdot 0.02585), 700)) + I_{\text{off}}$$

parameters $I_{\text{sub}}, n_s, I_{\text{off}}$

the exponential tail below threshold. The slope in decades per volt is the subthreshold swing, and $n = S/(60 \text{ mV/dec})$ at room temperature is the body factor: 60 mV/dec is the thermionic limit no ordinary MOSFET beats.

JFET / MESFET output (Curtice)

3 parameters

$$I_{\text{dss}} \cdot \tanh(\alpha_c \cdot x) \cdot (1 + \lambda_m \cdot x)$$

parameters $I_{\text{dss}}, \alpha_c, \lambda_m$

the hyperbolic-tangent model used in every microwave FET library. $\tanh$ handles the knee smoothly so the derivative stays continuous, which is what a harmonic-balance simulator needs.

Josephson junction I–V (RSJ)

2 parameters

$$R_n \cdot \text{sign}(x) \cdot \sqrt{\max(x^2 - I_c^2, 0)}$$

parameters R_n, I_c

the resistively shunted junction at zero temperature: no voltage at all below the critical current, then a square-root approach to the normal-state ohmic line. The rounding a real junction shows at the knee is thermal, and the Ambegaokar–Halperin form below describes it.

Josephson I–V with a thermally rounded knee

3 parameters

$$R_n \cdot \text{sign}(x) \cdot (\sqrt{\max(x^2 - I_c^2, 0)} + I_{\text{th}}^2) - I_{\text{th}}$$

parameters R_n, I_c, I_{th}

the RSJ square root with the sharp corner smoothed over a current scale I_{th} . Phase diffusion at finite temperature puts a small voltage below I_c and rounds the turn-on; a junction whose I–V is rounded at the knee is not a bad junction, it is a warm one. I_{th} is phenomenological here, of order $2ekT/\hbar$.

Superconductor E–J power law

3 parameters

$$E_c \cdot (x/J_c)^n$$

parameters E_c, J_c, n

the engineering definition of the critical current: $E = E_c(J/J_c)^n$ with E_c normally $1 \mu\text{V/cm}$. The exponent n is the figure of merit for a coated conductor — above 30 is excellent, below 10 means the current is finding its way round defects.

Flux creep (Anderson–Kim)

3 parameters

$$E_0 \cdot \exp(-U_c \cdot (1 - x/J_c))$$

parameters E_0, U_c, J_c

thermally activated vortex motion with a pinning barrier that closes linearly as the current approaches depinning. The logarithmic E–J slope gives U_0/kT directly.

Berezinskii–Kosterlitz–Thouless I–V ($V \propto I^a$)

2 parameters

$$kb \cdot x^{aT}$$

parameters kb, aT

in a two-dimensional superconductor the I–V is a pure power law whose exponent jumps from 1 to 3 at the BKT transition. Fit $a(T)$ across the transition and the temperature where $a = 3$ is T_{BKT} .

Coulomb blockade above threshold

3 parameters

$$kcb \cdot (\max(x/V_{th} - 1, 0))^{\zeta e}$$

parameters $kcb, V_{th}, \zeta e$

a one-dimensional array of tunnel junctions passes no current below a threshold set by the charging energy, then rises as a power law with ζ near 1 in one dimension and near 5/3 in two.

Coulomb gap (symmetric staircase foot)

3 parameters

$$G_{cb} \cdot \text{sign}(x) \cdot \max(\text{abs}(x) - V_{th}, 0) + G_l \cdot x$$

parameters G_{cb}, V_{th}, G_l

the simplest description of a blockaded junction: a hard gap of width $2V_{th}$, ohmic outside it. The offset of the extrapolated lines from the origin is the charging energy in volts.

Luttinger liquid tunnelling

3 parameters

$$k_l \cdot \text{sign}(x) \cdot \text{abs}(x)^{(\alpha_l + 1)} + G_l \cdot x$$

parameters k_l, α_l, G_l

tunnelling into a one-dimensional interacting conductor: the differential conductance is a power law in bias, $dI/dV \propto V^\alpha$, with α fixed by the Luttinger parameter. $\alpha = 0$ is a Fermi liquid.

Zero-bias anomaly (Altshuler–Aronov $\sqrt{V}$)

2 parameters

$$G_{z0} + cz \cdot \sqrt{\text{abs}(x)}$$

parameters G_{z0}, cz

the square-root dip in the density of states that electron–electron interaction opens at the Fermi level in a disordered three-dimensional metal. Shallow, symmetric and pinned to zero bias — which is how it is told from a Kondo resonance.

Hopping conduction at high field (sinh law)

3 parameters

$$I_h \cdot \sinh(\max(\min(x/V_h, 700), -700)) + G_l \cdot x$$

parameters I_h, V_h, G_l

when the field tilts the hopping landscape by more than kT the forward and backward rates stop cancelling and the characteristic becomes a hyperbolic sine. Symmetric in bias, unlike anything involving a junction.

Quantum point contact — conductance staircase

7 parameters

```
G0*(smearstep(x, V1, kTe) + smearstep(x, V2, kTe) + smearstep(x, V3, kTe) +
smearstep(x, V4, kTe)) + Gc
```

parameters `G0, V1, kTe, V2, V3, V4, Gc`

four thermally smeared conductance steps as sub-bands are pushed through the Fermi level. G_0 should refine to $2e^2/h = 7.748 \times 10^{-5}$ S per step if the contact is ballistic, and the smearing width is kT plus whatever the barrier adds.

Diode-connected series pair (two junctions, one sweep)

4 parameters

```
Isd*(exp(min((x - Vk)/(n1*0.02585), 700)) - 1)*step(x - Vk) + Gp*x
```

parameters `Isd, Vk, n1, Gp`

a rectifying junction in series with an ohmic element and a turn-on offset — the pragmatic model for a packaged device where the contact potential is not zero.

Quantum oscillations

29 MODELS

Everything periodic in inverse field, written in the field itself so no axis has to be inverted first: Shubnikov–de Haas and de Haas–van Alphen with the full Lifshitz–Kosevich damping, Landau fans and Berry phase, Aharonov–Bohm and Altshuler–Aronov–Spivak, Weiss and magnetophonon resonance, Fraunhofer and SQUID interference, and the real-space oscillations — Friedel and RKKY.

Shubnikov–de Haas — single frequency in B

4 parameters

$$A \cos(2\pi F / \text{abs}(x) + \text{ph}) + c0$$

parameters A , F , ph , $c0$

the bare oscillation, periodic in $1/B$ with period $1/F$. F is the extremal cross-section of the Fermi surface in tesla through the Onsager relation $F = (\hbar/2\pi e)A$. Fit this against B directly — there is no need to invert the axis first.

Shubnikov–de Haas with Dingle damping

5 parameters

$$A \exp(-aD / \text{abs}(x)) \cos(2\pi F / \text{abs}(x) + \text{ph}) + c0$$

parameters A , aD , F , ph , $c0$

oscillations die out at low field because scattering blurs the Landau levels. The Dingle factor $\exp(-\pi/\omega_c \tau)$ is exponential in $1/B$, and its coefficient gives the quantum lifetime — which is shorter than the transport lifetime, since small-angle scattering counts here and barely matters for resistivity.

Shubnikov–de Haas on a polynomial background

7 parameters

$$A \exp(-aD / \text{abs}(x)) \cos(2\pi F / \text{abs}(x) + \text{ph}) + b0 + b1x + b2x^2$$

parameters A , aD , F , ph , $b0$, $b1$, $b2$

the same oscillation riding on the smooth magnetoresistance, fitted together instead of subtracted first. Fitting the background separately biases the amplitude, because a polynomial fitted to oscillating data absorbs part of the oscillation.

Lifshitz–Kosevich (thermal and Dingle damping)

7 parameters

$$A (14.69 m^* T_k / \text{abs}(x)) / \sinh(14.69 m^* T_k / \text{abs}(x)) \exp(-14.69 m^* T_D / \text{abs}(x)) \cos(2\pi F / \text{abs}(x) + \text{ph}) + c0$$

parameters A , m^* , T_k , T_D , F , ph , $c0$

the full first-harmonic LK formula in the standard engineering units: $X = 14.69 \cdot m^* \cdot T/B$ with m^* in units of the free electron mass. Fix T_k to the measurement temperature and the fit returns the effective mass and the Dingle temperature together — the two numbers a quantum oscillation experiment exists to produce.

Lifshitz–Kosevich amplitude against temperature (mass plot)

4 parameters

$$A0 \cdot (14.69 \cdot m_s \cdot x / B_m) / \sinh(14.69 \cdot m_s \cdot x / B_m) + c0$$

parameters $A0$, m_s , B_m , $c0$

the conventional way to measure the effective mass: take the oscillation amplitude at several temperatures and fit $X/\sinh X$ against T at the mean inverse field B_m . Everything else in the LK formula is temperature-independent and drops out.

Dingle plot

2 parameters

$$aD0 - aDs \cdot x$$

parameters $aD0$, aDs

$\ln[A \cdot \sinh(X)/X]$ against $1/B$. A straight line confirms a single scattering rate and its slope is $14.69 \cdot m^* \cdot TD$; curvature means more than one frequency is present, or the background subtraction was wrong.

Two-frequency Shubnikov–de Haas

9 parameters

$$A1 \cdot \exp(-a1/abs(x)) \cdot \cos(2\pi \cdot F1/abs(x) + p1) + A2 \cdot \exp(-a2/abs(x)) \cdot \cos(2\pi \cdot F2/abs(x) + p2) + c0$$

parameters $A1$, $a1$, $F1$, $p1$, $A2$, $a2$, $F2$, $p2$, $c0$

two extremal orbits — an electron and a hole pocket, or the neck and belly of the same surface. Two frequencies close together produce a beat whose node spacing measures their difference far more precisely than either frequency alone.

Three-frequency Shubnikov–de Haas

12 parameters

$$A1 \cdot \exp(-aD/abs(x)) \cdot \cos(2\pi \cdot F1/abs(x) + p1) + A2 \cdot \exp(-aD/abs(x)) \cdot \cos(2\pi \cdot F2/abs(x) + p2) + A3 \cdot \exp(-aD/abs(x)) \cdot \cos(2\pi \cdot F3/abs(x) + p3) + c0 + b1 \cdot x$$

parameters $A1$, aD , $F1$, $p1$, $A2$, $F2$, $p2$, $A3$, $F3$, $p3$, $c0$, $b1$

three orbits sharing one Dingle temperature, which is the physically honest constraint when they belong to the same sample: the scattering rate is a property of the crystal, not of the orbit.

Shubnikov–de Haas with the second harmonic

6 parameters

$$A1 \cdot \exp(-aD/abs(x)) \cdot \cos(2\pi \cdot F/abs(x) + ph) + A2 \cdot \exp(-2 \cdot aD/abs(x)) \cdot \cos(4\pi \cdot F/abs(x) + 2 \cdot ph + pi) + c0$$

parameters $A1$, aD , F , ph , $A2$, $c0$

the $r = 2$ term of the LK sum, damped by exactly twice the Dingle exponent and phase-locked to the fundamental. Its presence is what makes clean oscillations look saw-toothed rather than sinusoidal.

Shubnikov–de Haas with spin splitting

7 parameters

$$A \cdot \cos(\pi \cdot g_s \cdot m_s / 2) \cdot \exp(-aD/abs(x)) \cdot \cos(2\pi \cdot F/abs(x) + ph) + c0$$

parameters A , g_s , m_s , aD , F , ph , $c0$

the spin-reduction factor $\cos(\pi g m^*/2)$ multiplies the whole oscillation and can drive it through zero: at a spin-zero the amplitude vanishes entirely. Finding the angles where that happens is the standard way to measure the g factor.

de Haas–van Alphen magnetisation

8 parameters

$$A * \text{abs}(x)^{0.5} * (14.69 * \text{ms} * \text{Tk} / \text{abs}(x)) / \sinh(14.69 * \text{ms} * \text{Tk} / \text{abs}(x)) * \exp(-14.69 * \text{ms} * \text{TD} / \text{abs}(x)) * \sin(2 * \pi * F / \text{abs}(x) + \text{ph}) + c0 + \text{chi} * x$$

parameters A, ms, Tk, TD, F, ph, c0, chi

the same LK physics seen in the magnetisation rather than the resistivity, with the $\sqrt{B}$ prefactor the thermodynamic derivation carries and a sine rather than a cosine. The linear term is the ordinary background susceptibility.

Landau level fan (1/B against index)

2 parameters

$$(x + \text{gam}) / F$$

parameters gam, F

plot the inverse field of each extremum against its integer index and the result is a straight line through $-\gamma$. The intercept is the phase offset: $1/2$ for a conventional parabolic band, 0 for a Dirac band carrying a π Berry phase. This one number is the standard evidence for topological carriers.

Onsager fan with a finite-thickness correction

3 parameters

$$(x + \text{gam}) / F + \text{dq} * x^2$$

parameters gam, F, dq

the fan line bends when the Fermi surface is not a perfect cylinder, or when the Zeeman energy is comparable to the cyclotron energy. Fitting the curvature keeps the intercept honest instead of letting a curved fan masquerade as a shifted one.

Aharonov–Bohm oscillations (h/e , periodic in B)

5 parameters

$$A * \cos(2 * \pi * x / B_p + \text{ph}) + c0 + b1 * x$$

parameters A, Bp, ph, c0, b1

a ring threaded by flux: the conductance is periodic in B with period h/eS , so the period measures the enclosed area. Unlike SdH this is periodic in B itself, not in $1/B$ — that is how the two are told apart.

Altshuler–Aronov–Spivak oscillations ($h/2e$)

5 parameters

$$A2 * \cos(4 * \pi * x / B_p + \text{ph}) + c0 + b1 * x$$

parameters A2, Bp, ph, c0, b1

the time-reversed pair of paths encircles the ring twice, so the period halves. AAS survives ensemble averaging where the h/e oscillation does not, which is why an array of rings shows $h/2e$ and a single ring shows both.

Aharonov–Bohm with both h/e and $h/2e$ components

7 parameters

$$A1 * \cos(2 * \pi * x / B_p + p1) + A2 * \cos(4 * \pi * x / B_p + p2) + c0 + b1 * x$$

parameters A1, Bp, p1, A2, p2, c0, b1

one ring, both harmonics, with the second period locked to exactly half the first — fitting them with independent periods lets noise decide the ratio, which is the whole point of the measurement.

Weiss commensurability oscillations

5 parameters

$$A w \sqrt{\text{abs}(x)} * \exp(-aD/\text{abs}(x)) * \cos(2\pi * Fw/\text{abs}(x) - \pi/4) + c0 + b1 * x$$

parameters $Aw, aD, Fw, c0, b1$

a two-dimensional gas in a weak periodic potential: the resistance oscillates when the cyclotron diameter is commensurate with the lattice period. Low field, $-\pi/4$ phase, and it dies out where SdH is only beginning — the two rarely overlap.

Magnetophonon resonance

5 parameters

$$A m \exp(-g m / \text{abs}(x)) * \cos(2\pi * wL0 / \text{abs}(x)) + c0 + b1 * x$$

parameters $Am, gm, wL0, c0, b1$

resonant scattering when the cyclotron energy divides the optical phonon energy. Periodic in $1/B$ like SdH but essentially temperature-independent up to well above 100 K, where SdH has long since washed out.

Chirped oscillation (non-parabolic band)

6 parameters

$$A \exp(-aD/\text{abs}(x)) * \cos(2\pi * (F/\text{abs}(x) + F^2/x^2) + ph) + c0$$

parameters $A, aD, F, F^2, ph, c0$

when the band is not parabolic the frequency drifts with field and the oscillation is no longer strictly periodic in $1/B$. A single-frequency fit that leaves a residual growing towards one end of the field range usually needs this rather than a second pocket.

Beating of two close frequencies

6 parameters

$$2 * A b * \cos(2\pi * dF/\text{abs}(x)) * \cos(2\pi * Fm/\text{abs}(x) + ph) * \exp(-aD/\text{abs}(x)) + c0$$

parameters $Ab, dF, Fm, ph, aD, c0$

the product form of a two-frequency spectrum, written so the beat frequency ΔF and the mean frequency are the fitted parameters. Far better conditioned than fitting two nearly equal frequencies separately, and ΔF is what warping or spin splitting actually sets.

Angular dependence of a two-dimensional orbit

2 parameters

$$F0 / \text{abs}(\cos(\pi * (x - th0) / 180))$$

parameters $F0, th0$

a cylindrical Fermi surface has $F(\theta) = F0/\cos\theta$. Measuring the frequency as the field is tilted and fitting this is how two-dimensionality is demonstrated; a three-dimensional pocket departs from it well before 60° .

Josephson junction Fraunhofer pattern

4 parameters

$$Ic * \text{abs}(\text{sinc}(\pi * (x - x0) / Bp)) + Ib$$

parameters $Ic, x0, Bp, Ib$

the critical current of a wide junction against flux: $|\sin(\pi\Phi/\Phi0)/(\pi\Phi/\Phi0)|$. A clean Fraunhofer pattern with zeros reaching the axis is the standard evidence that the supercurrent is uniform across the junction.

SQUID interference pattern

4 parameters

$$I_c \cdot \text{abs}(\cos(\pi \cdot (x - x_0)/B_p)) + I_b$$

parameters I_c , x_0 , B_p , I_b

two junctions in a loop: the critical current modulates with period Φ_0/A_{loop} . The envelope is set by the single-junction Fraunhofer pattern, which is far wider because the loop encloses much more area than the junctions do.

Asymmetric SQUID interference

5 parameters

$$\sqrt{(I_{c1} + I_{c2})^2 \cos(\pi \cdot (x - x_0)/B_p)^2 + (I_{c1} - I_{c2})^2 \sin(\pi \cdot (x - x_0)/B_p)^2} + I_b$$

parameters I_{c1} , I_{c2} , x_0 , B_p , I_b

when the two junctions do not carry the same critical current the minima no longer reach zero, and the residual current at the minimum measures the asymmetry directly.

Coulomb blockade conductance peaks

7 parameters

$$G_p \cdot (\text{sech}((x - x_1)/w_c)^2 + \text{sech}((x - x_2)/w_c)^2 + \text{sech}((x - x_3)/w_c)^2 + \text{sech}((x - x_4)/w_c)^2) + G_c$$

parameters G_p , x_1 , w_c , x_2 , x_3 , x_4 , G_c

a quantum dot in the classical blockade regime: evenly spaced conductance peaks with the $\cosh^{-2}$ line shape of a thermally broadened resonance. The spacing gives the charging energy, the width gives the electron temperature.

Fabry-Pérot conductance oscillations

5 parameters

$$A_f \cdot \cos(2\pi x/\lambda_p + \phi) + b_0 + b_1 x$$

parameters A_f , λ_p , ϕ , b_0 , b_1

ballistic carriers bouncing between two partially transmitting contacts. Periodic in gate voltage or in length rather than in field, and the period converts directly to a cavity length once the lever arm is known.

Friedel oscillations

5 parameters

$$A_f \cdot \cos(2k_F x + \phi) / \max(x, 1e-12)^d + c_0$$

parameters A_f , k_F , ϕ , d , c_0

the screening cloud round an impurity does not decay monotonically — the sharp Fermi surface imprints a $2k_F$ ripple with a power-law envelope, $d = 3$ in three dimensions and $d = 1$ in one. Fit the period and you have measured k_F in real space.

RKKY interaction against separation

3 parameters

$$A_r \cdot (2k_F x \cos(2k_F x) - \sin(2k_F x)) / \max(2k_F x, 1e-9)^4 + c_0$$

parameters A_r , k_F , c_0

the exact three-dimensional RKKY range function: an oscillating exchange coupling with a $1/r^3$ envelope. This is what makes the coupling between magnetic layers alternate between ferro- and antiferromagnetic as the spacer thickness grows.

Dirac Landau level against field

3 parameters

```
E0 + cl*sqrt(abs(nL*x))
```

parameters E0, cl, nL

$E_n = E_0 + v_F \sqrt{(2e\hbar |n| B)}$ for a massless Dirac band: levels spread as $\sqrt{B}$ instead of the linear fan of a parabolic band. Pin the level index nL; cl = $v_F \sqrt{(2e\hbar)}$ in your energy unit

Magnetic hysteresis

22 MODELS

Loops, branch by branch, plus the anhysteretic curves underneath them and the coercivity laws that govern how they change with temperature, time and grain size. Split a loop into its two branches first (Chapter~6) — these models describe one branch, not both.

Anhysteretic Langevin curve

3 parameters

```
Ms*langevin(x/aL) + chi*x
```

parameters M_s , a_L , χ

the loss-free backbone of any hysteresis loop: classical moments in thermal equilibrium. $a = kT/\mu_0 m$ sets the field scale, and the linear term is the paramagnetic or diamagnetic background of everything else in the sample holder.

Anhysteretic Brillouin curve

4 parameters

```
Ms*brillouin(x/aB, Jq) + chi*x
```

parameters M_s , a_B , J_q , χ

the quantum-mechanical version, for a well-defined total angular momentum J rather than a classical vector. $J \rightarrow \infty$ recovers Langevin; $J = 1/2$ gives a plain tanh.

Jiles–Atherton anhysteretic (with mean-field coupling)

4 parameters

```
Ms*langevin((x + alJ*Ms*langevin(x/aL))/aL) + chi*x
```

parameters M_s , αJ , a_L , χ

the Langevin curve with the inter-domain coupling αM added to the applied field, resolved by one Picard step — accurate wherever $\alpha M_s/a$ is not large, which covers most soft materials. α is what makes the anhysteretic curve steeper than the moments alone would allow.

Hysteresis branch — tanh (ascending)

4 parameters

```
Ms*tanh((x + Hc)/aH) + chi*x
```

parameters M_s , H_c , a_H , χ

the lower branch of a loop, swept from negative to positive field: shifted right by the coercivity. Fit the two branches separately — a loop is not a function of H , and any model fitted to both at once returns their average, which is the anhysteretic curve and not the loop.

Hysteresis branch — tanh (descending)

4 parameters

```
Ms*tanh((x - Hc)/aH) + chi*x
```

parameters M_s , H_c , a_H , χ

the upper branch, swept from positive to negative field. Fitting both branches with this pair and differencing the two H_c values gives the coercivity without ever reading it off a graph.

Hysteresis branch — arctangent (Talukdar–Bailey)

4 parameters

$$(2*Ms/\pi)*\text{atan}((x + H_c)/aH) + \chi*x$$

parameters M_s , H_c , aH , χ

the same shape with much heavier tails: an arctan loop approaches saturation as $1/H$ where a tanh approaches it exponentially. Materials with a broad switching-field distribution follow the arctan far better.

Hysteresis branch — error function (Gaussian switching-field distribution)

4 parameters

$$Ms*\text{erf}((x + H_c)/(aH*\sqrt{2})) + \chi*x$$

parameters M_s , H_c , aH , χ

a Preisach model with a Gaussian distribution of switching fields integrates to an error function. aH is then literally the standard deviation of the switching field, which is the number a recording medium is specified by.

Two-phase hysteresis branch (soft and hard)

7 parameters

$$M1*\tanh((x + H1)/a1) + M2*\tanh((x + H2)/a2) + \chi*x$$

parameters $M1$, $H1$, $a1$, $M2$, $H2$, $a2$, χ

an exchange-spring or a mechanical mixture: a soft phase that switches at low field and a hard phase that does not. A visible kink or waist in the loop is the signature, and the two magnetisations give the phase fractions.

Exchange-biased hysteresis branch

5 parameters

$$Ms*\tanh((x - H_{eb} + H_c)/aH) + \chi*x$$

parameters M_s , H_{eb} , H_c , aH , χ

a ferromagnet pinned by an antiferromagnet: the whole loop shifts along the field axis by the exchange bias field while keeping its width. H_{eb} and H_c are separate parameters here precisely because they are separate physics — unidirectional versus uniaxial anisotropy.

Initial magnetisation curve

4 parameters

$$Ms*\text{langevin}(x/aL)*(1 - \exp(-x/H_s)) + \chi*x$$

parameters M_s , aL , H_s , χ

the virgin curve from a demagnetised state, which lies inside the loop and has no counterpart on the way back. The extra factor suppresses the response at very low field, where domain walls are still pinned.

Rayleigh law (low-field magnetisation)

2 parameters

$$\chi_0*x + \nu R*x^2$$

parameters χ_0 , νR

below the coercive field the initial susceptibility is constant and the first correction is quadratic. The Rayleigh constant ν measures irreversible wall motion, and the loop it predicts has an area proportional to νH^3 .

Rayleigh loop branch

3 parameters

$$\chi_0 x + \nu_R x^2 + (\nu_R/2)(H_m^2 - x^2)$$

parameters χ_0 , ν_R , H_m

the closed-form Rayleigh minor loop: two parabolic arcs meeting at the tips $\pm H_m$. Fit the ascending branch with this and the descending one with the sign of the last term reversed.

Law of approach to saturation with a $\sqrt{H}$ term

4 parameters

$$M_s(1 - aA/\sqrt{\max(x, 1e-12)} - bA/x^2) + \chi x$$

parameters M_s , aA , bA , χ

the standard $1/H^2$ approach plus the $1/\sqrt{H}$ term that planar defects and surface anisotropy contribute. Fitting only the high-field tail and forcing the $\sqrt{H}$ coefficient to zero systematically overestimates M_s .

Stoner–Wohlfarth astroid (switching field against angle)

2 parameters

$$H_0/(\abs{\cos(\pi x/180)}^{2/3} + \abs{\sin(\pi x/180)}^{2/3})^{1.5} + H_b$$

parameters H_0 , H_b

the classic angular dependence of the switching field of a single-domain particle, with a minimum of $H_0/2$ at 45° . A measured astroid that is shallower than this is the usual evidence that reversal is not coherent rotation.

Switching-field distribution (Gaussian)

4 parameters

$$Asf \exp(-(x - H_c)^2/(2sSF^2)) + c_0$$

parameters Asf , H_c , sSF , c_0

the derivative of a loop branch. Its centre is the coercivity and its width is the spread of switching fields — both far better determined from the peak than from reading intercepts off the loop itself.

Kneller law (coercivity against temperature)

3 parameters

$$H_{c0}(1 - \sqrt{x/T_b}) + H_{cr}$$

parameters H_{c0} , T_b , H_{cr}

thermal activation over the anisotropy barrier drives the coercivity to zero at the blocking temperature as $1 - \sqrt{T/T_b}$. The residual term allows for a fraction that stays blocked.

Sharrock law (coercivity against temperature, free exponent)

4 parameters

$$H_{s0}(1 - (cSx)^{1/nS}) + H_{cr}$$

parameters H_{s0} , cS , nS , H_{cr}

the generalisation used in recording media: $n = 2$ for coherent rotation, $n = 3/2$ for domain-wall-assisted reversal. Fitting n rather than assuming it tells you which mechanism is operating.

Herzer random anisotropy (coercivity against grain size)

2 parameters

$$kH \cdot x^6 + Hc0$$

parameters kH , $Hc0$

below the exchange length the anisotropy averages out over many grains and the coercivity collapses as D^6 . This is the whole basis of nanocrystalline soft magnets — six orders of magnitude in Hc for one order in grain size.

Coercivity above the exchange length

2 parameters

$$aG/x + Hc0$$

parameters aG , $Hc0$

the other side of the Herzer curve: once the grains are larger than the exchange length each one has its own easy axis and the coercivity falls as $1/D$. The maximum sits at the crossover.

Magnetic viscosity (against $\ln t$)

3 parameters

$$M0 - Sv \cdot \ln(\max(x, 1e-12)/t0)$$

parameters $M0$, Sv , $t0$

thermally activated decay of the remanence: linear in $\ln t$ over many decades. The viscosity coefficient S divided by the irreversible susceptibility gives the fluctuation field, and thence the activation volume.

Steinmetz core loss

2 parameters

$$kSt \cdot x^{bSt}$$

parameters kSt , bSt

loss per cycle against peak induction, with β between 1.6 and 2.6 for laminated steel. Purely empirical, universally used, and the exponent is the number a transformer designer needs.

Barkhausen avalanche size distribution

4 parameters

$$Ab \cdot x^{(-taB)} \cdot \exp(-x/x0) + c0$$

parameters Ab , taB , $x0$, $c0$

domain-wall avalanches follow a power law cut off by the finite driving rate. The exponent is near $3/2$ in the mean-field universality class, and finding it is the standard evidence that magnetisation reversal is a critical phenomenon.

Part V

Reference

Numerical methods

What actually runs, in enough detail to reproduce it or to distrust it.

Linear least squares

For a model linear in its parameters — polynomials, the logarithm, the reciprocal, the square root, and the amplitudes of every composite and symbolic form — the solution is exact and immediate. The design matrix Φ is built, the weighted normal equations $(\Phi^T W \Phi) \mathbf{c} = \Phi^T W \mathbf{y}$ are formed, a Tikhonov ridge of 10^{-12} times the mean diagonal is added for conditioning, and the system is solved by Gauss–Jordan elimination with partial pivoting. The inverse of the normal matrix is the covariance matrix. No iteration, no starting values, no local minima.

Levenberg–Marquardt

Everything non-linear. Each iteration:

1. Build the Jacobian by central differences.
2. Form $\mathbf{A} = \mathbf{J}^T \mathbf{W} \mathbf{J}$ and $\mathbf{g} = \mathbf{J}^T \mathbf{W} \mathbf{r}$.
3. Solve $(\mathbf{A} + \lambda \text{diag } \mathbf{A}) \delta = \mathbf{g}$ and test $\mathbf{p} + \delta$. On improvement, accept and set $\lambda \leftarrow \max(0.3\lambda, 10^{-12})$; otherwise $\lambda \leftarrow 10\lambda$ and try again, up to fourteen times per iteration, giving up past $\lambda = 10^{12}$.
4. Stop when no step improves, or when a relative improvement falls below 10^{-12} .

λ starts at 10^{-3} , interpolating between Gauss–Newton when the quadratic approximation is good and gradient descent when it is not.

THE DERIVATIVE STEP, AND A BUG WORTH KNOWING ABOUT

The step for the numerical derivative is $h = |p_j| \times 10^{-6}$, scaled to each parameter, with a fallback to the mean magnitude of the others when a parameter is exactly zero.

It was previously $h = \max(10^{-8}, |p_j| \times 10^{-6})$. That floor looks harmless until you fit a relaxation time of 2×10^{-13} s: the "derivative" is then the model evaluated fifty thousand parameter-widths away, the Jacobian column is meaningless, and τ never moves off its starting value. On a Drude–Smith test this showed as $\chi^2/\nu = 1524$ with τ frozen exactly where it started. With the step scaled to the parameter, the same fit converges to $\chi^2/\nu = 0.87$ from any start, including no starting values at all. If you have ever fitted femtosecond lifetimes or small energies in other software and found a parameter suspiciously stuck at its initial value, this is a bug class worth checking for.

Nelder–Mead

The composite search needs a derivative-free method, because its objective involves an inner linear solve and returns infinity on rejected parameter values. The standard simplex is used, with reflection, expansion, contraction and shrink, stopping when the spread of function values across the simplex falls below 10^{-12} relative.

Variable projection

Described in Chapter 13. Amplitudes are eliminated analytically at every step, so only shape parameters are searched. This halves or better the dimension of every composite fit.

Multistart strategy

For a typed or library equation:

1. Exploration runs on an evenly spaced subsample of at most 250 points.
2. Starting points: the supplied or default values; a greedy coordinate sweep that tries data-derived candidate values one parameter at a time (skipped when every starting value was supplied, since it can then only make them worse); and up to 140 random restarts — reduced to 10 when all values are hinted, and to 1 for a model with more than twelve parameters, where no random draw in forty dimensions will beat a data-derived start.
3. Candidates are sorted by χ^2 , and among near-ties by the smallest total parameter magnitude, so an aliased frequency or a runaway branch does not consume every polish slot.
4. The best six — two when fully hinted, one for a big model — are polished properly.
5. The polish itself runs on at most 3000 points, then takes thirty final steps on every point so that the reported χ^2 , errors and covariance come from the whole dataset.

Keeping the page responsive

Everything above runs on the browser's single thread, so a long fit would freeze the tab. Three things prevent it:

- **Slicing.** Levenberg–Marquardt is run in slices of a few iterations, carrying the parameters *and the damping factor* across each break so the path is identical to one long run. The slice size is chosen from an operation budget of about 10^6 elementary operations, counted as $n \times (k+1) \times k$.
- **A free yield.** The hand-back uses a MessageChannel message, not `setTimeout(0)`. Once callbacks nest, browsers clamp `setTimeout` to 4 ms, and a large fit yields thousands of times — that clamp alone added half a minute to a fifty-thousand-point file.
- **Cancellation.** Starting a new fit stops the old one at its next yield. Pasting a three-column file starts a fit, then answering the σ question starts another; without this the abandoned search kept running and the two interleaved.

The measured result on a fifty-thousand-point file: the longest uninterrupted block is about one second, and the page paints and scrolls throughout.

Cost management on large files

STAGE	POINTS USED
Named non-linear models, coarse pass	1500
Named non-linear models, polish	all, 60 iterations
Equation fitting, exploration	250
Equation fitting, polish	3000, then all for 30 iterations
Composite triage / shortlist / polish	80 / 200 / 800
Composite scoring	all
Symbolic exhaustive stage	900
Symbolic rate refinement	800, rescored on all
Library ranking sweep	300
Library refit of the leaders	all
Frequency periodogram	4000

Every reported χ^2 is computed on every point you supplied. Thinning is used only to *find* a solution, never to score one.

Special functions

Implemented directly, to the accuracy noted:

erf

Abramowitz & Stegun 7.1.26 rational approximation, about 1.5×10^{-7} absolute.

lgamma, gamma

Lanczos approximation, $g = 7$, nine coefficients, with the reflection formula below $\frac{1}{2}$.

digamma

Recurrence up to argument 6, then the asymptotic series to five terms.

voigt

The real part of the Faddeeva function by Humlicek's w4 algorithm, four regions, about 10^{-4} relative. A true Voigt profile, not a pseudo-Voigt.

dynes

$|\operatorname{Re}[(E - i\Gamma)/\sqrt{(E - i\Gamma)^2 - \Delta^2}]|$, written out in real algebra.

frota

$\operatorname{Re} \sqrt{(i\Gamma/(i\Gamma + \epsilon))}$, likewise.

brillouin

The Brillouin function with the small-argument limit of coth handled explicitly.

softplus

$\ln(1 + e^z)$, switching to the linear and exponential limits beyond ± 30 to avoid overflow.

Baseline solvers

Asymmetric least squares gives symmetric pentadiagonal normal equations, factorised as LDL^T with two sub-diagonals in $O(N)$ time. The polynomial baseline uses the same linear solver as everything else, on x mapped to $-1 \dots 1$, with fourteen rounds of clipping.

The PDF writer

Assembled by hand: header, catalogue, pages, page, content stream, and an image XObject carrying the plot as DCTDecode (JPEG) data, then a correct cross-reference table and trailer. Five objects, written byte by byte. It exists so that saving a figure needs no library and no network.

Troubleshooting

Symptoms, causes and what to do — including the ones that look like bugs and are not.

Loading

"need at least 3 usable rows"

Almost always the column assignment. Check that x is not pointing at a column of text. If the file is long-format, the split should be set up automatically; if it is not, set **split into datasets** by yourself. Also check for decimal commas — 1,842 is not a number.

Half the rows are skipped

Either those rows have a missing x or y , or a σ column is in use and their σ is missing, zero or negative. A zero uncertainty cannot be used because it demands infinite weight.

The numbers loaded but the plot is empty

A log axis with no positive values on it. The plot says so explicitly. Switch to linear.

Some points are missing from the plot

A log axis cannot place a zero or a negative number. The status box counts them for you.

Fitting

The curve runs through the middle of my loop

Because a loop is not a function of x , and the curve between the branches is the anhysteretic curve — the average of the two, which was never measured. Tick **split the sweep into its two branches** under the column pickers; the checkbox appears on its own whenever x doubles back. Each branch is then fitted separately and both are drawn. See Chapter 6.

A polynomial won, and the card calls it an interpolation

It is telling you that nothing named fitted. A polynomial has no shape of its own, so when its residuals also come out structured — long stretches of one sign rather than random — the ranking has fallen back on the family that can approximate anything smooth. Do not quote the coefficients and do not extrapolate. Look at the residual strip: whatever shape is left in it is the model you need. If it is a spectrum, use the peak finder; if it is a physical curve, switch on the materials library or type the form you expect.

My oscillations were ignored and a smooth curve won

If x is a magnetic field, the oscillation is periodic in $1/x$ and the Shubnikov–de Haas and de Haas–van Alphen models in the **Quantum oscillations** group are the ones to use — they are in the automatic search and are seeded from a periodogram run against $1/x$. If the oscillation sits on a steeply rising background, that is now removed inside the frequency scan rather than competing with it. If it is still missed, subtract the background explicitly (Chapter 16) and fit what is left.

"the fit did not converge"

Give starting values. Order of magnitude is usually enough. Then pin what you know, fit the rest, and unpin one at a time.

"no parameter values make this expression finite over your data"

A pole, a logarithm of a non-positive number, or a square root of a negative one somewhere in your range. Restrict the range, or guard the term with a comparison — $(x > E_g) * \sqrt{x - E_g}$.

The winner is a quintic / a three-term composite / something absurd

First check the weighting. On data spanning decades with no σ column, equal weighting makes the largest values dominate, and a flexible model that fits those wins fairly. Switch to relative. Second, check that **charge for the search** is on. Third, turn off families you know do not apply.

A parameter is stuck exactly at its starting value

It is not affecting the model — check for a typo in the expression, or a term multiplied by something that is zero.

"at a physical limit"

See Chapter 10. It means the data wants a value the physics forbids: either the value really is at the limit, or the model is wrong.

Huge percentage errors on parameters that visibly fit

They are correlated: the data constrains a combination, not the parts. Pin one and refit to confirm.

"no error estimate"

The covariance matrix was singular. Two parameters are exactly degenerate, or the fit is on a boundary.

Results that look wrong

χ^2/ν is enormous but the curve looks perfect

Your σ is too small — or, if you generated the data yourself, your noise model gives some points a σ near zero. A relative σ on an oscillation does exactly that at every zero crossing, which is a singularity rather than a measurement.

χ^2/ν is tiny

σ is overestimated, or the model has as many parameters as the data has information.

The residuals have a clear shape

The model has the wrong form. This outranks any χ^2 . An arc is wrong curvature; a wave is a missing periodic term; a step is something that changed during the measurement.

Two models have nearly identical χ^2

Your data does not distinguish them. Tick both overlay boxes and look at where they differ — then measure there, if you can.

Speed and responsiveness

A large file takes a long time

Expected: about fifteen seconds at five thousand points, a minute or so at fifty thousand, with the page responsive throughout. To go faster, turn off **Combinations** and **Physics library**, which are the two expensive families.

"Auto multi-peak" makes everything slow

It is the slow family, which is why it is off by default. Use the peak finder panel instead: it gives

you control and produces an editable equation.

The page really does freeze

It should not. If it does, note the number of points and which families were on — that is a bug worth reporting.

Things that are not bugs

- **Everything is gone after reopening the page.** By design. Only History persists.
- **The equation box emptied itself.** You loaded a different dataset. An equation written for one dataset, silently refitted to different numbers, is a trap.
- **Axis scales went back to linear.** Also on loading new data, so that a log axis carried over from the previous file cannot hide half of this one.
- **Background subtraction switched off.** A baseline is specific to the measurement it was estimated from.
- **The best physics model differs from the top of the table.** They answer different questions. The table ranks everything; the physics card reports the best *named physical law*, chosen by BIC within the library.
- **A model is listed but greyed as "too few points".** It has fewer than three effective degrees of freedom and is not allowed to win.

Glossary

Terms used in this manual and in the program.

AICc

Akaike information criterion with the small-sample correction. Goodness of fit plus an explicit charge per parameter. Only differences between models on the same data are meaningful.

Asymmetric least squares

A baseline estimator that weights points above the current curve much less than points below it, so peaks are ignored and the baseline threads underneath them.

BIC

Bayesian information criterion. Like AICc but charging $\ln N$ per parameter, so it becomes stricter as the dataset grows. Used inside the physics library to choose its leader.

Bound

A physical constraint on a parameter, written $A > 0$ or $0 < \eta < 1$. Enforced as an infinitely steep wall during the fit.

Chi-square (χ^2)

The weighted sum of squared residuals. With error bars it counts each miss in units of that point's own uncertainty.

Composite / combination

A sum of two named components with an optional constant.

Covariance matrix

$(J^T W J)^{-1}$ at the optimum. Its diagonal gives the parameter variances.

Degrees of freedom (ν)

$N - k$: points minus fitted parameters.

Effective parameters (k_{eff})

Actual parameters plus a charge for the size of the search the model was selected from.

Long format

Data with the runs stacked vertically and named in a label column.

Matching pursuit

Greedy selection: score every candidate term against the current residual, keep the best, subtract it, repeat. Used to prescreen the symbolic dictionary.

MDL

Minimum description length. The principle behind charging for the search: transmitting a model costs both its parameters and the identity of its form.

Parsimony rule

The bench's tie-break: the top spot goes to the simplest model whose χ^2 the leader does not beat by more than 2.5 times the improvement chance alone would supply.

Pinning

Fixing a parameter with ! so it is held and not counted in k .

Prominence

How far you must descend from a local maximum before you can reach a higher one. The robust measure of whether a bump is a band.

Pull

A residual divided by that point's σ . What the residual strip shows when error bars are present.

Railed

A parameter that has ended up sitting on one of its bounds.

Reduced chi-square

χ^2/ν . With error bars, ≈ 1 for a correct model.

Residual

$y - f(x)$ at a point.

RMSE

Root-mean-square residual, in the units of y .

Robust fitting

Iteratively reweighted least squares with Tukey's biweight, which gives far-out points zero weight instead of letting them bend the curve.

Savitzky-Golay

Smoothing or differentiation by fitting a local polynomial. Its second derivative is what reveals a shoulder.

Selection charge

$\log_2(\text{number of candidates})$ effective parameters, added to a model found by searching.

Symbolic search

Finding the best small sum of terms from a fixed dictionary, rather than fitting a named model.

Variable projection

Eliminating linear parameters analytically so that only the non-linear ones are searched.

Wide format

Data with each run in its own column.

Colophon

Curve Fit Bench is one HTML file with no dependencies. The fitting engine, the expression parser, the model library, the plot renderer and the PDF writer are all plain JavaScript in that file; nothing is fetched at run time except two typefaces, and the program works with the network switched off.

Equations are compiled by a hand-written tokenizer and recursive-descent parser straight to closure trees. There is no `eval` and no `new Function` anywhere in the program, so nothing you type can be executed as code and nothing can be injected into it.

This manual's model catalogue is generated directly from the program's own model list, so the two cannot drift apart.

Set in Bitstream Charter, Nimbus Sans and Nimbus Mono.